

键落星辰

ACM 算法竞赛模板

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Contents

1 动态规划

1.1 四边形不等式与树形背包优化 3
1.1.1 树形背包优化 3
1.2 $O(n \cdot \max a_i)$ Subset Sum 3

2 字符串

2.1 最小表示法 3
2.2 Manacher 3
2.3 KMP, exKMP 3
2.4 AC 自动机 3
2.5 Lydon Word Decomposition 3
2.6 后缀自动机 3
2.7 广义后缀自动机 4
2.8 SAMSA & 后缀树 4
2.9 后缀数组 4
2.10 回文树 4
2.11 字符串 Hash 类 4
2.12 String Conclusions 4

3 数论（不含博弈论）

3.1 Long Long $O(1)$ 乘, Barrett 5
3.2 exgcd, 逆元 5
3.3 CRT 中国剩余定理 5
3.4 Miller Rabin, Pollard Rho 5
3.5 扩展卢卡斯 5
3.6 阶乘取模 5
3.7 类欧几里得直线下格点统计 5
3.8 万能欧几里德 5
3.9 平方剩余 6
3.10 原根 6
3.11 min25筛 6
3.12 Pell 方程 6
3.13 解一元三次方程 6
3.14 Formulas 公式表 6
3.14.1 Mobius Inversion 6
3.14.2 杜教筛 6
3.14.3 降幂公式 6
3.14.4 其他常用公式 6
3.14.5 单位根反演 6
3.14.6 Arithmetic Function 6
3.14.7 Binomial Coefficients 7
3.14.8 Fibonacci Numbers, Lucas Numbers 7
3.14.9 Sum of Powers 7
3.14.10 Catalan Numbers 1, 1, 2, 5, 14, 42, 132, 429, 1430... 7
3.14.11 Motzkin Numbers 1, 1, 2, 4, 9, 21, 51, 127, 323, 835... 7
3.14.12 Derangement 错排数 0, 1, 2, 9, 44, 265, 1854, 14833... 7
3.14.13 Bell Numbers 1, 1, 2, 5, 15, 52, 203, 877, 4140 ... 7
3.14.14 Stirling Numbers 7
3.14.15 Eulerian Numbers 8
3.14.16 Harmonic Numbers, 1, 3/2, 11/6, 25/12, 137/60... 8
3.14.17 卡迈克尔函数 8
3.14.18 求拆分数 8
3.14.19 Bernoulli Numbers 1, 1/2, 1/6, 0, -1/30, 0, 1/42 ... 8
3.14.20 kMAX-MIN反演 8
3.14.21 伍德伯里矩阵不等式 8
3.14.22 Sum of Squares 8
3.14.23 枚举勾股数 Pythagorean Triple 8
3.14.24 四面体体积 Tetrahedron Volume 8
3.14.25 杨氏矩阵与钩子公式 8
3.14.26 常见博弈游戏 8
3.14.27 概率相关 8
3.14.28 邻接矩阵行列式的意义 8
3.14.29 Others (某些近似数值公式在这里) 9
3.15 Calculus, Integration Table 导数积分表 9
3.16 Zeller 日期公式 9
3.17 有理数二分: Stern-Brocot 树, Farey 序列 9
3.18 Constant Table 常数表 9

4 多项式与生成函数

4.1 FFT 10
4.2 NTT 10
4.3 MTT 任意模数卷积 10
4.4 多项式运算 10
4.4.1 多项式求逆 开根 $\ln \exp$ 10
4.4.2 多项式除法 取模 11
4.4.3 多点求值 11
4.4.4 插值 11
4.5 线性递推 11
4.6 Berlekamp-Massey 最小多项式 12
4.7 FWT 12
4.8 K 进制 FWT 12

5 线性代数

5.1 Simplex 单纯形 12
5.2 高斯消元最小范数解 13

6 数据结构

6.1 bitset 13
6.2 FHQ treap 13
6.3 RangeBIT 13
6.4 李超线段树 13
6.5 线段树分裂 14
6.6 左偏树 14
6.7 非递归线段树 14

6.7.1 区间加, 区间求最大值 14
6.8 LCT 动态树 14
6.9 Hash Table 15
6.10 基数排序 15

7 树上问题

7.1 虚树 15
7.2 树哈希 15

8 计算几何

8.1 二维几何基础 15
8.1.1 声明与宏 15
8.1.2 点与向量 15
8.1.3 线 16
8.1.4 三角形 16
8.1.5 三角形相关公式: 三角形心、海伦公式、欧拉定理、内接球、外接圆 16
8.1.6 超球坐标系 17
8.1.7 三维旋转公式 17
8.1.8 立体角公式 17
8.1.9 常用体积公式 17
8.1.10 扇形与圆弧重心 17
8.1.11 高维球体积 17
8.2 多边形 17
8.2.1 多边形基础 17
8.2.2 简单多边形内最长线段 17
8.2.3 $O(n^2)$ Ear Clipping 三角剖分 18
8.3 圆 18
8.3.1 圆基础 18
8.3.2 圆反演与阿波罗尼茨圆 19
8.3.3 圆并 19
8.3.4 多边形与圆交 19
8.3.5 圆上整点 20
8.4 凸几何 20
8.4.1 凸包 20
8.4.2 凸包询问: 凸包内、切点、交点、最近点 20
8.4.3 单插入动态凸包 21
8.4.4 旋转卡壳 21
8.4.5 闵可夫斯基和 21
8.4.6 直线半平面交 21
8.4.7 整数半平面交 22
8.5 点集算法 22
8.5.1 最近点对 22
8.5.2 最小覆盖球 22
8.5.3 Delaunay 三角剖分 23
8.6 球面与三维几何 23
8.6.1 球面基础与经纬度球面距离 23
8.6.2 三维几何基础操作 24
8.6.3 三维凸包 24
8.7 数值工具 24
8.7.1 黄金三分 24
8.7.2 自适应 Simpson 24

9 图论

9.1 Hopcroft-Karp $O(\sqrt{VE})$ 25
9.2 Hungarian $O(VE/w)$ 25
9.3 Shuffle 一般图最大匹配 $O(VE)$ 25
9.4 极大团计数 25
9.5 KM 最大权匹配 $O(V^3)$ 25
9.6 欧拉回路 25
9.7 Blossom 一般图最大匹配 $O(|V||E|)$ 26
9.8 2-SAT, 强连通分量 / Bitset Kosaraju 26
9.9 Tarjan 点双, 边双 26
9.10 Dominator Tree 支配树 26
9.11 环计数 27
9.12 Dinic 最大流 27
9.13 原始对偶费用流 27
9.14 网络流总结 27
9.15 网络流总结 27
9.16 Gomory-Hu 无向图最小割树 $O(V^3E)$ 28
9.17 Stoer-Wagner 无向图最小割 $O(VE + V^2 \log V)$ 28
9.18 弦图 28
9.19 Minimum Mean Cycle 最小平均值环 $O(n^2)$ 28
9.20 斯坦纳树 28
9.21 图论结论 29
9.21.1 最小乘积问题原理 29
9.21.2 最小环 29
9.21.3 度序列的可图性 29
9.21.4 切比雪夫距离与曼哈顿距离转化 29
9.21.5 树链的交 29
9.21.6 带修改MST 29
9.21.7 差分约束 29
9.21.8 李超线段树 29
9.21.9 Segment Tree Beats 29
9.21.10 二分图 29
9.21.11 稳定婚姻问题 29
9.21.12 三元环 29
9.21.13 图同构 29
9.21.14 竞赛图 Landau's Theorem 29
9.21.15 Ramsey Theorem $R(3,3)=6, R(4,4)=18$ 29
9.21.16 树的计数 Prufer序列 29
9.21.17 有根树计数 1,1,2,4,9,20,48,115,286,719,1842,4766 29
9.21.18 无根树计数 29
9.21.19 生成树计数 Kirchhoff's Matrix-Tree Theorem 29
9.21.20 有向图欧拉回路计数 BEST Theorem 29
9.21.21 Tutte Matrix 29
9.21.22 Edmonds Matrix 29

9.21.23	有向图无环定向, 色多项式	30
9.21.24	拟阵交问题	30
9.21.25	双极定向	30
9.21.26	图中的环	30
10	离线算法 (如 CDQ、莫队)	30
10.1	莫队二次离线	30
11	随机化与博弈论	30
11.1	Python Hint	30
11.2	Hacks: 指令集, 读入优化, Bitset, builtin, 随机种子	30
11.3	试机赛与纪律文件	31

1. 动态规划

1.1 四边形不等式与树形背包优化

```
1 //  $a \leq b \leq c \leq d: w(b, c) \leq w(a, d)$ ,
2    $\hookrightarrow w(a, c) + w(b, d) \leq w(a, d) + w(b, c)$ 
3 for (int len = 2; len <= n; ++len) {
4     for (int l = 1, r = len; r <= n; ++l, ++r) {
5         f[l][r] = INF;
6         for (int k = m[l][r - 1]; k <= m[l + 1][r]; ++k) {
7             if (f[l][r] > f[l][k] + f[k + 1][r] + w(l, r)) {
8                 f[l][r] = f[l][k] + f[k + 1][r] + w(l, r);
9                 m[l][r] = k;
10            }
11        }
12    }
```

1.1.1 树形背包优化

限制: 必须取与根节点相连的一个连通块。

转化: 一个点的子树对应于DFS序中的一个区间. 则每个点的决策为, 取该点, 或者舍弃该点对应的区间. 从后往前dp, 设 $f(i, v)$ 表示从后往前考虑到i号点, 总体积为V的最优价值, 设i号点对应的区间为 $[i, i + \text{size}_i - 1]$, 转移为 $f(i, v) = \max\{f(i + 1, V - v_i) + w_i, f(i + \text{size}_i, v)\}$.

如果要求任意连通块, 则点分治后转为指定根的连通块问题即可。

1.2 $O(n \cdot \max a_i)$ Subset Sum

```
1 int SubsetSum(vector<int> &a, int t) {
2     if (a.empty() || t < 0) return 0;
3     int B = *max_element(a.begin(), a.end());
4     int n = (int) a.size(), s = 0, i = 0;
5     while (i < n && s + a[i] <= t) s += a[i++];
6     if (i == n) return s;
7     vector<int> f(2 * B + 1, -1), pre(B + 1, -1);
8     f[s - (t - B)] = i;
9     for (; i < n; i++) {
10        s += a[i];
11        for (int d = 0; d <= B; d++) pre[d] = max(0, f[d + B]);
12        for (int d = B; d >= 0; d--)
13            f[d + a[i]] = max(f[d + a[i]], f[d]);
14        for (int d = 2 * B; d > B; --d)
15            for (int j = pre[d - B]; j < f[d]; j++)
16                f[d - a[j]] = max(f[d - a[j]], j);
17        for (i = 0; i <= B; i++) if (f[B - i] >= 0) return t - i;
18        return 0;
19    }
```

2. 字符串

2.1 最小表示法

```
1 int minimal_string(){// 下标从0开始
2     int k = 0, i = 0, j = 1;
3     while(k < n && i < n && j < n){
4         if(a[(i+k)%n] == a[(j+k)%n]) {k++; continue;}
5         if(a[(i+k)%n] > a[(j+k)%n]) i = i+k+1;
6         else j = j+k+1;
7         i += (i == j); k = 0; }
8     return min(i, j); }
```

2.2 Manacher

```
1 void read(){ // 两倍空间
2     s[0] = '~', s[+n] = '|';
3     char c = getchar();
4     while(c < 'a' || c > 'z') c = getchar();
5     while('a' <= c && c <= 'z') s[+n] = c, s[+n] = '|', c =
6         getchar();
7 }
8 void manacher(){
9     int mx = 0, mid = -1, ans = 0;
10    for(int i=1; i<n; i++){
11        if(i <= mx) h[i] = min(h[(mid<<1)-i], mx-i+1);
12        else h[i] = 1;
13        while(s[i-h[i]] == s[i+h[i]]) ++h[i];
14        if(h[i] + i - 1 > mx) mx = h[i] + i - 1, mid = i; } }
```

2.3 KMP, exKMP

```
1 void kmp(const char *s, int n) { // 1-based
2     fail[0] = fail[1] = 0;
3     for (int i = 1; i < n; i++) { int j = fail[i];
4         while (j && s[i + 1] != s[j + 1]) j = fail[j];
5         if (s[i + 1] == s[j + 1]) fail[i + 1] = j + 1;
6         else fail[i + 1] = 0; } }
7 void exkmp(const char *s, int *a, int n) { // 1-based
8     int l = 0, r = 0; a[1] = n;
9     for (int i = 2; i <= n; i++) {
```

```
10 | | a[i] = i > r ? 0 : min(r - i + 1, a[i - 1 + 1]);
11 | | while (i+a[i] <= n && s[1+a[i]] == s[i+a[i]]) a[i]++;
12 | | if (i + a[i] - 1 > r) {l = i; r = i + a[i] - 1;}}
```

```
1 void init(){
2     int l = 0, r = 0;
3     z[1] = 0;
4     for(int i=2; i<=n; i++){
5         if(i <= r && z[i-l+1] < r-i+1) z[i] = z[i-l+1];
6         else {
7             z[i] = max(0, r-i+1);
8             while(i+z[i]<=n && b[i+z[i]]==b[z[i]+1]) z[i]++;
9         }
10        if(i + z[i] - 1 > r) l = i, r = i + z[i] - 1;
11    }
12 }
```

2.4 AC 自动机

注意代码是以 0 为根的, 如果要 1-base 的话要改一下没有儿子时的逻辑。

```
1 struct AC{
2     int ch[N][26], fail[N], tag[N], tot;
3     void ins(string s, int _idx){
4         int x = 0;
5         for(auto c : s){
6             int v = c - 'a';
7             if(!ch[x][v]) ch[x][v] = ++tot;
8             x = ch[x][v]; tag[x] = _idx; }
9         void build(){
10            queue<int> q;
11            for(int i=0; i<26; i++) if(ch[0][i]) q.push(ch[0][i]);
12            while(!q.empty()){
13                int x = q.front(); q.pop();
14                for(int i=0; i<26; i++) {
15                    if(ch[x][i]) fail[ch[x][i]] = ch[fail[x]][i],
16                        q.push(ch[x][i]);
17                    else ch[x][i] = ch[fail[x]][i]; } } }
18 }
```

2.5 Lydon Word Decomposition

```
1 //满足s的最小后缀等于s本身的串称为Lyndon串。
2 //等价于: s是它自己的所有循环移位中唯一最小的一个。
3 //任意字符串s可以分解为  $s = s_1 s_2 s_k$ , 其中  $s_i$  是Lyndon串,
4    $\hookrightarrow s_i \geq s_{i+1}$ . 且这种分解方法是唯一的。
5 //后缀排序后, 排名的所有前缀最小值构成了 Ly 分解的左端点。
6 void mnsuf(char *s, int *mn, int n){ // 每个前缀的最小后缀
7     //1 - base, 求 Lyndon 分解去掉 mn 即可
8     for(int i = 1; i <= n; i++){
9         int j = i + 1, k = i; mn[i] = i;
10        for(; j <= n && s[k] <= s[j]; j++){
11            if(s[k] < s[j]) k = mn[j] = i;
12            else mn[j] = mn[k] + j - k, k++;
13        } for(; i <= k; i += j - k) {} } //lyn+s[i..i+k-j-1]
14 void mxsuf(char *s, int *mx, int n){ // 每个前缀的最大后缀
15     fill(mx + 1, mx + n + 1, 0); // 1 - base
16     for(int i = 1; i <= n; i++){
17         int j = i + 1, k = i; !mx[i] ? mx[i] = i : 0;
18         for(; j <= n && s[k] <= s[j]; j++){
19             !mx[j] ? mx[j] = i : 0;
20             if(s[k] > s[j]) k = i : k++; }
21         for(; i <= k; i += j - k) {} }
```

2.6 后缀自动机

```
1 struct SAM{ // 别忘记空间是两倍 改字符集记得全改了
2     int last = 1, tot = 1;
3     int ch[MAXN<<1][26], len[MAXN<<1], f[MAXN<<1];
4     void ins(char c){
5         c -= 'a';
6         int p = last, cur = last = ++tot;
7         len[cur] = len[p] + 1;
8         for(; p && !ch[p][c]; p=f[p]) ch[p][c] = cur;
9         if(!p) {f[cur] = 1; return;}
10        int q = ch[p][c];
11        if(len[q] == len[p] + 1) {f[cur] = q; return;}
12        int clone = ++tot;
13        for(int i=0; i<26; i++) ch[clone][i] = ch[q][i];
14        f[clone] = f[q], len[clone] = len[p] + 1;
15        f[q] = f[cur] = clone;
16        for(; p && ch[p][c] == q; p=f[p]) ch[p][c] = clone;
17    }
18 }
```

2.7 广义后缀自动机

```

1 struct SAM{
2   int ch[MAXN<<1][26], f[MAXN<<1], len[MAXN<<1], tot = 1;
3   int ins(int c, int last){
4     c -= 'a';
5     int cur, p;
6     if(ch[last][c]){
7       p = last, cur = ch[p][c];
8       if(len[p] + 1 == len[cur]) return cur;
9       int q = cur, clone = ++tot;
10      for(int i=0;i<26;i++) ch[clone][i] = ch[q][i];
11      len[clone] = len[p] + 1;
12      f[clone] = f[q], f[q] = clone;
13      for(;p&&ch[p][c]==q;p=f[p]) ch[p][c] = clone;
14      return clone;
15    }
16    p = last, cur = ++tot;
17    len[cur] = len[p] + 1;
18    for(;p&&!ch[p][c];p=f[p]) ch[p][c] = cur;
19    if(!p) {f[cur] = 1; return cur;}
20    int q = ch[p][c];
21    if(len[p] + 1 == len[q]) {f[cur] = q; return cur;}
22    int clone = ++tot;
23    for(int i=0;i<26;i++) ch[clone][i] = ch[q][i];
24    len[clone] = len[p] + 1;
25    f[clone] = f[q], f[q] = f[cur] = clone;
26    for(;p&&ch[p][c]==q;p=f[p]) ch[p][c] = clone;
27    return cur;
28  }
29  void ins(char* s){
30    int last = 1;
31    for(int i=0;s[i];i++) last = ins(s[i], last);
32  }
33 } sam;

```

2.8 SAMSA & 后缀树

```

1 bool vis[MAXN * 2]; char s[MAXN];
2 int id[MAXN * 2], ch[MAXN * 2][26], height[MAXN], stamp = 0;
3 void dfs(int x) {
4   if (id[x]) { height[stamp++] = val[last];
5   | sa[stamp] = id[x]; last = x; }
6   for (int c = 0; c < 26; c++)
7   | if (ch[x][c]) dfs(ch[x][c]);
8   last = par[x]; }
9 int main() { last = ++cnt; scanf("%s", s + 1);
10 | int n = strlen(s + 1); for (int i = n; i--;) {
11 | | expand(s[i] - 'a'); id[last] = i; }
12 | vis[1] = true; for (int i = 1; i <= cnt; i++) if (id[i])
13 | | for (int x = i, pos = n; x && !vis[x]; x = par[x]){
14 | | | vis[x] = true; pos -= val[x] - val[par[x]];
15 | | | ch[par[x]][s[pos + 1] - 'a'] = x; }
16 | dfs(1); for (int i = 1; i <= n; i++)
17 | | printf("%d%c", sa[i], i < n ? ' ' : '\n');
18 | for (int i = 1; i < n; i++) printf("%d%c", height[i],
19 | | i < n ? ' ' : '\n'); return 0; }

```

2.9 后缀数组

```

1 // height[i] = lcp(sa[i], sa[i - 1])
2 // 如果有多组数据, 全部都清空到 max(n, m)
3 constexpr int MAXN = 1000005;
4 void get_sa(char *s, int n, int *sa,
5 | int *rnk, int *height) { // 1-based
6   static int buc[MAXN], id[MAXN], p[MAXN], t[MAXN];
7   int m = 300;
8   for (int i = 1; i <= n; i++) buc[rnk[i] = s[i]]++;
9   for (int i = 1; i <= m; i++) buc[i] += buc[i - 1];
10  for (int i = n; i--;) sa[buc[rnk[i]]--] = i;
11  memset(buc, 0, sizeof(int) * (m + 1));
12  for (int k = 1, cnt = 0; cnt != n; k *= 2, m = cnt) {
13    cnt = 0;
14    for (int i = n; i > n - k; i--) id[++cnt] = i;
15    for (int i = 1; i <= n; i++)
16    | if (sa[i] > k) id[++cnt] = sa[i] - k;
17    for (int i = 1; i <= n; i++) buc[p[i]=rnk[id[i]]]++;
18    for (int i = 1; i <= m; i++) buc[i] += buc[i - 1];
19    for (int i = n; i--;) sa[buc[p[i]]--] = id[i];
20    memset(buc, 0, sizeof(int) * (m + 1));
21    memcpy(t, rnk, sizeof(int) * (n + 1));
22    t[n + 1] = 0; // 记得清空 n + 1
23    cnt = 0; for (int i = 1; i <= n; i++) {
24    | if (t[sa[i]] != t[sa[i - 1]] ||
25    | | t[sa[i] + k] != t[sa[i - 1] + k]) cnt++;

```

```

26 | | rnk[sa[i]] = cnt; } }
27 | for (int i = 1; i <= n; i++) sa[rnk[i]] = i;
28 | for (int i = 1, k = 0; i <= n; i++) { if (k) k--;
29 | | if (rnk[i] > 1) while (sa[rnk[i] - 1] + k <= n &&
30 | | | s[i + k] == s[sa[rnk[i] - 1] + k]) k++;
31 | | height[rnk[i]] = k; } } // 两个都要判, 否则会左/右越界
32 char s[MAXN]; int sa[MAXN], rnk[MAXN], height[MAXN];
33 int main() { scanf("%s", s + 1); int n = strlen(s + 1);
34 | get_sa(s, n, sa, rnk, height); }

```

2.10 回文树

0 的子树是长为偶数的串, 1 的子树是长为奇数的。0 代表空串, 1 没有意义。

```

1 struct PAM{ // sum[i]: 以i结尾的回文串的个数
2   int ch[N][26], len[N], fail[N], tot = 1, now = 1, sum[N];
3   void build(){
4     fail[0] = 1, len[1] = -1;
5     s[0] = '#';
6     for(int i=1;s[i];i++){
7       int j, v = (s[i] - 'a');
8       while(s[i-len[now]-1] != s[i]) now = fail[now];
9       if(!ch[now][v]){
10        len[++tot] = len[now] + 2;
11        j = fail[now];
12        while(s[i-len[j]-1] != s[i]) j = fail[j];
13        fail[tot] = ch[j][v];
14        ch[now][v] = tot;
15        sum[tot] = sum[fail[tot]] + 1;
16      }
17      now = ch[now][v];
18    }
19  }
20 } pam; // 定理: 长度为n的字符串S最多有n个本质不同回文串

```

2.11 字符串 Hash 类

Random primes generated at Mon Aug 31 21:26:51 2026
1e12 1012341230111 1034563452511 1045674564269 1045674568643
1e13 9912341238929 9934563450143 9934563451873 10345634560397
2e13 19123412342747 19123412346611 20123412341083 20567856782549
1e15 994567456743373 1012341234125821 1056785678566141
1e17 99345634563452129 101234123412346807 104567456745674651
1e18 1012341234123419393 1034563456345634041 1045674567456747479

```

1 static constexpr u128 inv = []() {
2   | u128 ret = P;
3   | for (int i = 0; i < 6; i++) ret *= 2 - ret * P;
4   | return ret; }();
5 constexpr u128 chk = u128(-1) / P;
6 bool check(i128 a, i128 b) { // 注意: i128
7   | if (a < b) swap(a, b);
8   | return (a - b) * inv <= chk; }

```

2.12 String Conclusions

双回文串

如果 $s = x_1x_2 = y_1y_2 = z_1z_2, |x_1| < |y_1| < |z_1|, x_2, y_2, z_2$ 是回文串, 则 x_1 和 z_2 也是回文串。

Border 和周期

如果 r 是 S 的一个 border, 则 $|S| - r$ 是 S 的一个周期。

如果 p 和 q 都是 S 的周期, 且满足 $p + q \leq |S| + \gcd(p, q)$, 则 $\gcd(p, q)$ 也是一个周期。

字符串匹配与 Border

若字符串 S, T 满足 $2|S| \geq |T|$, 则 S 在 T 中所有匹配位置成等差数列。

若 S 的匹配次数大于 2, 则等差数列的周期恰好等于 S 的最小周期。

Border 的结构

字符串 S 的所有不小于 $|S|/2$ 的 border 长度组成一个等差数列。

字符串 S 的所有 border 按长度排序后可分成 $O(\log |S|)$ 段, 每段是一个等差数列。

回文串 Border

回文串长度为 t 的后缀是一个回文后缀, 等价于 t 是该串的后缀。因此回文后缀的长度也可以划分成 $O(\log |S|)$ 段。

子串最小后缀

设 $s[p..n]$ 是 $s[i..n], (l \leq i \leq r)$ 中最小者, 则 $\text{minisuf}(l, r)$ 等于 $s[p..r]$ 的最短非空 border。 $\text{minisuf}(l, r) = \min\{s[p..r], \text{minisuf}(r - 2^k + 1, r)\}, (2^k < rl + 1 \leq 2^{k+1})$ 。

子串最大后缀

从左往右扫, 用 set 维护后缀的字典序递减的单调队列, 并在对应时刻添加“小于事件”点以便在之后修改队列; 查询直接在 set 里 lower_bound 。

3. 数论（不含博弈论）

3.1 Long Long $O(1)$ 乘, Barrett

```

1 LL modmul(LL a, LL b, LL M) { // skip2004, M < 63bit
2   LL ret = (LL)((__int128)a * b % M);
3   return ret < 0 ? ret + M : ret; }
4 ULL modmul(ULL a, ULL b, LL M) { // orz@CF, M in 63 bit
5   return (ULL)((u128)a * b % (ULL)M); }
6 // use int128 instead if M > 63 bit
7 struct DIV {
8   ULL p, ip;
9   void init (ULL _p) { p = _p; ip = -1llu / p; }
10  ULL mod (ULL x) { // x < 2 ^ 64
11    ULL q = ULL((u128)ip * x) >> 64;
12    ULL r = x - q * p;
13    return r >= p ? r - p : r;
14 } }; // speedup only when mod is not const

```

3.2 exgcd, 逆元

```

1 LL exgcd(LL a, LL b, LL &x, LL &y) {
2   if (b == 0) return x = 1, y = 0, a;
3   LL t = exgcd(b, a % b, y, x);
4   y -= a / b * x; return t; }
5 LL inv(LL x, LL m) {
6   LL a, b; exgcd(x, m, a, b); return (a % m + m) % m; }

```

递推逆元: $\text{inv}(i) \equiv (P - P/i) \cdot \text{inv}(P \bmod i)$

3.3 CRT 中国剩余定理

```

1 bool crt_merge(LL a1, LL m1, LL a2, LL m2, LL &A, LL &M) {
2   LL c = a2 - a1, d = __gcd(m1, m2); //合并两个模方程
3   if(c % d) return 0; // gcd(m1, m2) | (a2 - a1) 时才有解
4   c = (c % m2 + m2) % m2; c /= d; m1 /= d; m2 /= d;
5   c = c * inv(m1 % m2, m2) % m2; //0逆元可任意值
6   M = m1*m2*d; A = (c * m1 % M * d % M + a1) % M; return 1; } //有解

```

3.4 Miller Rabin, Pollard Rho

```

1 mt19937_64 rng(123);
2 #define rand() (LL)(rng() & LONGLONG_MAX)
3 const LL BASE[] =
4   { 2, 325, 9375, 28178, 450775, 9780504, 1795265022 }; //
5   ↳ deterministic for 64 bit
6 struct miller_rabin {
7   bool check (LL M, LL base) {
8     LL a = M - 1;
9     while (~a & 1) a >>= 1;
10    LL w = qpow (base, a, M); // qpow should use mul
11    for (; a != M - 1 && w != 1 && w != M - 1; a <= 1)
12      w = mul (w, M);
13    return w == M - 1 || (a & 1) == 1; }
14 bool solve (LL a) { // O((3 or 7) · log n · mul)
15   if (a < 4) return a > 1;
16   if (~a & 1) return false;
17   for (LL base : BASE) {
18     LL chk = base % a;
19     if (chk == 0) continue;
20     if (!check (a, chk)) return false;
21   }
22   return true; } };
23 miller_rabin is_prime;
24 LL get_factor (LL a, LL seed) { // O(n^{1/4} · log n · mul)
25   LL x = rand() % (a - 1) + 1, y = x;
26   for (int head = 1, tail = 2; ; ) {
27     x = mul (x, x, a); x = (x + seed) % a;
28     if (x == y) return a;
29     LL ans = gcd (abs (x - y), a);
30     if (ans > 1 && ans < a) return ans;
31     if (++head == tail) { y = x; tail <= 1; } } }
32 void factor (LL a, vector<LL> &d) {
33   if (a <= 1) return;
34   if (is_prime.solve (a)) d.push_back (a);
35   else {
36     LL f = a;
37     for (; f >= a; f = get_factor (a, rand() % (a - 1) +
38       ↳ 1));
39     factor (a / f, d);
40     factor (f, d); } }

```

3.5 扩展卢卡斯

```

1 int l, M, A[33], p[33], P[33];

```

```

2 U fac(int k, LL n) { // 求 n! mod pk^tk, 返回值 U{ 不包含 pk 的
3   ↳ 值, pk 出现的次数 }
4   if (!n) return U{1, 0};
5   LL x = n/p[k], y = n/P[k], ans = 1, i;
6   if (y) { // 求出循环节的答案
7     for (i = 2; i < P[k]; i++) if (i%p[k]) ans = ans * i % P[k];
8     ans = Pw(ans, y, P[k]);
9   } for (i = y * P[k]; i < n; i++) if (i%p[k]) ans = ans * i % P[k]; // 求零
10  ↳ 散部分
11  U z = fac(k, x); return U{ans * z.x % P[k], x + z.z};
12 } LL get(int k, LL n, LL m) { // 求 C(n, m) mod pk^tk
13   U a = fac(k, n), b = fac(k, m), c = fac(k, n - m); // 分三部分求解
14   return Pw(p[k], a.z - b.z - c.z, P[k]) * a.x % P[k] *
15     ↳ inv(b.x, P[k]) % P[k] * inv(c.x, P[k]) % P[k];
16 } LL CRT() { // CRT 合并答案
17   LL w, y, x, ans = 0;
18   fr(i, 1, l) w = M/P[i], exgcd(w, P[i], x, y),
19     ↳ ans = (ans + w * x % M * A[i]) % M;
20   return (ans + M) % M;
21 } LL C(LL n, LL m) { // 求 C(n, m)
22   fr(i, 1, l) A[i] = get(i, n, m);
23   return CRT();
24 } LL exLucas(LL n, LL m, int MM) {
25   if (m < 0 || m > n) return 0;
26   l = 0; M = MM; LL jj = M; // 求 C(n, m) mod M, M = prod(pi^ki),
27     ↳ O(pi^kilg^2n)
28   for (LL i = 2; i * i <= jj; i++) if (jj % i == 0)
29     for (p[++l] = i, P[l] = 1; jj % i == 0; P[l] *= p[l]) jj /= i;
30   if (jj > 1) l++, p[l] = P[l] = jj;
31   return C(n, m); }

```

3.6 阶乘取模

```

1 // n! mod p^q Time : O(pq^2 log^2 n / log p)
2 // Output : {a, b} means a * p^b
3 using Val = unsigned long long; // Val 需要 mod p^q 意义下 + *
4 typedef vector<Val> poly;
5 poly polymul(const poly &a, const poly &b) {
6   int n = (int) a.size(); poly c (n, Val(0));
7   for (int i = 0; i < n; ++i) {
8     for (int j = 0; i + j < n; ++j) {
9       c[i + j] = c[i + j] + a[i] * b[j]; } }
10  return c; } Val choo[70][70];
11 poly polyshift(const poly &a, Val delta) {
12   int n = (int) a.size(); poly res (n, Val(0));
13   for (int i = 0; i < n; ++i) { Val d = 1;
14     for (int j = 0; j <= i; ++j) {
15       res[i - j] = res[i - j] + a[i] * choo[i][j] * d;
16       d = d * delta; } } return res; }
17 void prepare(int q) {
18   for (int i = 0; i < q; ++i) { choo[i][0] = Val(1);
19     for (int j = 1; j <= i; ++j)
20       choo[i][j] = choo[i-1][j-1] + choo[i-1][j]; } }
21 pair<Val, LL> fact(LL n, LL p, LL q) {
22   assert(1 <= q && q <= 70); prepare((int)q); Val ans = 1;
23   for (int r = 1; r < p; ++r) {
24     poly x (q, Val(0)), res (q, Val(0));
25     res[0] = 1; LL _res = 0; x[0] = r; LL _x = 0;
26     if (q > 1) x[1] = p, _x = 1;
27     LL m = (n - r + p) / p;
28     while (m) { if (m & 1) {
29       res = polymul(res, polyshift(x, _res)); _res += _x; }
30     m >>= 1; x = polymul(x, polyshift(x, _x)); _x += _x;
31     ↳ }
32     ans = ans * res[0]; }
33   LL cnt = n / p; if (n >= p) { auto tmp = fact(n / p, p, q);
34     ↳ ans = ans * tmp.first; cnt += tmp.second; }
35   return {ans, cnt}; }

```

3.7 类欧几里得直线下格点统计

```

1 // Σ_{i=0}^{n-1} ⌊ (a+bi)/m ⌋, n, m, a, b > 0
2 LL solve(LL n, LL a, LL b, LL m) {
3   if (b == 0) return n * (a / m);
4   if (a >= m) return n * (a / m) + solve(n, a % m, b, m);
5   if (b >= m) return (n-1) * n / 2 * (b/m) + solve(n, a, b % m, m);
6   return solve((a + b * n) / m, (a + b * n) % m, m, b); }

```

3.8 万能欧几里德

```

1 Val work(LL P, LL R, LL Q, LL n, Val VU, Val VR) {
2   // (Px+R)/Q, 1<=x<=i, 经过整数先U再R
3   if (!(((i128)n * P + R) / Q)) return ksm(VR, n);

```

```

4 | if(P>=Q) return work(P%Q,R,Q,n, VU, ksm(VU, P/Q) * VR);
5 | Val res; swap(VU,VR);
6 | res = ksm(VU, (Q-R-1)/P)*VR;
7 | LL m = ((i128)n * P + R) / Q;
8 | res = res * work(Q, (Q-R-1)%P, P, m-1, VU, VR);
9 | return res * ksm(VU, n - ((i128)m*Q - R - 1) / P); }

```

3.9 平方剩余

```

1 // x^2=a (mod p), 0 <=a<p, 返回 true or false 代表是否存在解
2 // p必须是质数, 若是多个单质数的乘积, 可以分别求解再用CRT合并
3 // 复杂度为 O(log n)
4 void multiply(ll &c, ll &d, ll a, ll b, ll w) {
5 | int cc = (a * c + b * d % MOD * w) % MOD;
6 | int dd = (a * d + b * c) % MOD; c = cc, d = dd; }
7 bool solve(int n, int &x) {
8 | if (n==0) return x=0,true; if (MOD==2) return x=1,true;
9 | if (power(n, MOD / 2, MOD) == MOD - 1) return false;
10 | ll c = 1, d = 0, b = 1, a, w;
11 | // finding a such that a^2 - n is not a square
12 | do { a = rand() % MOD; w = (a * a - n + MOD) % MOD;
13 | | if (w == 0) return x = a, true;
14 | | while (power(w, MOD / 2, MOD) != MOD - 1);
15 | for (int times = (MOD + 1) / 2; times; times >>= 1) {
16 | | if (times & 1) multiply(c, d, a, b, w);
17 | | multiply(a, b, a, b, w); }
18 | // x = (a + sqrt(w)) ^ ((p + 1) / 2)
19 | return x = c, true; }

```

3.10 原根

定义 使得 $a^x \bmod m = 1$ 的最小的 x , 记作 $\delta_m(a)$. 若 $a \equiv g^s \bmod m$, 其中 g 为 m 的一个原根. 则虽然 s 随 g 的不同取值有所不同, 但是必然满足 $\delta_m(a) = \gcd(s, \varphi(m))$.

性质 $\delta_m(a^k) = \frac{\delta_m(a)}{\gcd(\delta_m(a), k)}$

k 次剩余 给定方程 $x^k \equiv a \bmod m$, 求所有解. 若 k 与 $\varphi(m)$ 互质, 则可以直接求出 k 对 $\varphi(m)$ 的逆元. 否则, 将 k 拆成两部分, $k = uv$, 其中 $u \perp \varphi(m)$, $v | \varphi(m)$, 先求 $x^v \equiv a \bmod m$, 则 $ans = x^{u^{-1}}$. 下面讨论 $k | \varphi(m)$ 的问题. 任取一原根 g , 对两侧取离散对数, 设 $x = g^s$, $a = g^t$, 其中 t 可以用BSGS求出, 则问题转化为求出所有的 s 满足 $ks \equiv t \bmod \varphi(m)$, exgcd 即可求解, 显然有解的条件是 $k | \delta_m(a)$.

3.11 min25筛

```

1 typedef long long ll;
2
3 ll n;
4 ll pri[N], sta[M], g[M]; int tot, mod;
5 int getid(ll x) {
6 | if(x <= sqrt(n)) return x;
7 | return tot + 1 - (n / x);
8 }
9
10 int solve(ll n, int id) {
11 | if(n < pri[id]) return 0;
12 | if(n / pri[id] < pri[id])
13 | | return (g[getid(n)] - (id > 1 ? g[pri[id - 1]] : 0) +
14 | | mod) % mod;
15 | ll e = n / pri[id]; int d = pri[id], Sum = 0;
16 | while(e) {
17 | | Sum = (Sum + (ll)d * (d - 1 + mod) % mod * (solve(e,
18 | | id + 1) + 1)) % mod;
19 | | d = (ll)d * pri[id] % mod; e /= pri[id];
20 | }
21 | return (Sum + solve(n, id + 1)) % mod;
22 }
23
24 void init(){
25 | for(int i=1; i<=pri[0]; i++){
26 | | for(int j=tot; j>=1; j--){
27 | | | if(sta[j] / pri[i] < pri[i]) break;
28 | | | int num = (g[getid(sta[j] / pri[i])] - g[pri[i] -
29 | | | 1] + mod) % mod;
30 | | | g[j] = (g[j] - (ll)pri[i] * pri[i] % mod * num %
31 | | | mod + mod) % mod;
32 | | }
33 }

```

3.12 Pell 方程

```

1 // x^2 - n * y^2 = 1 最小正整数根, n 为完全平方数时无解

```

```

2 // x_{k+1} = x_0 x_k + n y_0 y_k
3 // y_{k+1} = x_0 y_k + y_0 x_k
4 pair<LL, LL> peLL(LL n) {
5 | static LL p[N], q[N], g[N], h[N], a[N];
6 | p[1] = q[0] = h[1] = 1; p[0] = q[1] = g[1] = 0;
7 | a[2] = (LL)(floor(sqrt(1(n) + 1e-7L)));
8 | for(int i = 2; ; i++) {
9 | | g[i] = -g[i - 1] + a[i] * h[i - 1];
10 | | h[i] = (n - g[i] * g[i]) / h[i - 1];
11 | | a[i + 1] = (g[i] + a[2]) / h[i];
12 | | p[i] = a[i] * p[i - 1] + p[i - 2];
13 | | q[i] = a[i] * q[i - 1] + q[i - 2];
14 | | if(p[i] * p[i] - n * q[i] * q[i] == 1)
15 | | | return {p[i], q[i]}; }

```

3.13 解一元三次方程

```

1 double a(p[3]), b(p[2]), c(p[1]), d(p[0]);
2 double k(b / a), m(c / a), n(d / a);
3 double p(-k * k / 3. + m);
4 double q(2. * k * k * k / 27 - k * m / 3. + n);
5 Complex omega[3] = {Complex(1, 0), Complex(-0.5, 0.5 *
6 | sqrt(3)), Complex(-0.5, -0.5 * sqrt(3))};
7 Complex r1, r2; double delta(q * q / 4 + p * p * p / 27);
8 if (delta > 0) {
9 | r1 = cubrt(-q / 2. + sqrt(delta));
10 | r2 = cubrt(-q / 2. - sqrt(delta));
11 } else {
12 | r1 = pow(-q / 2. + pow(Complex(delta), 0.5), 1. / 3);
13 | r2 = pow(-q / 2. - pow(Complex(delta), 0.5), 1. / 3); }
14 for(int _ (0); _ < 3; _++) {
15 | Complex x = -k/3. + r1*omega[_] + r2*omega[_* 2 % 3]; }

```

3.14 Formulas 公式表

3.14.1 Mobius Inversion

$$F(n) = \sum_{d|n} f(d) \Rightarrow f(n) = \sum_{d|n} \mu(d) F\left(\frac{n}{d}\right)$$

$$F(n) = \sum_{n|d} f(d) \Rightarrow f(n) = \sum_{n|d} \mu\left(\frac{d}{n}\right) F(d) \quad \text{引理}$$

$$[x=1] = \sum_{d|x} \mu(d), \quad x = \sum_{d|x} \varphi(d), \quad \frac{1}{k} \sum_{i=0}^{k-1} \omega_k^{in} = [k | n]$$

3.14.2 杜教筛

$$S_\varphi(n) = \frac{n(n+1)}{2} - \sum_{d=2}^n S_\varphi\left(\left\lfloor \frac{n}{d} \right\rfloor\right)$$

$$S_\mu(n) = 1 - \sum_{d=2}^n S_\mu\left(\left\lfloor \frac{n}{d} \right\rfloor\right)$$

3.14.3 降幂公式

$$a^k \equiv a^{k \bmod \varphi(p) + \varphi(p)}, \quad k \geq \varphi(p)$$

3.14.4 其他常用公式

$$\sum_{i=1}^n [(i, n) = 1] i = n \frac{\varphi(n) + e(n)}{2}$$

$$\sum_{i=1}^n \sum_{j=1}^i [(i, j) = d] = S_\varphi\left(\left\lfloor \frac{n}{d} \right\rfloor\right)$$

$$\sum_{i=1}^n \sum_{j=1}^m [(i, j) = d] = \sum_{d|k} \mu\left(\frac{k}{d}\right) \left\lfloor \frac{n}{k} \right\rfloor \left\lfloor \frac{m}{k} \right\rfloor$$

$$\sum_{i=1}^n f(i) \sum_{j=1}^{\left\lfloor \frac{n}{i} \right\rfloor} g(j) = \sum_{i=1}^n g(i) \sum_{j=1}^{\left\lfloor \frac{n}{i} \right\rfloor} f(j)$$

3.14.6 Arithmetic Function

$$(p-1)! \equiv -1 \pmod{p}$$

$$a > 1, m, n > 0, \text{ then } \gcd(a^m - 1, a^n - 1) = a^{\gcd(m, n)} - 1$$

$$\mu^2(n) = \sum_{d^2|n} \mu(d)$$

$$a > b, \gcd(a, b) = 1, \text{ then } \gcd(a^m - b^m, a^n - b^n) = a^{\gcd(m, n)} - b^{\gcd(m, n)}$$

$$\prod_{k=1, \gcd(k, m)=1}^m k \equiv \begin{cases} -1 & \pmod{m, m=4, p^q, 2p^q} \\ 1 & \pmod{m, \text{otherwise}} \end{cases}$$

$$\sigma_k(n) = \sum_{d|n} d^k = \prod_{i=1}^{\omega(n)} \frac{p_i^{(a_i+1)k} - 1}{p_i^k - 1}$$

$$J_k(n) = n^k \prod_{p|n} \left(1 - \frac{1}{p^k}\right)$$

3.14.5 单位根反演

$$\sum_{i=0}^{\left\lfloor \frac{n}{k} \right\rfloor} C_n^{ik}$$

$$\frac{1}{k} \sum_{i=0}^{k-1} \omega_k^{in} = [k | n]$$

反演

$$Ans = \sum_{i=0}^n C_n^i [k | i]$$

$$= \sum_{i=0}^n C_n^i \left(\frac{1}{k} \sum_{j=0}^{k-1} \omega_k^{ij}\right)$$

$$= \frac{1}{k} \sum_{i=0}^n C_n^i \sum_{j=0}^{k-1} \omega_k^{ij}$$

$$= \frac{1}{k} \sum_{j=0}^{k-1} \left(\sum_{i=0}^n C_n^i (\omega_k^j)^i\right)$$

$$= \frac{1}{k} \sum_{j=0}^{k-1} (1 + \omega_k^j)^n$$

另, 如果要求的是 $[n \% k = t]$, 其实就是 $[k | (n - t)]$. 同理推式子即可.

$J_k(n)$ is the number of k -tuples of positive integers all less than or equal to n that form a coprime $(k+1)$ -tuple together with n .

$$\sum_{\delta|n} J_k(\delta) = n^k$$

$$\sum_{i=1}^n \sum_{j=1}^n [\gcd(i, j) = 1] ij = \sum_{i=1}^n i^2 \varphi(i)$$

$$\sum_{\delta|n} \delta^s J_r(\delta) J_s\left(\frac{n}{\delta}\right) = J_{r+s}(n)$$

$$\sum_{\delta|n} \varphi(\delta) d\left(\frac{n}{\delta}\right) = \sigma(n), \sum_{\delta|n} |\mu(\delta)| = 2^{\omega(n)}$$

$$\sum_{\delta|n} 2^{\omega(\delta)} = d(n^2), \sum_{\delta|n} d(\delta^2) = d^2(n)$$

$$\sum_{\delta|n} d\left(\frac{n}{\delta}\right) 2^{\omega(\delta)} = d^2(n), \sum_{\delta|n} \frac{\mu(\delta)}{\delta} = \frac{\varphi(n)}{n}$$

$$\sum_{\delta|n} \frac{\mu(\delta)}{\varphi(\delta)} = \prod_{p|n} \frac{p-2}{p-1}, \sum_{\delta|n} \frac{\mu^2(\delta)}{\varphi(\delta)} = \frac{n}{\varphi(n)} \quad (n=3: \frac{1}{2}, \frac{3}{2})$$

$$n|(\varphi(a^n) - 1)$$

$$\sum_{\substack{1 \leq k \leq n \\ \gcd(k, n)=1}} f(\gcd(k-1, n)) = \varphi(n) \sum_{d|n} \frac{(\mu * f)(d)}{\varphi(d)}$$

$$\varphi(\text{lcm}(m, n)) \varphi(\gcd(m, n)) = \varphi(m) \varphi(n)$$

$$\sum_{\delta|n} d^3(\delta) = \left(\sum_{\delta|n} d(\delta)\right)^2$$

$$d(uv) = \sum_{\delta|\gcd(u, v)} \mu(\delta) d\left(\frac{u}{\delta}\right) d\left(\frac{v}{\delta}\right)$$

$$\sigma_k(u) \sigma_k(v) = \sum_{\delta|\gcd(u, v)} \delta^k \sigma_k\left(\frac{uv}{\delta^2}\right)$$

$$\mu(n) = \sum_{k=1}^n [\gcd(k, n) = 1] \cos 2\pi \frac{k}{n}$$

$$\varphi(n) = \sum_{k=1}^n [\gcd(k, n) = 1] = \sum_{k=1}^n \gcd(k, n) \cos 2\pi \frac{k}{n}$$

$$\begin{cases} S(n) = \sum_{k=1}^n (f * g)(k) \\ \sum_{k=1}^n S\left(\lfloor \frac{n}{k} \rfloor\right) = \sum_{i=1}^n f(i) \sum_{j=1}^{\lfloor \frac{n}{i} \rfloor} (g * 1)(j) \end{cases}$$

$$\begin{cases} S(n) = \sum_{k=1}^n (f \cdot g)(k), g \text{ completely multiplicative} \\ \sum_{k=1}^n S\left(\left\lfloor \frac{n}{k} \right\rfloor\right) g(k) = \sum_{k=1}^n (f * 1)(k) g(k) \end{cases}$$

3.14.7 Binomial Coefficients

C	0	1	2	3	4	5	6	7	8	9	10
0	1										
1	1	1									
2	1	2	1								
3	1	3	3	1							
4	1	4	6	4	1						
5	1	5	10	10	5	1					
6	1	6	15	20	15	6	1				
7	1	7	21	35	35	21	7	1			
8	1	8	28	56	70	56	28	8	1		
9	1	9	36	84	126	126	84	36	9	1	
10	1	10	45	120	210	252	210	120	45	10	1

$$\binom{n}{k} \equiv [n \& k = k] \pmod{2}$$

$$\sum_{k=0}^n \binom{k}{m} = \binom{n+1}{m+1}$$

$$\sqrt{1+z} = 1 + \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \times 2^{2k-1} \binom{2k-2}{k-1} z^k$$

$$\sum_{k=0}^r \binom{r-k}{m} \binom{s+k}{n} = \binom{r+s+1}{m+n+1}$$

$$C_{n,m} = \binom{n+m}{m} - \binom{n+m}{m-1}, n \geq m$$

$$\binom{n}{k} = (-1)^k \binom{k-n-1}{k}, \sum_{k \leq n} \binom{r+k}{k} = \binom{r+n+1}{n}$$

$$\binom{n_1 + \dots + n_p}{m} = \sum_{k_1 + \dots + k_p = m} \binom{n_1}{k_1} \dots \binom{n_p}{k_p}$$

3.14.8 Fibonacci Numbers, Lucas Numbers

$$F(z) = \frac{z}{1-z-z^2}$$

$$\hat{\phi} = \frac{1-\sqrt{5}}{2}$$

$$\sum_{k=1}^n f_k = f_{n+2} - 1, \sum_{k=1}^n f_k^2 = f_n f_{n+1}$$

$$\sum_{k=0}^n f_k f_{n-k} = \frac{1}{5}(n-1)f_n + \frac{2}{5}n f_{n-1}$$

$$\frac{f_{2n}}{f_n} = f_{n-1} + f_{n+1}$$

$$f_1 + 2f_2 + 3f_3 + \dots + n f_n = n f_{n+2} - f_{n+3} + 2$$

$$\gcd(f_m, f_n) = f_{\gcd(m, n)}$$

$$f_n^2 + (-1)^n = f_{n+1} f_{n-1}$$

$$f_{n+k} = f_n f_{k+1} + f_{n-1} f_k$$

$$f_{2n+1} = f_n^2 + f_{n+1}^2$$

$$(-1)^k f_{n-k} = f_n f_{k-1} - f_{n-1} f_k$$

```
def fib(n): # F(n), F(n+1)
    if not n: return (0, 1)
    a, b = fib(n >> 1)
    c = a * (2 * b - a)
    d = a * a + b * b
    if n & 1:
        return (d, c + d)
    else:
        return (c, d)
```

$$\text{Modulo } f_n, f_{mn+r} \equiv \begin{cases} f_r, & m \bmod 4 = 0; \\ (-1)^{r+1} f_{n-r}, & m \bmod 4 = 1; \\ (-1)^n f_r, & m \bmod 4 = 2; \\ (-1)^{r+1+n} f_{n-r}, & m \bmod 4 = 3. \end{cases}$$

$$L_0 = 2, L_1 = 1, L_n = L_{n-1} + L_{n-2} = \left(\frac{1+\sqrt{5}}{2}\right)^n + \left(\frac{1-\sqrt{5}}{2}\right)^n$$

$$L(x) = \frac{2-x}{1-x-x^2}$$

除了 $n=0, 4, 8, 16$, L_n 是素数, 则 n 是素数.

$$\phi = \frac{1+\sqrt{5}}{2}, \hat{\phi} = \frac{1-\sqrt{5}}{2}$$

$$F_n = \frac{\phi^n - \hat{\phi}^n}{\sqrt{5}}, L_n = \phi^n + \hat{\phi}^n$$

$$\frac{L_n + F_n \sqrt{5}}{2} = \left(\frac{1+\sqrt{5}}{2}\right)^n$$

3.14.9 Sum of Powers

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2}\right)^2$$

$$\sum_{i=1}^n i^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

$$\sum_{i=1}^n i^5 = \frac{n^2(n+1)^2(2n^2+2n-1)}{12}$$

3.14.10 Catalan Numbers 1, 1, 2, 5, 14, 42, 132, 429, 1430...

$$c_0 = 1, c_n = \sum_{i=0}^{n-1} c_i c_{n-1-i} = c_{n-1} \frac{4n-2}{n+1} = \frac{\binom{2n}{n}}{n+1} = \binom{2n}{n} - \binom{2n}{n-1}$$

$$c(x) = \frac{1 - \sqrt{1-4x}}{2x}$$

Usage: n 对话框序列; n 个点满二叉树; $n \times n$ 的方格左下到右上不过对角线方案数; 凸 $n+2$ 边形三角形分割数; n 个数的出栈方案数; $2n$ 个顶点连接, 线段两两不交的的方案数.

类卡特兰数 从 $(1, 1)$ 出发走到 (n, m) , 只能向右或者向上走, 不能越过 $y = x$ 这条线 (即保证 $x \geq y$). 当 $n \geq m$ 时, 合法方案数是 $\binom{n+m-2}{m-1} - \binom{n+m-2}{m-2}$ (例如 $(3, 2)$ 时为 $3-1=2$); 当 $n < m$ 时, 合法方案数为 0.

3.14.11 Motzkin Numbers 1, 1, 2, 4, 9, 21, 51, 127, 323, 835...

圆上 n 点间画不相交弦的方案数. 选 n 个数 $k_1, k_2, \dots, k_n \in \{-1, 0, 1\}$, 保证 $\sum_i k_i (1 \leq a \leq n)$ 非负且所有数总和为 0 的方案数.

$$M_{n+1} = M_n + \sum_i^{n-1} M_i M_{n-1-i} = \frac{(2n+3)M_n + 3nM_{n-1}}{n+3}$$

$$M_n = \sum_{i=0}^{\lfloor \frac{n}{2} \rfloor} \binom{n}{2k} \text{Catalan}(k)$$

$$M(X) = \frac{1-x-\sqrt{1-2x-3x^2}}{2x^2}$$

3.14.12 Derangement 错排数 0, 1, 2, 9, 44, 265, 1854, 14833...

$$D_1 = 0, D_2 = 1, D_n = n! \left(\frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + \frac{(-1)^n}{n!} \right)$$

$$D_n = (n-1)(D_{n-1} + D_{n-2})$$

3.14.13 Bell Numbers 1, 1, 2, 5, 15, 52, 203, 877, 4140 ...

n 个元素集合划分的方案数.

$$B_n = \sum_{k=1}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\}, B_{n+1} = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} B_k$$

$$B_p^{m+n} \equiv m B_n + B_{n+1} \pmod{p}$$

$$B(x) = \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n = e^{e^x - 1}$$

3.14.14 Stirling Numbers

第一类 n 个元素集合分作 k 个非空轮换方案数.

$$\begin{bmatrix} n+1 \\ k \end{bmatrix} = n \begin{bmatrix} n \\ k \end{bmatrix} + \begin{bmatrix} n \\ k-1 \end{bmatrix}$$

$$s(n, k) = (-1)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}$$

n\k	0	1	2	3	4	5	6
0	1						
1	0	1					
2	0	1	1				
3	0	2	3	1			
4	0	6	11	6	1		
5	0	24	50	35	10	1	
6	0	120	274	225	85	15	1
7	0	720	1764	1624	735	175	21

第二类 把 n 个元素集合分作 k 个非空子集方案数.

$$\begin{Bmatrix} n+1 \\ k \end{Bmatrix} = k \begin{Bmatrix} n \\ k \end{Bmatrix} + \begin{Bmatrix} n \\ k-1 \end{Bmatrix}$$

$$m! \begin{Bmatrix} n \\ m \end{Bmatrix} = \sum_k \binom{m}{k} k^n (-1)^{m-k}$$

n\k	0	1	2	3	4	5	6
0	1						
1	0	1					
2	0	1	1				
3	0	1	3	1			
4	0	1	7	6	1		
5	0	1	15	25	10	1	
6	0	1	31	90	65	15	1
7	0	1	63	301	350	140	21

3.14.15 Eulerian Numbers

n\k	0	1	2	3	4	5	6
1	1						
2	1	1					
3	1	4	1				
4	1	11	11	1			
5	1	26	66	26	1		
6	1	57	302	302	57	1	
7	1	120	1191	2416	1191	120	1

3.14.16 Harmonic Numbers, 1, 3/2, 11/6, 25/12, 137/60...

$$H_n = \sum_{k=1}^n \frac{1}{k}, \sum_{k=1}^n H_k = (n+1)H_n - n$$

$$\sum_{k=1}^n k H_k = \frac{n(n+1)}{2} H_n - \frac{n(n-1)}{4}$$

$$\sum_{k=1}^n \binom{k}{m} H_k = \binom{n+1}{m+1} \left(H_{n+1} - \frac{1}{m+1} \right)$$

3.14.18 求拆分数

```
def penta(k):
    return k*(3*k-1)//2
def compute_partition(goal):
    p = [1]
    for n in range(1, goal+1):
        p.append(0)
        for k in range(1, n+1):
            c = (-1)**(k+1)
            for t in [penta(k), penta(-k)]:
                if (n-t) >= 0:
                    p[n] = p[n] + c*p[n-t]
    return p
```

$$\Phi(x) = \prod_{n=1}^{\infty} (1 - x^n) = \sum_{n=-\infty}^{\infty} (-1)^k x^{k(3k-1)/2}$$

3.14.19 Bernoulli Numbers 1, 1/2, 1/6, 0, -1/30, 0, 1/42 ...

$$B(x) = \sum_{i \geq 0} \frac{B_i x^i}{i!} = \frac{x}{e^x - 1}$$

$$B_n = [n=0] - \sum_{i=0}^{n-1} \binom{n}{i} \frac{B_i}{n-k+1}, \sum_{i=0}^n \binom{n+1}{i} B_i = 0$$

$$S_n(m) = \sum_{i=0}^{m-1} i^n = \sum_{i=0}^n \binom{n}{i} B_{n-i} \frac{m^{i+1}}{i+1}$$

$$B_0 = 1, B_1 = -\frac{1}{2}, B_4 = -\frac{1}{30}, B_6 = \frac{1}{42}, B_8 = -\frac{1}{30}, \dots$$

(除了 $B_1 = -\frac{1}{2}$ 以外, 伯努利数的奇数项都是 0.)

自然数幂次和关于次数的EGF:

$$F(x) = \sum_{k=0}^{\infty} \frac{\sum_{i=0}^n i^k}{k!} x^k$$

$$= \sum_{i=0}^n e^{ix} = \frac{e^{(n+1)x} - 1}{e^x - 1}$$

$$\begin{bmatrix} n+1 \\ 2 \end{bmatrix} = n! H_n \text{ (see 3.14.16)}$$

$$x^n = \sum_k \begin{bmatrix} n \\ k \end{bmatrix} (-1)^{n-k} x^k$$

$$x^{\bar{n}} = \sum_k \begin{bmatrix} n \\ k \end{bmatrix} x^k$$

For fixed k , EGF:

$$\sum_{n=0}^{\infty} \begin{bmatrix} n \\ k \end{bmatrix} \frac{x^n}{n!} = \frac{x^k}{k!} \left(\frac{-\ln(1-x)}{x} \right)^k$$

$$(k = 1 : -\ln(1-x))$$

$$x^n = \sum_k \begin{bmatrix} n \\ k \end{bmatrix} x^{\bar{k}}$$

$$= \sum_k \begin{bmatrix} n \\ k \end{bmatrix} (-1)^{n-k} x^{\bar{k}}$$

For fixed k , EGF and OGF:

$$\sum_{n=0}^{\infty} \begin{Bmatrix} n \\ k \end{Bmatrix} \frac{x^n}{n!} = \frac{x^k}{k!} \left(\frac{e^x - 1}{x} \right)^k$$

$$\sum_{n=0}^{\infty} \begin{Bmatrix} n \\ k \end{Bmatrix} x^n = x^k \prod_{i=1}^k (1 - ix)^{-1}$$

$$\begin{Bmatrix} n \\ k \end{Bmatrix} = (k+1) \begin{Bmatrix} n-1 \\ k \end{Bmatrix} + (n-k) \begin{Bmatrix} n-1 \\ k-1 \end{Bmatrix}$$

$$x^n = \sum_k \begin{Bmatrix} n \\ k \end{Bmatrix} \binom{x+k}{n}$$

$$\begin{Bmatrix} n \\ m \end{Bmatrix} = \sum_{k=0}^m \binom{n+1}{k} (m+1-k)^n (-1)^k$$

3.14.17 卡迈克尔函数

卡迈克尔函数表示模 m 剩余系下最大的阶, 即 $\lambda(m) = \max_{a \perp m} \delta_m(a)$.

容易看出, 若 $\lambda(m) = \varphi(m)$, 则 m 存在原根.

该函数可由下述方法计算: 分解质因数 $m = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_t^{\alpha_t}$. 则 $\lambda(m) = \text{lcm}(\lambda(p_1^{\alpha_1}), \lambda(p_2^{\alpha_2}), \dots, \lambda(p_t^{\alpha_t}))$. 其中对奇质数 p , $\lambda(p^\alpha) = (p-1)p^{\alpha-1}$. 对 2 的幂, $\lambda(2^k) = 2^{k-2}$, $s.t. k \geq 3$. $\lambda(4) = \lambda(2) = 2$.

记 $p(n)$ 表示 n 的拆分数, $f(n, k)$ 表示将 n 拆分且每种数字使用次数必须小于等于 k 的拆分数. 则

$$P(x)\Phi(x) = 1, F(x^k)\Phi(x) = 1$$

暴力拆开卷积, 可以得到将 1, -1, 2, -2... 带入五边形数 $(-1)^k x^{k(3k-1)/2}$ 中, 由于小于 n 的五边形数只有 $\sqrt{n}$ 个, 可以 $O(n\sqrt{n})$ 计算答案:

$$p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) + \dots,$$

$$f(n, k) = p(n) - p(n-k) - p(n-2k) + p(n-5k) + p(n-7k) - \dots$$

3.14.20 kMAX-MIN反演

$$k\text{MAX}(S) = \sum_{T \subset S, T \neq \emptyset} (-1)^{|T|-k} C_{|T|-1}^{k-1} \text{MIN}(T)$$

代入 $k = 1$ 即为 MAX-MIN 反演:

$$\text{MAX}(S) = \sum_{T \subset S, T \neq \emptyset} (-1)^{|T|-1} \text{MIN}(T)$$

3.14.21 伍德伯里矩阵不等式

$$(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$$

该等式可以动态维护矩阵的逆, 令 $C = [1]$, U, V 分别为 $1 \times n$ 和 $n \times 1$ 的向量, 这样可以构造出 UCV 为只有某行或者某列不为 0 的矩阵, 一次修改复杂度为 $O(n^2)$.

3.14.22 Sum of Squares

$r_k(n)$ 表示用 k 个平方数组成 n 的方案数. 假设:

$$n = 2^{a_0} p_1^{2a_1} \dots p_r^{2a_r} q_1^{b_1} \dots q_s^{b_s}$$

其中 $p_i \equiv 3 \pmod{4}$, $q_i \equiv 1 \pmod{4}$, 那么

$$r_2(n) = \begin{cases} 0 & \text{if any } a_i \text{ is a half-integer} \\ 4 \prod_{i=1}^r (b_i + 1) & \text{if all } a_i \text{ are integers} \end{cases}$$

$r_3(n) > 0$ 当且仅当 n 不满足 $4^a(8b+7)$ 的形式 (a, b 为整数).

3.14.23 枚举勾股数 Pythagorean Triple

枚举 $x^2 + y^2 = z^2$ 的三元组: 可令 $x = m^2 - n^2$, $y = 2mn$, $z = m^2 + n^2$, 枚举 m 和 n 即可 $O(n)$ 枚举勾股数. 判断素勾股数方法: m, n 至少一个为偶数并且 m, n 互质, 那么 x, y, z 就是素勾股数.

3.14.24 四面体体积 Tetrahedron Volume

If U, V, W, u, v, w are lengths of edges of the tetrahedron (first three form a triangle; u opposite to U and so on)

$$V = \frac{\sqrt{4u^2v^2w^2 - \sum_{cyc} u^2(v^2 + w^2 - U^2)^2 + \prod_{cyc} (v^2 + w^2 - U^2)}}{12}$$

3.14.25 杨氏矩阵与钩子公式

满足: 格子 (i, j) 没有元素, 则它右边和上边相邻格子也没有元素; 格子 (i, j) 有元素 $a[i][j]$, 则它右边和上边相邻格子要么没有元素, 要么有元素且比 $a[i][j]$ 大.

计数: $F_1 = 1, F_2 = 2, F_n = F_{n-1} + (n-1)F_{n-2}$, $F(x) = e^{x + \frac{x^2}{2}}$

钩子公式: 对于给定形状 λ , 不同杨氏矩阵的个数为:

$$d_\lambda = \frac{n!}{\prod h_\lambda(i, j)}$$

$h_\lambda(i, j)$ 表示该格子右边和上边的格子数量加 1.

3.14.26 常见博弈游戏

Nim-K游戏 n 堆石子轮流拿, 每次最多

可以拿 k 堆石子, 谁走最后一步输. 结论: 把每一堆石子的 sg 值 (即石子数量) 二进制分解, 先手必败当且仅当每一位二进制位上 1 的个数是 $(k+1)$ 的倍数.

Anti-Nim游戏 n 堆石子轮流拿, 谁走最后一步输. 结论: 先手胜当且仅当 1. 所有堆石子数都为 1 且游戏的 SG 值为 0 (即有偶数个孤单堆-每堆只有 1 个石子数) 2. 存在某堆石子数大于 1 且游戏的 SG 值不为 0.

斐波那契博弈 有一堆物品, 两人轮流取物品, 先手最少取一个, 至多无上限, 但不能把物品取完, 之后每次取的物品数不能超过上一次取的物品数的二倍且至少为一件, 取走最后一件物品的人获胜. 结论: 先手胜当且仅当物品数 n 不是斐波那契数.

威佐夫博弈 有两堆石子, 博弈双方每次可以取一堆石子中的任意个, 不能不取, 或

者取两堆石子中的相同个. 先取完者赢. 结论: 求出两堆石子 A 和 B 的差值 C , 如果 $\left\lfloor C * \frac{\sqrt{5}+1}{2} \right\rfloor = \min(A, B)$ 那么后手赢, 否则先手赢.

约瑟夫环 n 个人 $0, \dots, n-1$, 令 $f_{i,m}$ 为 i 个人报 m 的胜利者, 则 $f_{i,m} = 0, f_{i,m} = (f_{i-1,m} + m) \bmod i$.

阶梯Nim 在一个阶梯上, 每次选一个台阶上任意个式子移到下一个台阶上, 不可移动者输. 结论: SG 值等于奇数层台阶上石子数的异或和. 对于树形结构也适用, 奇数层节点上所有石子数异或起来即可.

图上博弈 给定无向图, 先手从某点开始走, 只能走相邻且未走过的点, 无法移动者输. 对该图求最大匹配, 若某个点不一定在最大匹配中则先手必败, 否则先手必胜.

最大最小定理求纳什均衡点 在二人零和博弈中, 可以用以下方式求出一个纳什均衡点: 在博弈双方中任选一方, 求混合策略 p 使得对方选择任意一个纯策略时, 己方的最小收益最大 (等价于对方的最大收益最小). 据此可以求出双方在此局面下的最优期望得分, 分别等于己方最大的最小收益和对方最小的最大收益. 一般而言, 可以得到形如

$$\max_p \min_i \sum_{p_j \in p} p_j w_{i,j}, \text{ s.t. } p_j \geq 0, \sum p_j = 1$$

的形式. 当 $\sum p_j w_{i,j}$ 可以表示成只与 i 有关的函数 $f(i)$ 时, 可以令初始时 $p_i = 0$, 不断调整 $\sum p_j w_{i,j}$ 最小的那个 i 的概率 p_i , 直至无法调整或者 $\sum p_j = 1$ 为止.

3.14.27 概率相关

$$D(X) = E(X - E(X))^2 = E(X^2) - (E(X))^2, D(X+Y) = D(X) + D(Y), D(aX) = a^2 D(X)$$

$$E[x] = \sum_{i=1}^{\infty} P(X \geq i)$$

m 个数的方差:

$$s^2 = \frac{\sum_{i=1}^m x_i^2}{m} - \bar{x}^2$$

3.14.28 邻接矩阵行列式的意义

在无向图中取若干个环, 一种取法权值就是边权的乘积, 对行列式的贡献是 $(-1)^{\text{even}}$, 其中 even 是偶环的个数.

3.14.29 Others (某些近似数值公式在这里)

$$S_j = \sum_{k=1}^n x_k^j$$

$$h_m = \sum_{1 \leq j_1 < \dots < j_m \leq n} x_{j_1} \cdots x_{j_m}, \quad H_m = \sum_{1 \leq j_1 \leq \dots \leq j_m \leq n} x_{j_1} \cdots x_{j_m}$$

$$h_n = \frac{1}{n} \sum_{k=1}^n (-1)^{k+1} S_k h_{n-k}$$

$$H_n = \frac{1}{n} \sum_{k=1}^n S_k H_{n-k}$$

$$\sum_{k=0}^n k c^k = \frac{n c^{n+2} - (n+1) c^{n+1} + c}{(c-1)^2}$$

$$\sum_{i=1}^n \frac{1}{n} \approx \ln\left(n + \frac{1}{2}\right) + \frac{1}{24(n+0.5)^2} + \Gamma, \quad (\Gamma \approx 0.5772156649015328606065)$$

$$n! = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \left(1 + \frac{1}{12n} + \frac{1}{288n^2} + O\left(\frac{1}{n^3}\right)\right)$$

$$\max\{x_a - x_b, y_a - y_b, z_a - z_b\} - \min\{x_a - x_b, y_a - y_b, z_a - z_b\}$$

$$= \frac{1}{2} \sum_{cyc} |(x_a - y_a) - (x_b - y_b)|$$

$$(a+b)(b+c)(c+a) = \frac{(a+b+c)^3 - a^3 - b^3 - c^3}{3}$$

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2), \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$n \bmod 2 = 1:$$

$$a^n + b^n = (a+b)(a^{n-1} - a^{n-2}b + a^{n-3}b^2 - \dots - ab^{n-2} + b^{n-1})$$

划分问题: n 个 $k-1$ 维向量最多把 k 维空间分为 $\sum_{i=0}^k C_n^i$ 份.

3.15 Calculus, Integration Table 导数积分表

$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$	$(\arctan x)' = \frac{1}{1+x^2}$	$(\operatorname{arcsinh} x)' = \frac{1}{\sqrt{1+x^2}}$
$(a^x)' = (\ln a) a^x$	$(\operatorname{arccot} x)' = -\frac{1}{1+x^2}$	$(\operatorname{arcosh} x)' = \frac{1}{\sqrt{x^2-1}}$
$(\tan x)' = \sec^2 x$	$(\operatorname{arccsc} x)' = -\frac{1}{x\sqrt{1-x^2}}$	$(\operatorname{arctanh} x)' = \frac{1}{1-x^2}$
$(\cot x)' = \csc^2 x$	$(\operatorname{arcsec} x)' = \frac{1}{x\sqrt{1-x^2}}$	$(\operatorname{arcoth} x)' = \frac{1}{x^2-1}$
$(\sec x)' = \tan x \sec x$	$(\operatorname{arcsec} x)' = \frac{1}{x\sqrt{1-x^2}}$	$(\operatorname{arcsch} x)' = -\frac{1}{ x \sqrt{1+x^2}}$
$(\csc x)' = -\cot x \csc x$	$(\tanh x)' = \operatorname{sech}^2 x$	$(\operatorname{arcsech} x)' = -\frac{1}{x\sqrt{1-x^2}}$
$(\operatorname{arcsin} x)' = \frac{1}{\sqrt{1-x^2}}$	$(\coth x)' = -\operatorname{csch}^2 x$	
$(\operatorname{arccos} x)' = -\frac{1}{\sqrt{1-x^2}}$	$(\operatorname{sech} x)' = -\operatorname{sech} x \tanh x$	
	$(\operatorname{csch} x)' = -\operatorname{csch} x \coth x$	

$$\cdot ax^2 + bx + c (a > 0)$$

$$\int \frac{dx}{ax^2 + bx + c} = \begin{cases} \frac{2}{\sqrt{4ac-b^2}} \arctan \frac{2ax+b}{\sqrt{4ac-b^2}} + C & (b^2 < 4ac) \\ \frac{1}{\sqrt{b^2-4ac}} \ln \left| \frac{2ax+b-\sqrt{b^2-4ac}}{2ax+b+\sqrt{b^2-4ac}} \right| + C & (b^2 > 4ac) \end{cases}$$

$$\int \frac{x}{ax^2+bx+c} dx = \frac{1}{2a} \ln |ax^2 + bx + c| - \frac{b}{2a} \int \frac{dx}{ax^2+bx+c}$$

$$\cdot \sqrt{\pm ax^2 + bx + c} (a > 0)$$

$$\int \frac{dx}{\sqrt{ax^2+bx+c}} = \frac{1}{\sqrt{a}} \ln |2ax + b + 2\sqrt{a}\sqrt{ax^2+bx+c}| + C$$

$$\int \sqrt{ax^2+bx+c} dx = \frac{2ax+b}{4a} \sqrt{ax^2+bx+c} + \frac{4ac-b^2}{8\sqrt{a^3}} \ln |2ax + b + 2\sqrt{a}\sqrt{ax^2+bx+c}| + C$$

$$\int \frac{x}{\sqrt{ax^2+bx+c}} dx = \frac{1}{a} \sqrt{ax^2+bx+c} - \frac{b}{2\sqrt{a^3}} \ln |2ax + b + 2\sqrt{a}\sqrt{ax^2+bx+c}| + C$$

$$\int \frac{dx}{\sqrt{c+bx-ax^2}} = -\frac{1}{\sqrt{a}} \arcsin \frac{2ax-b}{\sqrt{b^2+4ac}} + C$$

$$\int \sqrt{c+bx-ax^2} dx = \frac{2ax-b}{4a} \sqrt{c+bx-ax^2} + \frac{b^2+4ac}{8\sqrt{a^3}} \arcsin \frac{2ax-b}{\sqrt{b^2+4ac}} + C$$

$$\int \frac{x}{\sqrt{c+bx-ax^2}} dx = -\frac{1}{a} \sqrt{c+bx-ax^2} + \frac{b}{2\sqrt{a^3}} \arcsin \frac{2ax-b}{\sqrt{b^2+4ac}} + C$$

$$\cdot \sqrt{\pm \frac{x-a}{x-b}} \text{ 或 } \sqrt{(x-a)(x-b)}, (a < b)$$

$$\int \frac{dx}{\sqrt{(x-a)(b-x)}} = 2 \arcsin \sqrt{\frac{x-a}{b-x}} + C$$

$$\int \sqrt{(x-a)(b-x)} dx = \frac{2x-a-b}{4} \sqrt{(x-a)(b-x)} + \frac{(b-a)^2}{4} \arcsin \sqrt{\frac{x-a}{b-x}} + C$$

三角函数的积分 (省略 +C)

$\int \tan x dx = -\ln \cos x $	$\int \sec^2 x dx = \tan x$
$\int \cot x dx = \ln \sin x $	$\int \csc^2 x dx = -\cot x$
$\int \sec x dx = \ln \left \tan \left(\frac{x}{4} + \frac{\pi}{2} \right) \right $	$\int \sec x \tan x dx = \sec x$
$= \ln \sec x + \tan x $	$\int \csc x \cot x dx = -\csc x$
$\int \csc x dx = \ln \left \tan \frac{x}{2} \right $	$\int \sin^2 x dx = \frac{x}{2} - \frac{1}{4} \sin 2x$
$= \ln \csc x - \cot x $	$\int \cos^2 x dx = \frac{x}{2} + \frac{1}{4} \sin 2x$

$$\int \sin^n x dx = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x dx$$

$$\int \cos^n x dx = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int \cos^{n-2} x dx$$

$$\int \frac{dx}{\sin^n x} = -\frac{1}{n-1} \frac{\cos x}{\sin^{n-1} x} + \frac{n-2}{n-1} \int \frac{dx}{\sin^{n-2} x}$$

$$\int \frac{dx}{\cos^n x} = \frac{1}{n-1} \frac{\sin x}{\cos^{n-1} x} + \frac{n-2}{n-1} \int \frac{dx}{\cos^{n-2} x}$$

$$\int \cos^m x \sin^n x dx$$

$$= \frac{1}{m+n} \cos^{m-1} x \sin^{n+1} x + \frac{m-1}{m+n} \int \cos^{m-2} x \sin^n x dx$$

$$= -\frac{1}{m+n} \cos^{m+1} x \sin^{n-1} x + \frac{n-1}{m+1} \int \cos^m x \sin^{n-2} x dx$$

$$\int \frac{dx}{a+b \sin x} = \begin{cases} \frac{2}{\sqrt{a^2-b^2}} \arctan \frac{a \tan \frac{x}{2} + b}{\sqrt{a^2-b^2}} & (a^2 > b^2) \\ \frac{1}{\sqrt{b^2-a^2}} \ln \left| \frac{a \tan \frac{x}{2} + b - \sqrt{b^2-a^2}}{a \tan \frac{x}{2} + b + \sqrt{b^2-a^2}} \right| & (a^2 < b^2) \end{cases}$$

$$\int \frac{dx}{a+b \cos x} = \begin{cases} \frac{2}{a+b} \sqrt{\frac{a+b}{a-b}} \arctan \left(\sqrt{\frac{a-b}{a+b}} \tan \frac{x}{2} \right) & (a^2 > b^2) \\ \frac{1}{a+b} \sqrt{\frac{a+b}{a-b}} \ln \left| \frac{\tan \frac{x}{2} + \sqrt{\frac{a+b}{a-b}}}{\tan \frac{x}{2} - \sqrt{\frac{a+b}{a-b}}} \right| & (a^2 < b^2) \end{cases}$$

$$\int \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x} = \frac{1}{ab} \arctan \left(\frac{b}{a} \tan x \right)$$

$$\int \frac{dx}{a^2 \cos^2 x - b^2 \sin^2 x} = \frac{1}{2ab} \ln \left| \frac{b \tan x + a}{b \tan x - a} \right|$$

$$\int x \sin ax dx = \frac{1}{a^2} \sin ax - \frac{1}{a} x \cos ax$$

$$\int x^2 \sin ax dx = -\frac{1}{a} x^2 \cos ax + \frac{2}{a^2} x \sin ax + \frac{2}{a^3} \cos ax$$

$$\int x \cos ax dx = \frac{1}{a^2} \cos ax + \frac{1}{a} x \sin ax$$

$$\int x^2 \cos ax dx = \frac{1}{a} x^2 \sin ax + \frac{2}{a^2} x \cos ax - \frac{2}{a^3} \sin ax$$

反三角函数的积分 (其中 $a > 0$)

$$\int \arcsin \frac{x}{a} dx = x \arcsin \frac{x}{a} + \sqrt{a^2 - x^2} + C$$

$$\int x \arcsin \frac{x}{a} dx = \left(\frac{x^2}{2} - \frac{a^2}{4} \right) \arcsin \frac{x}{a} + \frac{x}{4} \sqrt{x^2 - x^2} + C$$

$$\int x^2 \arcsin \frac{x}{a} dx = \frac{x^3}{3} \arcsin \frac{x}{a} + \frac{1}{9} (x^2 + 2a^2) \sqrt{a^2 - x^2} + C$$

$$\int \arccos \frac{x}{a} dx = x \arccos \frac{x}{a} - \sqrt{a^2 - x^2} + C$$

$$\int x \arccos \frac{x}{a} dx = \left(\frac{x^2}{2} - \frac{a^2}{4} \right) \arccos \frac{x}{a} - \frac{x}{4} \sqrt{a^2 - x^2} + C$$

$$\int x^2 \arccos \frac{x}{a} dx = \frac{x^3}{3} \arccos \frac{x}{a} - \frac{1}{9} (x^2 + 2a^2) \sqrt{a^2 - x^2} + C$$

$$\int \arctan \frac{x}{a} dx = x \arctan \frac{x}{a} - \frac{a}{2} \ln(a^2 + x^2) + C$$

$$\int x \arctan \frac{x}{a} dx = \frac{1}{2} (a^2 + x^2) \arctan \frac{x}{a} - \frac{a}{2} x + C$$

$$\int x^2 \arctan \frac{x}{a} dx = \frac{x^3}{3} \arctan \frac{x}{a} - \frac{a}{6} x^2 + \frac{a^3}{6} \ln(a^2 + x^2) + C$$

指数函数的积分

$$\int a^x dx = \frac{1}{\ln a} a^x + C$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C$$

$$\int x e^{ax} dx = \frac{1}{a^2} (ax - 1) e^{ax} + C$$

$$\int x^n e^{ax} dx = \frac{1}{a} x^n e^{ax} - \frac{n}{a} \int x^{n-1} e^{ax} dx$$

$$\int x a^x dx = \frac{x}{\ln a} a^x - \frac{1}{(\ln a)^2} a^x + C$$

$$\int x^n a^x dx = \frac{1}{\ln a} x^n a^x - \frac{n}{\ln a} \int x^{n-1} a^x dx$$

$$\int e^{ax} \sin bxdx = \frac{1}{a^2+b^2} e^{ax} (a \sin bx - b \cos bx) + C$$

$$\int e^{ax} \cos bxdx = \frac{1}{a^2+b^2} e^{ax} (b \sin bx + a \cos bx) + C$$

$$\int e^{ax} \sin^n bxdx = \frac{1}{a^2+b^2 n^2} e^{ax} \sin^{n-1} bx (a \sin bx - nb \cos bx) +$$

$$\frac{n(n-1)b^2}{a^2+b^2 n^2} \int e^{ax} \sin^{n-2} bxdx$$

$$\int e^{ax} \cos^n bxdx = \frac{1}{a^2+b^2 n^2} e^{ax} \cos^{n-1} bx (a \cos bx + nb \sin bx) +$$

$$\frac{n(n-1)b^2}{a^2+b^2 n^2} \int e^{ax} \cos^{n-2} bxdx$$

对数函数的积分

$$\int \ln x dx = x \ln x - x + C$$

$$\int \frac{dx}{x \ln x} = \ln |\ln x| + C$$

$$\int x^n \ln x dx = \frac{1}{n+1} x^{n+1} \left(\ln x - \frac{1}{n+1} \right) + C$$

$$\int (\ln x)^n dx = x (\ln x)^n - n \int (\ln x)^{n-1} dx$$

$$\int x^m (\ln x)^n dx = \frac{1}{m+1} x^{m+1} (\ln x)^n - \frac{n}{m+1} \int x^m (\ln x)^{n-1} dx$$

3.16 Zeller 日期公式

```
1 // weekday=(id+1)%7;{Sun=0,Mon=1,...}   getId(1, 1, 1) = 0
2 int getId(int y, int m, int d) {
3   | if (m < 3) { y --; m += 12; }
4   | return 365 * y + y / 4 - y / 100 + y / 400 + (153 * (m -
   |   ↳ 3) + 2) / 5 + d - 307; }
5 // y<0: 统一加400的倍数年
6 auto date(int id) {
7   | int x=id+1789995, n, i, j, y, m, d;
8   | n = 4 * x / 146097; x -= (146097 * n + 3) / 4;
9   | i = (4000 * (x + 1)) / 1461001; x -= 1461 * i / 4 - 31;
10  | j = 80 * x / 2447; d = x - 2447 * j / 80; x = j / 11;
11  | m = j + 2 - 12 * x; y = 100 * (n - 49) + i + x;
12  | return make_tuple(y, m, d); }
```

3.17 有理数二分: Stern-Brocot 树, Farey 序列

```
1 def build(a, b, c, d, level = 1):
2   | x = a + c; y = b + d
3   | build(a, b, x, y, level + 1)
4   | build(x, y, c, d, level + 1)
5 build(0, 1, 1, 0) # 最简分数, Stern-Brocot
6 build(0, 1, 1, 1) # 最简真分数, Farey
```

3.18 Constant Table 常数表

Random primes generated at Mon Aug 31 21:26:55 2026
 5e2 367 379 397 409 433 457 461 479 509 541 577 587 613 617
 1e3 839 877 881 883 929 941 977 1021 1117 1123 1151 1163

```

3e4 27803 29009 29063 29209 30091 30119 30983 31741 32051
1e5 94117 94951 95773 95957 97501 99371 100363 101741 102859
1e6 938071 938713 939061 959689 1028189 1041239 1063661
5e6 4688503 4951907 5155277 5162893 5259407 5309039 5343581
1e7 9480661 9671687 9783223 9927119 10094621 10355717 10638269
2e7 18884269 19517177 19997581 20029939 20503517 20722283
1e9 971010829 991582313 1006335607 1054547983 1055600129
2e9 1933212287 1948119493 2001265333 2034750049 2034781423
NTT 976224257 r=3 (20) 985661441 r=3 (22) 998244353 r=3 (23)
1004535809 r=3 (21) 1007681537 r=3 (20) 1012924417 r=5 (21)
1045430273 r=3 (20) 1051721729 r=6 (20) 1053818881 r=7 (20)

```

4. 多项式与生成函数

4.1 FFT

```

1 using cp = complex<double>; const double PI = acos(-1.0);
2 vector<cp> omega[25]; // 单位根
3 // n 是 DFT 的最大长度, 例如如果最多有两个长为 m 的多项式相乘,
4 // 或者求逆的长度为 m, 那么 n 需要 >= 2m
5 void fft_init(int n) { // n = 2^k
6     for (int k = 2, d = 0; k <= n; k *= 2, d++) {
7         omega[d].resize(k + 1);
8         for (int i = 0; i <= k; i++) // polar 是用模和辐角求复
            ↪ 数
9             omega[d][i] = polar(1.0, 2 * PI * i / k); } }
10 void fft(cp* a, int n, int t) {
11     for (int i = 1, j = 0; i < n - 1; i++) {
12         int k = n; do j ^= (k >= 1); while (j < k);
13         if (i < j) swap(a[i], a[j]);
14         for (int k = 1, d = 0; k < n; k *= 2, d++)
15             for (int i = 0; i < n; i += k * 2)
16                 for (int j = 0; j < k; j++) {
17                     cp w = omega[d][t > 0 ? j : k * 2 - j];
18                     cp u = a[i + j], v = w * a[i + j + k];
19                     a[i + j] = u + v; a[i + j + k] = u - v; }
20     if (t < 0) for (int i = 0; i < n; i++) a[i] /= n; }

```

4.2 NTT

```

1 vector<int> omega[25]; // 单位根
2 // n 是 DFT 的最大长度, 例如如果最多有两个长为 m 的多项式相乘,
3 // 或者求逆的长度为 m, 那么 n 需要 >= 2m
4 void ntt_init(int n) { // n = 2^k
5     for (int k = 2, d = 0; k <= n; k *= 2, d++) {
6         omega[d].resize(k + 1);
7         int wn = qpow(3, (p - 1) / k), tmp = 1;
8         for (int i = 0; i <= k; i++) { omega[d][i] = tmp;
9             tmp = (LL)tmp * wn % p; } } }
10 // 传入的数必须是 [0, p) 范围内, 不能有负的
11 // 否则把 d == 16 改成 d % 8 == 0 之类, 多取几次模
12 void ntt(int *c, int n, int tp) {
13     static ULL a[N];
14     for (int i = 0; i < n; i++) a[i] = c[i];
15     for (int i = 1, j = 0; i < n - 1; i++) {
16         int k = n; do j ^= (k >= 1); while (j < k);
17         if (i < j) swap(a[i], a[j]);
18         for (int k = 1, d = 0; k < n; k *= 2, d++) {
19             if (d == 16) for (int i = 0; i < n; i++) a[i] %= p;
20             for (int i = 0; i < n; i += k * 2)
21                 for (int j = 0; j < k; j++) {
22                     int w = omega[d][tp > 0 ? j : k * 2 - j];
23                     ULL u = a[i + j], v = w * a[i + j + k] % p;
24                     a[i + j] = u + v;
25                     a[i + j + k] = u - v + p; } }
26     if (tp > 0) { for (int i = 0; i < n; i++) c[i] = a[i] % p; }
27     else { int inv = qpow(n, p - 2);
28         for (int i = 0; i < n; i++) c[i] = a[i] * inv % p; } }

```

4.3 MTT 任意模数卷积

```

1 void dft(cp* a, cp* b, int n) { static cp c[MXN];
2     for (int i = 0; i < n; i++)
3         c[i] = cp(a[i].real(), b[i].real());
4     fft(c, n, 1);
5     for (int i = 0; i < n; i++) { int j = (n - i) & (n - 1);
6         a[i] = (c[i] + conj(c[j])) * 0.5;
7         b[i] = (c[i] - conj(c[j])) * -0.5i; } }
8 void idft(cp* a, cp* b, int n) { static cp c[MXN];
9     for (int i = 0; i < n; i++) c[i] = a[i] + 1i * b[i];
10    fft(c, n, -1);
11    for (int i = 0; i < n; i++) {
12        a[i] = c[i].real(); b[i] = c[i].imag(); } }
13 vector<int> multiply(const vector<int>& u,
14                    const vector<int>& v, int mod) { // 任意模数卷积
15     static cp a[2][MXN], b[2][MXN], c[3][MXN];
16     int base = ceil(sqrt(mod));
17     int n = (int)u.size(), m = (int)v.size();

```

```

18 | int fft_n = 1; while (fft_n < n + m - 1) fft_n *= 2;
19 | for (int i = 0; i < 2; i++) {
20 |     fill(a[i], a[i] + fft_n, 0);
21 |     fill(b[i], b[i] + fft_n, 0); }
22 | for (int i = 0; i < 3; i++)
23 |     fill(c[i], c[i] + fft_n, 0);
24 | for (int i = 0; i < n; i++) { // 一定要取模!
25 |     a[0][i] = (u[i] % mod) % base;
26 |     a[1][i] = (u[i] % mod) / base; }
27 | for (int i = 0; i < m; i++) { // 一定要取模!
28 |     b[0][i] = (v[i] % mod) % base;
29 |     b[1][i] = (v[i] % mod) / base; }
30 | dft(a[0], a[1], fft_n); dft(b[0], b[1], fft_n);
31 | for (int i = 0; i < fft_n; i++) {
32 |     c[0][i] = a[0][i] * b[0][i];
33 |     c[1][i] = a[0][i] * b[1][i] + a[1][i] * b[0][i];
34 |     c[2][i] = a[1][i] * b[1][i]; }
35 | fft(c[1], fft_n, -1); idft(c[0], c[2], fft_n);
36 | int base2 = base * base % mod;
37 | vector<int> ans(n + m - 1);
38 | for (int i = 0; i < n + m - 1; i++)
39 |     ans[i] = ((LL)(c[0][i].real() + 0.5) +
40 |              ((LL)(c[1][i].real() + 0.5) % mod * base +
41 |              ((LL)(c[2][i].real() + 0.5) % mod * base2) % mod;
42 | return ans; }

```

4.4 多项式运算

4.4.1 多项式求逆 开根 ln exp

```

1 using poly = vector<int>;
2 poly poly_calc(const poly& u, const poly& v,
3               function<int(int, int)> op) { // 返回长度补齐到 2 的幂
4     if (u.empty() || v.empty()) return {};
5     static int a[MXN], b[MXN], c[MXN];
6     int n = 1, need = (int)u.size() + (int)v.size() - 1;
7     while (n < need) n *= 2;
8     assert(n <= MXN);
9     memcpy(a, u.data(), sizeof(int) * u.size());
10    fill(a + u.size(), a + n, 0);
11    memcpy(b, v.data(), sizeof(int) * v.size());
12    fill(b + v.size(), b + n, 0);
13    ntt(a, n, 1); ntt(b, n, 1);
14    for (int i = 0; i < n; i++) c[i] = op(a[i], b[i]);
15    ntt(c, n, -1); return poly(c, c + n); }
16 poly poly_mul(const poly& u, const poly& v) { // 乘法
17     return poly_calc(u, v, [](int a, int b)
18         | { return (LL)a * b % p; }); } // 返回长度补齐到 2 的幂
19 poly poly_inv(const poly& a) { // 求逆, 返回长度不变
20     assert(!a.empty() && a[0] % p != 0);
21     int size = (int)a.size(), limit = 1; while (limit < size)
        ↪ limit *= 2;
22     poly aa = a; aa.resize(limit); poly c{qpow(a[0], p - 2)};
23     for (int k = 2; k <= limit; k *= 2) {
24         c.resize(k); poly b(aa.begin(), aa.begin() + k);
25         c = poly_calc(b, c, [](int bi, int ci) {
26             | return ((2 - (LL)bi * ci) % p + p) * ci % p; });
27         memset(c.data() + k, 0, sizeof(int) * k); }
28     c.resize(size); return c; }
29 poly poly_sqrt(const poly& a) { // 开根, 返回长度不变
30     int size = (int)a.size(), limit = 1; while (limit < size)
        ↪ limit *= 2;
31     poly aa = a; aa.resize(limit); poly c{1}; // 常数项不是 1
        ↪ 的话要写二次剩余
32     for (int k = 2; k <= limit; k *= 2) {
33         c.resize(k); poly b(aa.begin(), aa.begin() + k);
34         b = poly_mul(b, poly_inv(c));
35         for (int i = 0; i < k; i++) // inv_2 是 2 的逆元
36             c[i] = (LL)(c[i] + b[i]) * inv_2 % p; }
37     c.resize(size); return c; }
38 poly poly_derivative(const poly& a) { poly c(a.size());
39     for (int i = 1; i < (int)a.size(); i++) // 求导
40         c[i - 1] = (LL)a[i] * i % p;
41     return c; }
42 poly poly_integrate(const poly& a) { poly c(a.size());
43     for (int i = 1; i < (int)a.size(); i++) // 不定积分
44         c[i] = (LL)a[i - 1] * inv[i] % p;
45     return c; }
46 poly poly_ln(const poly& a) { // ln, 常数项非 0, 返回长度不变
47     auto c = poly_mul(poly_derivative(a), poly_inv(a));
48     c.resize(a.size()); return poly_integrate(c); }
49 // exp, 常数项必须是 0, 返回长度不变
50 // 常数很大并且总代码很长, 一般可以改用分治 FFT
51 // 依据: 设  $G(x) = \exp F(x)$ , 则  $g_i = \frac{1}{i} \sum_{k=1}^{i-1} g_{i-k} k f_k$ 
52 poly poly_exp(const poly& a) { poly c{1};

```

```

53 | int size = (int)a.size(), limit = 1; while (limit < size)
    |   limit *= 2;
54 | poly aa = a; aa.resize(limit);
55 | for (int k = 2; k <= limit; k *= 2) {
56 |   c.resize(k); auto b = poly_ln(c);
57 |   for (int i = 0; i < k; i++)
58 |     b[i] = (aa[i] - b[i] + p) % p;
59 |   (++b[0]) %= p; c = poly_mul(b, c);
60 |   memset(c.data() + k, 0, sizeof(int) * k); }
61 | c.resize(size); return c; }

```

4.4.2 多项式除法 取模

需要抄求逆。

```

1 | poly poly_auto_mul(poly a, poly b) { // 自动判断长度的乘法
2 |   int res_len = (int)a.size() + (int)b.size() - 1;
3 |   int ntt_n = 1; while (ntt_n < res_len) ntt_n *= 2;
4 |   a.resize(ntt_n); b.resize(ntt_n);
5 |   ntt(a.data(), ntt_n, 1); ntt(b.data(), ntt_n, 1);
6 |   for (int i = 0; i < ntt_n; i++)
7 |     a[i] = (LL)a[i] * b[i] % p;
8 |   ntt(a.data(), ntt_n, -1); a.resize(res_len); return a; }
9 | // 多项式除法, a 和 b 长度可以任意
10 | // 商的长度是 n - m + 1, 余数的长度是 m - 1
11 | poly poly_div(const poly& a, const poly& b) {
12 |   int n = (int)a.size(), m = (int)b.size();
13 |   if (n < m) return {};
14 |   int ntt_n = 1; while (ntt_n < n - m + 1) ntt_n *= 2;
15 |   poly f(ntt_n), g(ntt_n);
16 |   for (int i = 0; i < n - m + 1; i++) f[i] = a[n - i - 1];
17 |   for (int i = 0; i < m && i < n - m + 1; i++)
18 |     g[i] = b[m - i - 1];
19 |   auto g_inv = poly_inv(g);
20 |   fill(g_inv.begin() + n - m + 1, g_inv.end(), 0);
21 |   auto c = poly_mul(f, g_inv); c.resize(n - m + 1);
22 |   reverse(c.begin(), c.end()); return c; }
23 | // 多项式取模, a 和 b 长度可以任意, 返回 (余数, 商)
24 | pair<poly, poly> poly_mod(const poly& a, const poly& b) {
25 |   int n = (int)a.size(), m = (int)b.size();
26 |   if (n < m) return {a, {}};
27 |   auto d = poly_div(a, b); auto c = poly_auto_mul(b, d);
28 |   poly r(m - 1);
29 |   for (int i = 0; i < m - 1; i++)
30 |     r[i] = (a[i] - c[i] + p) % p;
31 |   return {r, d}; }

```

4.4.3 多点求值

需要抄取模。

```

1 | struct poly_eval { poly f; vector<int> x; int query_count;
    |   // 函数和询问点
2 |   vector<poly> gs; vector<int> ans; // gs 是预处理数组
3 |   poly_eval(poly f_, vector<int> x_) : f(f_), x(x_),
    |     query_count(x_.size()) {}
4 |   void pretreat(int l, int r, int o) { poly& g = gs[o];
5 |     if (l == r) { g = poly{p - x[l], 1}; return; }
6 |     int mid = (l + r) / 2; pretreat(l, mid, o * 2);
7 |     pretreat(mid + 1, r, o * 2 + 1);
8 |     if (o > 1)
9 |       g = poly_auto_mul(gs[o * 2], gs[o * 2 + 1]); }
10 |   void solve(int l, int r, int o, const poly& rem) {
11 |     if (l == r) { ans[l] = rem[0]; return; }
12 |     int mid = (l + r) / 2;
13 |     solve(l, mid, o * 2, poly_mod(rem, gs[o * 2]).first);
14 |     solve(mid + 1, r, o * 2 + 1,
15 |       poly_mod(rem, gs[o * 2 + 1]).first); }
16 |   vector<int> operator() () { // 包装好的接口
17 |     int n = (int)f.size(), m = (int)x.size();
18 |     if (m <= n) x.resize(m = n + 1);
19 |     else if (n < m - 1) f.resize(n = m - 1);
20 |     int bit_ceil = 1; while (bit_ceil < m) bit_ceil *= 2;
21 |     ntt_init(bit_ceil * 2); // 注意这里 ntt_init 过了
22 |     gs.resize(2 * bit_ceil + 1); pretreat(0, m - 1, 1);
23 |     ans.resize(m); solve(0, m - 1, 1, f);
24 |     ans.resize(query_count); return ans; } };

```

4.4.4 插值

牛顿插值 实现时可以用 k 次差分替代右边的式子, 也可以卷积。

$$f(n) = \sum_{i=0}^k \binom{n}{i} r_i \iff r_i = \sum_{j=0}^i (-1)^{i-j} \binom{i}{j} f(j)$$

拉格朗日插值

$$f(x) = \sum_i f(x_i) \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}$$

4.5 线性递推

$O(k^2 \log n)$

```

1 | // Complexity: init  $O(n^2 \log)$  query  $O(n^2 \log k)$ 
2 | // Requirement: const LOG const MOD
3 | // Example: In: {1, 3} {2, 1} an = 2an-1 + an-2
4 | // | | Out: calc(3) = 7
5 | typedef vector<int> poly;
6 | struct LinearRec {
7 |   int n; poly first, trans; vector<poly> bin;
8 |   poly add(poly &a, poly &b) {
9 |     poly res(n * 2 + 1, 0);
10 |    // 不要每次新开 vector, 可以使用矩阵乘法优化
11 |    for (int i = 0; i <= n; ++i) {
12 |      for (int j = 0; j <= n; ++j) {
13 |        (res[i+j] += (LL)a[i] * b[j] % MOD) %= MOD;
14 |      }
15 |    }
16 |    for (int i = 2 * n; i > n; --i) {
17 |      for (int j = 0; j < n; ++j) {
18 |        (res[i-1-j] += (LL)res[i]*trans[j]%MOD) %= MOD; }
19 |      res[i] = 0; }
20 |    res.erase(res.begin() + n + 1, res.end());
21 |    return res; }
22 | LinearRec(poly &first_, poly &trans_): first(first_),
    |   trans(trans_) {
23 |   n = first.size(); poly a(n + 1, 0); a[1] = 1;
24 |   bin.push_back(a); for (int i = 1; i < LOG; ++i)
25 |     bin.push_back(add(bin[i - 1], bin[i - 1])); }
26 | int calc(long long k) { poly a(n + 1, 0); a[0] = 1;
27 |   for (int i = 0; i < LOG; ++i)
28 |     if (k >> i & 1) a = add(a, bin[i]);
29 |   int ret = 0; for (int i = 0; i < n; ++i)
30 |     if ((ret += (LL)a[i + 1] * first[i] % MOD) >= MOD)
31 |       ret -= MOD;
32 |   return ret; } };

```

$O(k \log k \log n)$ - Bostan-Mori

```

1 | int bostan_mori(long long k, poly a, poly b) {
2 |   int n = (int)a.size(); while (k) { poly c = b;
3 |     for (int i = 1; i < n; i += 2) c[i] = (p - c[i]) % p;
4 |     a = poly_mul(a, c); b = poly_mul(b, c);
5 |     for (int i = 0; i < n; i++) {
6 |       a[i] = a[i * 2 + k % 2]; b[i] = b[i * 2]; }
7 |     a.resize(n); b.resize(n); k /= 2; }
8 |   return (LL)a[0] * qpow[b[0], p - 2] % p; }
9 | //  $a_n = \sum_{i=1}^m f_i a_{n-i}$  ( $f_0 = 0$ ), f.size() = a.size() + 1
10 | int linear_recurrence(long long n, poly f, poly a) {
11 |   int m = (int)a.size(), ntt_n = 1;
12 |   while (ntt_n <= m) ntt_n *= 2;
13 |   ntt_init(ntt_n * 2);
14 |   f.resize(ntt_n); a.resize(ntt_n); f[0] = 1;
15 |   for (int i = 1; i <= m; i++) f[i] = (p - f[i]) % p;
16 |   a = poly_mul(a, f); a.resize(ntt_n);
17 |   fill(a.data() + m, a.data() + ntt_n, 0);
18 |   return bostan_mori(n, a, f); }

```

$O(k \log k \log n)$ - 多项式取模

需要抄前面的多项式取模。预处理 $O(k \log k \log n)$, 固定 n 和系数只改变初始值的话, 询问一次 $O(k)$ 。注意只询问一次的话不如 Bostan-Mori 快。

```

1 | poly poly_power_mod(LL k, const poly& m) { //  $x^k \bmod m$ 
2 |   poly ans{1}, a{0, 1}; while (k) { if (k & 1)
3 |     ans = poly_mod(poly_auto_mul(ans, a), m).first;
4 |     a = poly_mod(poly_auto_mul(a, a), m).first; k /= 2; }
5 |   return ans; }
6 | //  $a_n = \sum_{i=1}^m c_i a_{n-i}$  ( $c_0 = 0$ )
7 | struct linear_recurrence { poly f; // f 是预处理结果
8 |   linear_recurrence(const poly& c, LL n) {
9 |     assert(c[0] == 0); // c[0] 是没有用的
10 |    int m = (int)c.size() - 1;
11 |    int ntt_n = 1; while (ntt_n < m * 2) ntt_n *= 2;
12 |    ntt_init(ntt_n); // 图省事就直接 ntt_init(1 << 18)
13 |    poly t(m + 1); t[m] = 1;
14 |    for (int i = 0; i < m; i++) t[i] = (p - c[m - i]) % p;
15 |    f = poly_power_mod(n, t); f.resize(m, 0); }
16 |   int operator()(const vector<int>& a) { // 0~m-1 项初始值
17 |     assert(a.size() == f.size()); int ans = 0;

```

```

14 |         for (int j = 0; j < n; j++)
15 |             a[i] = (a[i] + (long long)f[i][j] * w[j]) % p;
16 |     auto c = berlekamp_massey(a); int m = (int)c.size();
17 |     vector<int> ans(n);
18 |     for (int i = 0; i < m - 1; i++)
19 |         for (int j = 0; j < n; j++)
20 |             ans[j] = (ans[j] +
21 |                 (long long)c[m - 2 - i] * f[i][j]) % p;
22 |     int inv = qpow(p - c[m - 1], p - 2);
23 |     for (int i = 0; i < n; i++)
24 |         ans[i] = (long long)ans[i] * inv % p;
25 |     return ans; }

```

如果不加最后的处理的话，代码返回的结果会变成 $a_i = \sum_{j=0}^{m-1} c_{j-1} a_{i-j}$ ，有时候这样会方便接着跑递推，需要的话就删掉最后的处理。

```

1  /* |
2  And:  $\begin{pmatrix} 1,1 \\ 0,1 \end{pmatrix}$   $\begin{pmatrix} 1,-1 \\ 0,1 \end{pmatrix}$  Or:  $\begin{pmatrix} 1,0 \\ 1,1 \end{pmatrix}$   $\begin{pmatrix} -1,0 \\ -1,1 \end{pmatrix}$  Xor:  $\begin{pmatrix} 1,1 \\ 1,-1 \end{pmatrix}$   $\begin{pmatrix} 0.5,0.5 \\ 0.5,-0.5 \end{pmatrix}$ 
3  IFFT的矩阵时FFT的逆，对于任意运算  $\oplus$ ，满足FFT的矩阵需要：
4   $C[i][j] \times C[i][k] = C[i][j \oplus k]$ 
5  对于不存在FFT矩阵的运算：通过映射01变成另外一个可行的运算。*/
6  const LL XOR[2][2] = {{1, 1}, {1, M-1}};
7  const LL i2 = (M+1)/2, iXOR[2][2] = {{i2, i2}, {i2, M-i2}};
8  void FWT(LL f[], const LL C[2][2], int n) {
9      for (int t = 1; t < n; t <= 1) {
10         for (int l = 0; l < n; l += t + t) {
11             for (int i = 0; i < t; i++) {
12                 LL x = f[l + i], y = f[l + t + i];
13                 f[l + i] = (C[0][0] * x + C[0][1] * y) % M;
14                 f[l + t + i] = (C[1][0] * x + C[1][1] * y) % M;
15             } } }

```

```

1 // n : power of k, omega[i] : (primitive kth root) ^ i
2 void fwt(int* a, int k, int type) {
3     static int tmp[K];
4     for (int i = 1; i < n; i *= k)
5         for (int j = 0, len = i * k; j < n; j += len)
6             for (int low = 0; low < i; low++) {
7                 for (int t = 0; t < k; t++)
8                     tmp[t] = a[j + t * i + low];
9                 for (int t = 0; t < k; t++){
10                     int x = j + t * i + low;
11                     long long s = 0;
12                     for (int y = 0; y < k; y++)
13                         s = (s + 1ll * tmp[y] * omega[(k + type) * t * y %
14                             ↳ k]) % MOD;
15                     a[x] = (int)s;
16                 }
17             }
18     if (type == -1)
19         for (int i = 0, invn = inv(n); i < n; i++)
20             a[i] = int(1ll * a[i] * invn % MOD); }

```

```

1 const LD eps = 1e-9, INF = 1e9; const int N = 105;
2 namespace Simplex {
3 int n, m, id[2*N], tp[N]; LD a[N][N];
4 void pivot(int r, int c) {
5     swap(id[r + n], id[c]);
6     LD t = -a[r][c]; a[r][c] = -1;
7     for (int i = 0; i <= n; i++) a[r][i] /= t;
8     for (int i = 0; i <= m; i++) if (a[i][c] && r != i) {
9         t = a[i][c]; a[i][c] = 0;
10        for (int j = 0; j <= n; j++) a[i][j] += t*a[r][j];}
11 bool solve() {
12     for (int i = 1; i <= n; i++) id[i] = i;
13     for ( ; ; ) {
14         int i = 0, j = 0; LD w = -eps;
15         for (int k = 1; k <= m; k++)
16             if (a[k][0] < w || (a[k][0] < -eps && rand() & 1))
17                 w = a[i = k][0];
18         if (!i) break;
19         for (int k = 1; k <= n; k++)
20             if (a[i][k] > eps) {j = k; break;}
21         if (!j) { printf("Infeasible"); return 0;}
22         pivot(i, j);}
23     for ( ; ; ) {
24         int i = 0, j = 0; LD w = eps, t;
25         for (int k = 1; k <= n; k++)

```

```

4 int merge(int x, int y){
5     if(!x || !y) return x+y;
6     if(dat[x] < dat[y]) {push_down(x); rs[x] = merge(rs[x],
7         ↪ y); push_up(x); return x;}
8     else {push_down(y); ls[y] = merge(x, ls[y]); push_up(y);
9         ↪ return y;} }
10 void splitVal(int p, int k, int& x, int& y){
11     if(!p) {x = y = 0; return;}
12     push_down(p);
13     if(val[p] <= k) {x = p; splitVal(rs[x], k, rs[x], y);}
14     else {y = p; splitVal(ls[y], k, x, ls[y]);}
15     push_up(p);
16 }
17 void splitSiz(int p, int k, int& x, int& y){
18     if(!p) {x = y = 0; return;}
19     push_down(p);
20     if(siz[ls[p]] < k) {x = p; splitSiz(rs[p],
21         ↪ k-siz[ls[p]]-1, rs[x], y);}
22     else {y = p; splitSiz(ls[p], k, x, ls[y]);}
23     push_up(p);
24 }
25 int new_node(int v){
26     siz[++tot] = 1; val[tot] = v;
27     dat[tot] = rd(); return tot;}
28 void ins(int v){
29     int x, y; splitVal(rt, v, x, y);
30     rt = merge(merge(x, new_node(v)), y); }
31 void del(int v){
32     int x, y, z;
33     splitVal(rt, v, x, z), splitVal(x, v-1, x, y);
34     y = merge(ls[y], rs[y]); rt = merge(merge(x, y), z); }
35 int rk(int v){
36     int x, y; splitVal(rt, v-1, x, y);
37     int t = siz[x] + 1; rt = merge(x, y); return t; }
38 int kth(int p, int k){
39     if(!p || k <= 0) return 0;
40     while(p){
41         if(siz[ls[p]] >= k) p = ls[p];
42         else if(siz[ls[p]] + 1 == k) return val[p];
43         else k -= siz[ls[p]] + 1, p = rs[p]; }
44     return 0; }
45 int pre(int v){
46     int x, y; splitVal(rt, v-1, x, y);
47     int t = kth(x, siz[x]); rt = merge(x, y); return t; }
48 int nxt(int v){
49     int x, y; splitVal(rt, v, x, y);
50     int t = kth(y, 1); rt = merge(x, y); return t; }
51 int s[MAXN]; // 线性建立FHQ
52 int build(int n){
53     int top = 0;
54     for(int i=1;i<=n;i++){
55         int tmp = new_node(a[i]), last = 0;
56         while(top && dat[s[top]] > dat[tmp])
57             push_up(s[top]), last = s[top--];
58         if(top) rs[s[top]] = tmp; ls[tmp] = last, s[++top] =
59             ↪ tmp;
60     }
61     while(top) push_up(s[top--]); return s[1]; }
62 } tr;

```

```

1 typedef vector <LD> vec; /* sum a[i][0..d] = 0 */
2 pair<vec, vector<vec>> gauss(vector<vec> &a, int n, int d) {
3     vector<int> pivot(d, -1);
4     for (int i = 0, w = 0; i < d; i++) {
5         int j = 0; while (j < n && abs(a[j][i]) < eps) j++;
6         if (j == n) continue;
7         swap(a[j], a[0]); LD w = a[0][i];
8         for (int k = 0; k <= d; k++) a[0][k] /= w;
9         for (int x = 0; x < n; x++)
10             if (x != 0 && abs(a[x][i]) > eps) {
11                 w = a[x][i];
12                 for (int k = 0; k <= d; k++)
13                     a[x][k] -= a[0][k] * w; }
14         pivot[i] = 0++;
15     } vec x0(d); vector<vec> t; int free = 0;
16     for (int i = 0; i < d; i++)
17         if (pivot[i] != -1) x0[i] = -a[pivot[i]][d];
18         else free++;
19     for (int i = 0; i < d; i++) if (pivot[i] == -1) {
20         vec x(d); x[i] = -1;
21         for (int j = 0; j < d; j++)
22             if (pivot[j] != -1) x[j] = a[pivot[j]][i];
23         t.push_back(x);
24     } if (t.size()) {
25         vector<vec> f;
26         for (int u = 0; u < free; u++) {
27             vec x(free + 1);
28             for (int i = 0; i < free; i++)
29                 for (int j = 0; j < d; j++)
30                     x[i] += t[u][j] * t[i][j];
31             for (int j = 0; j < d; j++)
32                 x[free] += t[u][j] * x0[j];
33             f.push_back(x);
34         }
35         auto [k, tt] = gauss(f, free, free);
36         assert(tt.size() == 0);
37         for (int x = 0; x < free; x++)
38             for (int i = 0; i < d; i++)
39                 x0[i] += k[x] * t[x][i];
40     } return {x0, t}; }

```

```
1 template<int len = 1> // 动态size的bitset
2 void solve(int x) {
3     if(siz[x] > len && len < MAXN) return (void)
4         ↳ solve<min(len*2, MAXN)>(x);
5     bitset<len> dp; }
```

```
1 struct FHQ{
2 int val[N], dat[N], siz[N], ls[N], rs[N], tot = 0, rt;
3 void push up(int x) {siz[x] = siz[ls[x]] + siz[rs[x]] + 1;}
```

```

1 struct BIT{
2     | #define lowbit(x) (x & (-x))
3     | ll d[MAXN], d1[MAXN];
4     | void upd(int x, ll k) {
5     |     | for (int pos = x; x <= n; x += lowbit(x))
6     |     | | d[x] += k, d1[x] += (pos - 1) * k;
7     |     | }
8     | void upd(int l, int r, ll k) {upd(l, k), upd(r+1, -k);}
9     | ll qry(int x) {
10    |     | ll ret = 0, pos = x;
11    |     | for (int pos = x; x; x -= lowbit(x))
12    |     | | ret += d[x] * pos - d1[x];
13    |     | return ret;
14    | }
15    | ll qry(int l, int r) {return qry(r) - qry(l-1);}
16 };

```

```
1 namespace Li{ // 默认取max 可持久化李超记得从x转移而非p
2 | int tag[MAXN];
3 | double k[MAXN], b[MAXN];
```

```

4 | const double EPS = 1e-10;
5 | double calc(int i, double x) {return x * k[i] + b[i];}
6 | void update(int nl, int nr, int l, int r, int p, int x){
7 |     if(l == r) {if(!tag[p] || calc(tag[p], l) < calc(x,
8 |         ↪ l)) tag[p] = x; return;}
9 |     int mid = (l+r)>>1;
10 |     if(nl <= l && r <= nr){
11 |         if(!tag[p]) {tag[p] = x; return;}
12 |         double l1 = calc(tag[p], l), l2 = calc(x, l), r1 =
13 |             ↪ calc(tag[p], r), r2 = calc(x, r);
14 |         if(abs(l1 - l2) < EPS && abs(r1 - r2) < EPS)
15 |             ↪ return;
16 |         if(l1 > l2 && r1 > r2) return;
17 |         if(l2 > l1 && r2 > r1) tag[p] = x;
18 |         if(calc(tag[p], mid) < calc(x, mid)) swap(x,
19 |             ↪ tag[p]);
20 |         if(k[x] > k[tag[p]]) update(nl, nr, mid+1, r, rs,
21 |             ↪ x);
22 |         if(k[x] < k[tag[p]]) update(nl, nr, l, mid, ls, x);
23 |         ↪ return;
24 |     }
25 |     if(nl <= mid) update(nl, nr, l, mid, ls, x);
26 |     if(nr > mid) update(nl, nr, mid+1, r, rs, x);
27 | }
28 | double query(int pos, int l, int r, int p){
29 |     double cur = tag[p] ? calc(tag[p], pos) : -1e100;
30 |     if(l == r) return cur;
31 |     int mid = (l+r)>>1;
32 |     if(pos <= mid) return max(cur, query(pos, l, mid,
33 |         ↪ ls));
34 |     else return max(cur, query(pos, mid+1, r, rs));
35 | }

```

6.5 线段树分裂

```

1 | void split(int& x, int& y, int nl, int nr, int l, int r){
2 |     if(!x) {y = 0; return;}
3 |     if(nl <= l && r <= nr) {y = x, x = 0; return;}
4 |     y = new_node();
5 |     int mid = (l+r)>>1;
6 |     if(nl <= mid) split(ls[x], ls[y], nl, nr, l, mid);
7 |     if(nr > mid) split(rs[x], rs[y], nl, nr, mid+1, r);
8 |     push_up(x), push_up(y);
9 | }

```

6.6 左偏树

```

1 | namespace Leftist{
2 |     int a[MAXN], ls[MAXN], rs[MAXN], dis[MAXN], fa[MAXN];
3 |     bool vis[MAXN];
4 |     void init() {dis[0] = -1; for(int i=1;i<=n;i++) fa[i] =
5 |         ↪ i;}
6 |     int find(int x) {return fa[x] == x ? x : fa[x] =
7 |         ↪ find(fa[x]);}
8 |     int _merge(int x, int y){
9 |         if(!x || !y) return x + y;
10 |         if(a[y] < a[x]) swap(x, y);
11 |         rs[x] = _merge(rs[x], y);
12 |         if(dis[ls[x]] < dis[rs[x]]) swap(ls[x], rs[x]);
13 |         dis[x] = dis[rs[x]] + 1;
14 |         return x;
15 |     }
16 |     void merge(int x, int y){
17 |         if(vis[x] || vis[y]) return;
18 |         x = find(x), y = find(y);
19 |         if(x != y) fa[x] = fa[y] = _merge(x, y);
20 |     }
21 |     int del(int x){
22 |         if(vis[x]) return -1; // deleted flag
23 |         x = find(x); vis[x] = true;
24 |         int ret = a[x];
25 |         fa[ls[x]] = fa[rs[x]] = fa[x] = _merge(ls[x], rs[x]);
26 |         ls[x] = rs[x] = dis[x] = 0;
27 |         return ret;
28 |     }
29 | }

```

6.7 非递归线段树

6.7.1 区间加，区间求最大值

```

1 | // t,mark 开 2*M+5; M=2*k 且 M>=n+2
2 | void update(int l, int r, int d) {

```

```

3 |     for (l += M-1, r += M+1; l^r^1; l >>= 1, r >>= 1) {
4 |         if (l < M) {
5 |             t[l] = max(t[l*2], t[l*2+1]) + mark[l];
6 |             t[r] = max(t[r*2], t[r*2+1]) + mark[r]; }
7 |         if (~l & 1) { t[l ^ 1] += d; mark[l ^ 1] += d; }
8 |         if (r & 1) { t[r ^ 1] += d; mark[r ^ 1] += d; } }
9 |     for (; l >>= 1, r >>= 1)
10 |         if (l < M) t[l] = max(t[l*2], t[l*2+1]) + mark[l],
11 |             ↪ t[r] = max(t[r*2], t[r*2+1]) + mark[r]; }
12 | int query(int l, int r) {
13 |     int maxl = -INF, maxr = -INF;
14 |     for (l += M-1, r += M+1; l^r^1; l >>= 1, r >>= 1) {
15 |         maxl += mark[l]; maxr += mark[r];
16 |         if (~l & 1) maxl = max(maxl, t[l ^ 1]);
17 |         if (r & 1) maxr = max(maxr, t[r ^ 1]); }
18 |     while (l) { maxl += mark[l]; maxr += mark[r];
19 |         l >>= 1; r >>= 1; }
20 |     return max(maxl, maxr); }

```

6.8 LCT 动态树

```

1 | namespace LCT{
2 |     #define ls(x) c[x][0]
3 |     #define rs(x) c[x][1]
4 |     int val[N], ans[N], c[N][2], f[N], rev[N];
5 |     void push_up(int x){
6 |         ans[x] = val[x];
7 |         if (ls(x)) ans[x] = max(ans[x], ans[ls(x)]);
8 |         if (rs(x)) ans[x] = max(ans[x], ans[rs(x)]); }
9 |     void opt(int x) { if (!x) return; swap(ls(x), rs(x)), rev[x]
10 |         ↪ ^= 1;}
11 |     void push_down(int x) {
12 |         if (rev[x]) { if (ls(x)) opt(ls(x)); if (rs(x))
13 |             ↪ opt(rs(x)); rev[x] = 0; } }
14 |     bool nrt(int x) {return x && (ls(f[x]) == x || rs(f[x]) ==
15 |         ↪ x);}
16 |     bool get(int x) {return x == rs(f[x]);}
17 |     void rot(int x){
18 |         int y = f[x], z = f[y], t = get(x), w = c[x][t^1];
19 |         if(nrt(y)) c[z][get(y)] = x;
20 |         c[x][t^1] = y, c[y][t] = w;
21 |         if(w) {f[w] = y; f[y] = x, f[x] = z; push_up(y); }
22 |         void push_all(int x) {if(nrt(x)) push_all(f[x]);
23 |             ↪ push_down(x);}
24 |     void splay(int x){
25 |         push_all(x);
26 |         while(nrt(x)){
27 |             int y = f[x];
28 |             if(nrt(y)) rot(get(x) == get(y) ? y : x);
29 |             rot(x);
30 |         } push_up(x); }
31 |     void access(int x) {for(int y=0;y=x,x=f[x]) splay(x),
32 |         ↪ rs(x) = y, push_up(x);}
33 |     void mkrt(int x) {access(x); splay(x); opt(x);}
34 |     int findrt(int x){
35 |         access(x), splay(x);
36 |         while(ls(x)) push_down(x), x = ls(x);
37 |         splay(x); return x; }
38 |     void split(int x, int y) {mkrt(x), access(y), splay(y);}
39 |     bool reach(int x, int y) {mkrt(x); return findrt(y) == x;}
40 |     void link(int x, int y) {if(!reach(x, y)) f[x] = y;}
41 |     void cut(int x, int y) {if(reach(x, y) && f[y] == x &&
42 |         ↪ !ls(y)) f[y] = rs(x) = 0;}
43 |     // 维护子树信息树的直径，F[x]:splay子树深度最小的点往下最大深度
44 |     // 直径版须自备结构操作，全部走 opt_diam / push_up_diam，不能
45 |         ↪ 复用上面的 splay/mkrt
46 |     void push_up_diam(int x){
47 |         if (!x) return;
48 |         siz[x] = siz[ls(x)] + siz[rs(x)] + 1;
49 |         F[x] = max(F[ls(x)], F[rs(x)] + siz[ls(x)] + 1);
50 |         G[x] = max(G[rs(x)], G[ls(x)] + siz[rs(x)] + 1);
51 |         if(s[x].size())
52 |             F[x] = max(F[x], (--s[x].end()) + 1 + siz[ls(x)]),
53 |             ↪ G[x] = max(G[x], (--s[x].end()) + 1 + siz[rs(x)]); }
54 |     void opt_diam(int x) { if (!x) return; swap(ls(x), rs(x)),
55 |         ↪ swap(F[x], G[x]), rev[x] ^= 1;}
56 |     void push_down_diam(int x) {
57 |         if (rev[x]) { if (ls(x)) opt_diam(ls(x)); if (rs(x))
58 |             ↪ opt_diam(rs(x)); rev[x] = 0; } }
59 |     void rot_diam(int x){
60 |         int y = f[x], z = f[y], t = get(x), w = c[x][t^1];
61 |         if(nrt(y)) c[z][get(y)] = x;
62 |         c[x][t^1] = y, c[y][t] = w;
63 |         if(w) {f[w] = y; f[y] = x, f[x] = z; push_up_diam(y); }

```

```

55 void push_all_diam(int x) {if(nrt(x)) push_all_diam(f[x]);
   ↳ push_down_diam(x);}
56 void splay_diam(int x){
57 | push_all_diam(x);
58 | while(nrt(x)){
59 | | int y = f[x];
60 | | if(nrt(y)) rot_diam(get(x) == get(y) ? y : x);
61 | | rot_diam(x);
62 | } push_up_diam(x); }
63 void access_diam(int x) {
64 | for(int y=0;x;y=x, x=f[x]) {
65 | | splay_diam(x);
66 | | if(rs(x)) s[x].insert(F[rs(x)]);
67 | | int oldF = y ? F[y] : 0;
68 | | if(rs(x) = y) { auto it = s[x].find(oldF); if (it !=
   ↳ s[x].end()) s[x].erase(it); }
69 | | push_up_diam(x); } }
70 void mkrt_diam(int x) {access_diam(x); splay_diam(x);
   ↳ opt_diam(x);}
71 int findrt_diam(int x){
72 | access_diam(x), splay_diam(x);
73 | while(ls(x)) push_down_diam(x), x = ls(x);
74 | splay_diam(x); return x; }
75 void split_diam(int x, int y) {mkrt_diam(x), access_diam(y),
   ↳ splay_diam(y);}
76 bool reach_diam(int x, int y) {mkrt_diam(x); return
   ↳ findrt_diam(y) == x;}
77 void link_diam(int x, int y) {mkrt_diam(x);if(findrt_diam(y)
   ↳ != x) mkrt_diam(y), f[x] = y, s[y].insert(F[x]),
   ↳ push_up_diam(y);}
78 void cut_diam(int x, int y) {if(reach_diam(x, y) && f[y] ==
   ↳ x && !ls(y)) f[y] = rs(x) = 0, push_up_diam(x);}
79 #undef ls
80 #undef rs
81 }

```

6.9 Hash Table

```

1 template <class T,int P = 314159/*,451411,1141109,2119969*/>
2 struct hashmap {
3 ULL id[P]; T val[P];
4 int R[P + 1]; // del: few clears; R[0] 为占用数
5 hashmap() {memset(id, -1, sizeof id); memset(val, 0, sizeof
   ↳ val); R[0] = 0;}
6 T get(const ULL &x) const {
7 | for (int i = int(x % P), j = 1, steps = 0; ~id[i]; i = (i
   ↳ + j) % P, j = (j + 2) % P) {
8 | | if (id[i] == x) return val[i];
9 | | if (++steps >= P) break; }
10 | return 0; }
11 T& operator [] (const ULL &x) {
12 | for (int i = int(x % P), j = 1, steps = 0; ; i = (i + j)
   ↳ % P, j = (j + 2) % P) {
13 | | if (id[i] == x) return val[i];
14 | | else if (id[i] == -1llu) {
15 | | | id[i] = x; val[i] = T{};
16 | | | R[++R[0]] = i; // del: few clears
17 | | | return val[i]; }
18 | | if (++steps >= P) { assert(false); static T dummy{};
   ↳ return dummy; } } } // 表满: 不写其它 key 的槽
19 void clear() { // del: few clears
20 for (int &x = R[0]; x; id[R[x]] = -1, val[R[x]] = 0, --x); }
21 void fullclear() {memset(id, -1, sizeof id); memset(val, 0,
   ↳ sizeof val); R[0] = 0; } };

```

6.10 基数排序

```

1 const int SZ = 1 << 8; // almost always fit in L1 cache
2 void SORT(int a[], int c[], int n, int w) {
3 | for(int i=0; i<SZ; i++) b[i] = 0;
4 | for(int i=1; i<n; i++) b[(a[i]>>w) & (SZ-1)]++;
5 | for(int i=1; i<SZ; i++) b[i] += b[i - 1];
6 | for(int i=n; i; i--) c[b[(a[i]>>w) & (SZ-1)]--] = a[i];}
7 void Sort(int *a, int n){ // signed int, 从小到大
8 | for(int i=1;i<n;i++) a[i] ^= 1u<<31;
9 | SORT(a, c, n, 0); SORT(c, a, n, 8);
10 | SORT(a, c, n, 16); SORT(c, a, n, 24);
11 | for(int i=1;i<n;i++) a[i] ^= 1u<<31; }

```

7. 树上问题

7.1 虚树

```

1 bool cmp(int x,int y) {return id[x] < id[y];}

```

```

2 void build(vector<int> v){
3 | sort(v.begin(), v.end(), cmp);
4 | v.erase(unique(v.begin(), v.end()), v.end());
5 | int top = 0;
6 | s[top=1] = 1; vt::e[1].clear();
7 | for(auto x : v){
8 | | if(x == 1) continue;
9 | | int l = lca(x, s[top]);
10 | | if(l != s[top]){
11 | | | while(id[l] < id[s[top-1]]) vt::push(s[top-1],
   ↳ s[top]), top--;
12 | | | if(l != s[top-1]) vt::e[1].clear(), vt::push(l,
   ↳ s[top]), s[top] = l;
13 | | | else vt::push(l, s[top--]);
14 | | }
15 | | s[++top] = x; vt::e[x].clear();
16 | }
17 | for(int i=1;i<top;i++) vt::push(s[i], s[i+1]);
18 }

```

7.2 树哈希

```

1 ULL get(vector <ULL> ha) {
2 | sort(ha.begin(), ha.end());
3 | ULL ret = 0xdeadbeef;
4 | for (auto i : ha) {
5 | | ret = ret * P + i, ret ^= ret << 17;
6 | } return ret * 997; }

```

8. 计算几何

8.1 二维几何基础

8.1.1 声明与宏

```

1 #define cp const P &
2 #define cl const L &
3 #define cc const C &
4 #define vp vector<P>
5 #define cvp const vector<P> &
6 #define D double
7 #define LD long double
8
9 mt19937 rnd(time(nullptr));
10 const LD eps = 1e-12L;
11 const LD pi = acosl(-1.0L);
12 const LD INF = 1e18L;
13
14 int sgn(LD x) {
15 | return x > eps ? 1 : (x < -eps ? -1 : 0);
16 }

```

8.1.2 点与向量

```

1 struct P {
2 | LD x, y;
3
4 | P(LD x_ = 0, LD y_ = 0) : x(x_), y(y_) {}
5 | P(cp a) : x(a.x), y(a.y) {}
6 | P &operator=(cp a) { x = a.x, y = a.y; return *this; }
7
8 | P rot(LD t) const { // 逆时针旋转 t 弧度
9 | | LD c = cosl(t), s = sinl(t);
10 | | return {x * c - y * s, x * s + y * c};
11 | }
12 | P rot90() const { return {-y, x}; }
13 | P _rot90() const { return {y, -x}; }
14 | LD len() const { return sqrtl(x * x + y * y); }
15 | LD len2() const { return x * x + y * y; }
16 | P unit() const { // 零向量的单位向量约定为零向量
17 | | LD d = len();
18 | | return sgn(d) ? P{x / d, y / d} : P{};
19 | }
20
21 | void read() { if (scanf("%Lf%Lf", &x, &y) != 2) x = y =
   ↳ 0; }
22 | void print() const { printf("%.9Lf,%.9Lf", x, y); }
23 };
24
25 bool operator<(cp a, cp b) { return a.x == b.x ? a.y < b.y :
   ↳ a.x < b.x; }
26 bool operator>(cp a, cp b) { return b < a; }
27 bool operator==(cp a, cp b) { return !sgn(a.x - b.x) &&
   ↳ !sgn(a.y - b.y); }
28 bool operator!=(cp a, cp b) { return !(a == b); }

```

```

29 P operator+(cp a, cp b) { return {a.x + b.x, a.y + b.y}; }
30 P operator-(cp a, cp b) { return {a.x - b.x, a.y - b.y}; }
31 P operator~(cp a) { return {-a.x, -a.y}; }
32 P operator*(cp a, LD k) { return {a.x * k, a.y * k}; }
33 P operator*(LD k, cp a) { return a * k; }
34 P operator/(cp a, LD k) { return {a.x / k, a.y / k}; }
35 LD operator*(cp a, cp b) { return a.x * b.x + a.y * b.y; }
    ↪ // 点积
36 LD operator^(cp a, cp b) { return a.x * b.y - a.y * b.x; }
    ↪ // 叉积
37
38 LD rad(cp a, cp b) { // 夹角 [0, pi]
39 | LD d = a.len() * b.len();
40 | if (!sgn(d)) return 0;
41 | return acos1(clamp((a * b) / d, -1.0L, 1.0L));
42 }
43 bool left(cp a, cp b) { return sgn(a ^ b) > 0; }
44 int turn(cp a, cp b, cp c) { return sgn((b - a) ^ (c - a)); }
    ↪
45 int half(cp a) { return a.y > 0 || (a.y == 0 && a.x > 0); }

```

8.1.3 线

```

1 struct L {
2 | P s, t;
3
4 | L(cp s_ = P(), cp t_ = P()) : s(s_), t(t_) {}
5 | L(cl a) : s(a.s), t(a.t) {}
6 | L &operator=(cl a) { s = a.s, t = a.t; return *this; }
7 | void read() { s.read(), t.read(); }
8 };
9
10 bool point_on_line(cp a, cl l) {
11 | return !sgn((a - l.s) ^ (l.t - l.s));
12 }
13 bool point_on_segment(cp a, cl l) {
14 | return point_on_line(a, l) && sgn((a - l.s) * (a - l.t))
    ↪ <= 0;
15 }
16 bool two_side(cp a, cp b, cl l) {
17 | return sgn((a - l.s) ^ (l.t - l.s))
18 | | * sgn((b - l.s) ^ (l.t - l.s)) < 0;
19 }
20 bool intersection_judge_strict(cl a, cl b) {
21 | return two_side(a.s, a.t, b) && two_side(b.s, b.t, a);
22 }
23 bool intersection_judge(cl a, cl b) {
24 | return point_on_segment(a.s, b) || point_on_segment(a.t,
    ↪ b)
25 | | point_on_segment(b.s, a) || point_on_segment(b.t,
    ↪ a)
26 | | intersection_judge_strict(a, b);
27 }
28 P ll_intersection(cl a, cl b) { // 两直线需不平行
29 | LD s1 = (a.t - a.s) ^ (b.s - a.s);
30 | LD s2 = (a.t - a.s) ^ (b.t - a.s);
31 | return (b.s * s2 - b.t * s1) / (s2 - s1);
32 }
33 bool point_on_ray(cp a, cl b) {
34 | return point_on_line(a, b) && sgn((a - b.s) * (b.t -
    ↪ b.s)) >= 0;
35 }
36 bool ray_intersection_judge(cl a, cl b) {
37 | if (!sgn((a.t - a.s) ^ (b.t - b.s))) {
38 | | if (!point_on_line(b.s, a)) return false;
39 | | return point_on_ray(b.s, a) || point_on_ray(a.s, b);
40 | }
41 | P p = ll_intersection(a, b);
42 | return point_on_ray(p, a) && point_on_ray(p, b);
43 }
44 LD point_to_line(cp a, cl b) {
45 | if (b.s == b.t) return (a - b.s).len();
46 | return fabs1((b.t - b.s) ^ (a - b.s)) / (b.t -
    ↪ b.s).len();
47 }
48 P project_to_line(cp a, cl b) {
49 | P d = b.t - b.s;
50 | if (b.s == b.t) return b.s;
51 | return b.s + d * ((a - b.s) * d / d.len2());
52 }
53 array<L, 2> offset_lines(cl l, LD d) { // 非退化直线向两侧平
    ↪ 移距离 d
54 | P v = (l.t - l.s).unit().rot90() * d;
55 | return {L(l.s + v, l.t + v), L(l.s - v, l.t - v)};
56 }

```

```

57 LD point_to_segment(cp a, cl b) {
58 | if (b.s == b.t) return (a - b.s).len();
59 | if (sgn((a - b.s) * (b.t - b.s)) >= 0
60 | | && sgn((a - b.t) * (b.s - b.t)) >= 0)
61 | | return point_to_line(a, b);
62 | return min((a - b.s).len(), (a - b.t).len());
63 }
64 LD segment_to_segment(cl a, cl b) {
65 | if (intersection_judge(a, b)) return 0;
66 | return min({point_to_segment(a.s, b),
    ↪ point_to_segment(a.t, b),
67 | | point_to_segment(b.s, a), point_to_segment(b.t, a)});
68 }

```

8.1.4 三角形

```

1 P incenter(cp a, cp b, cp c) {
2 | LD p = (a-b).len() + (b-c).len() + (c-a).len();
3 | return (a*(b-c).len() + b*(c-a).len() + c*(a-b).len()) /
    ↪ p; }
4 P circumcenter(cp a, cp b, cp c) { return C(a, b, c).c; }
5 P orthocenter(cp a, cp b, cp c) {
6 | return a + b + c - circumcenter(a, b, c) * 2; }
7 P fermit_point(cp a, cp b, cp c) {
8 | if (a == b) return a;
9 | if (b == c) return b;
10 | if (c == a) return c;
11 | LD ab = (a-b).len(), bc = (b-c).len(), ca = (c-a).len();
12 | LD cosa = ((b-a)*(c-a)) / ab / ca;
13 | LD cosb = ((a-b)*(c-b)) / ab / bc;
14 | LD cosc = ((b-c)*(a-c)) / ca / bc;
15 | LD sq3 = pi / 3; P mid;
16 | if (sgn(cosa + 0.5) < 0) mid = a;
17 | else if (sgn(cosb + 0.5) < 0) mid = b;
18 | else if (sgn(cosc + 0.5) < 0) mid = c;
19 | else if (sgn((b-a)^(c-a)) < 0)
20 | | mid = ll_intersection({a, b+(c-b).rot(sq3)}, {b,
    ↪ c+(a-c).rot(sq3)});
21 | else mid = ll_intersection({a, c+(b-c).rot(sq3)}, {c,
    ↪ b+(a-b).rot(sq3)});
22 | return mid; } // minimize(|A-x|+|B-x|+|C-x|)

```

8.1.5 三角形相关公式：三角形心、海伦公式、欧拉定理、内接球、外接圆

Heron's Formula

$$p = \frac{a+b+c}{2}, a \geq b \geq c,$$

$$S = \sqrt{p(p-a)(p-b)(p-c)},$$

$$S = \frac{1}{2} \sqrt{a^2 c^2 - \left(\frac{a^2 + c^2 - b^2}{2} \right)^2}.$$

四面体内接球球心

假设 s_i 是第 i 个顶点相对面的面积，则有

$$\begin{cases} x = \frac{s_1 x_1 + s_2 x_2 + s_3 x_3 + s_4 x_4}{s_1 + s_2 + s_3 + s_4} \\ y = \frac{s_1 y_1 + s_2 y_2 + s_3 y_3 + s_4 y_4}{s_1 + s_2 + s_3 + s_4} \\ z = \frac{s_1 z_1 + s_2 z_2 + s_3 z_3 + s_4 z_4}{s_1 + s_2 + s_3 + s_4} \end{cases}$$

体积可以使用 $1/6$ 混合积求，内接球半径为

$$r = \frac{3V}{s_1 + s_2 + s_3 + s_4}$$

三角形内心

$$\vec{I} = \frac{a\vec{A} + b\vec{B} + c\vec{C}}{a + b + c}$$

三角公式

$$\sin(a) + \sin(b) = 2 \sin\left(\frac{a+b}{2}\right) \cos\left(\frac{a-b}{2}\right)$$

$$\sin(a) - \sin(b) = 2 \cos\left(\frac{a+b}{2}\right) \sin\left(\frac{a-b}{2}\right)$$

$$\cos(a) + \cos(b) = 2 \cos\left(\frac{a+b}{2}\right) \cos\left(\frac{a-b}{2}\right)$$

$$\cos(a) - \cos(b) = -2 \sin\left(\frac{a+b}{2}\right) \sin\left(\frac{a-b}{2}\right)$$

$$\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b$$

$$\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b$$

$$\tan(a \pm b) = \frac{\tan(a) \pm \tan(b)}{1 \mp \tan(a) \tan(b)}$$

$$\tan(a) \pm \tan(b) = \frac{\sin(a \pm b)}{\cos(a) \cos(b)}$$

$$\sin(na) = n \cos^{n-1} a \sin a - \binom{n}{3} \cos^{n-3} a \sin^3 a + \binom{n}{5} \cos^{n-5} a \sin^5 a - \dots$$

$$\cos(na) = \cos^n a - \binom{n}{2} \cos^{n-2} a \sin^2 a + \binom{n}{4} \cos^{n-4} a \sin^4 a - \dots$$

三角形外心

$$\vec{O} = \frac{\vec{A} + \vec{B} - \frac{\vec{BC} \cdot \vec{CA}}{\vec{AB} \times \vec{BC}} \vec{AB}^T}{2}$$

三角形垂心

$$\vec{H} = 3\vec{G} - 2\vec{O}$$

三角形偏心

$\frac{-a\vec{A} + b\vec{B} + c\vec{C}}{-a + b + c}$
内角的平分线和对边的两个外角平分线交点，外切圆圆心。剩余两点的同理。
三角形内接外接圆半径

$$r = \frac{2S}{a + b + c}, R = \frac{abc}{4S}$$

Pick's Theorem 格点多边形面积

$$S = I + \frac{B}{2} - 1. I \text{ 内部点}, B \text{ 边界点}.$$

Euler's Formula 多面体与平面图

的点、边、面
Convex polyhedron: $V - E + F = 2$.
Planar graph: $|F| = |E| - |V| + n + 1$,
 n : #(connected components).

8.1.6 超球坐标系

$$\begin{aligned}
 x_1 &= r \cos(\phi_1) \\
 x_2 &= r \sin(\phi_1) \cos(\phi_2) \\
 &\dots \\
 x_{n-1} &= r \sin(\phi_1) \cdots \sin(\phi_{n-2}) \cos(\phi_{n-1}) \\
 x_n &= r \sin(\phi_1) \cdots \sin(\phi_{n-2}) \sin(\phi_{n-1}) \\
 \phi_{n-1} &\in [0, 2\pi] \\
 \forall i = 1..n-1 \phi_i &\in [0, \pi]
 \end{aligned}$$

8.1.7 三维旋转公式

绕着 $(0, 0, 0) - (ux, uy, uz)$ 旋转 θ , (ux, uy, uz) 是单位向量. $\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = R \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$$R = \begin{pmatrix} \cos\theta + u_x^2(1-\cos\theta) & u_x u_y(1-\cos\theta) - u_z \sin\theta & u_x u_z(1-\cos\theta) + u_y \sin\theta \\ u_y u_x(1-\cos\theta) + u_z \sin\theta & \cos\theta + u_y^2(1-\cos\theta) & u_y u_z(1-\cos\theta) - u_x \sin\theta \\ u_z u_x(1-\cos\theta) - u_y \sin\theta & u_z u_y(1-\cos\theta) + u_x \sin\theta & \cos\theta + u_z^2(1-\cos\theta) \end{pmatrix}$$

8.1.8 立体角公式

ϕ : 二面角

$$\Omega = (\phi_{ab} + \phi_{bc} + \phi_{ac}) \text{ rad} - \pi \text{ sr}$$

$$\tan\left(\frac{1}{2}\Omega/\text{rad}\right) = \frac{|\vec{a} \vec{b} \vec{c}|}{abc + (\vec{a} \cdot \vec{b})c + (\vec{a} \cdot \vec{c})b + (\vec{b} \cdot \vec{c})a}$$

$$\theta_s = \frac{\theta_a + \theta_b + \theta_c}{2}$$

8.1.9 常用体积公式

- 棱锥 Pyramid $V = \frac{1}{3}Sh$.
- 球 Sphere $V = \frac{4}{3}\pi R^3$.
- 棱台 Frustum $V = \frac{1}{3}h(S_1 + \sqrt{S_1 S_2} + S_2)$.
- 椭球 Ellipsoid $V = \frac{4}{3}\pi abc$.

8.1.10 扇形与圆弧重心

扇形重心与圆心距离为 $\frac{4r \sin(\theta/2)}{3\theta}$,
圆弧重心与圆心距离为 $\frac{4r \sin^3(\theta/2)}{3(\theta - \sin\theta)}$.

8.1.11 高维球体积

$$\begin{aligned}
 V_2 &= \pi R^2, S_2 = 2\pi R \\
 V_3 &= \frac{4}{3}\pi R^3, S_3 = 4\pi R^2 \\
 V_4 &= \frac{1}{2}\pi^2 R^4, S_4 = 2\pi^2 R^3 \\
 \text{Generally, } V_n &= \frac{2\pi}{n} V_{n-2}, S_{n-1} = \frac{2\pi}{n-2} S_{n-3} \\
 \text{Where, } S_0 &= 2, V_1 = 2, S_1 = 2\pi, V_2 = \pi
 \end{aligned}$$

8.2 多边形

8.2.1 多边形基础

```

1  vp polygon_cut(cvp p, cl l) { // 保留有向直线 l 左侧 (含边界)
2  | vp ret;
3  | int n = (int)p.size();
4  | if (!n) return ret;
5  | for (int i = 0; i < n; ++i) {
6  | | P a = p[i], b = p[(i + 1) % n];
7  | | int sa = turn(l.s, l.t, a), sb = turn(l.s, l.t, b);
8  | | if (sa >= 0) ret.push_back(a);
9  | | if (sa * sb < 0) ret.push_back(ll_intersection(l, {a,
10 | | | b}));
11 | }
12 | return ret;
13 }
14 LD polygon_signed_area2(cvp p) { // 逆时针为正, 返回两倍有向面
15 | LD ret = 0;
16 | for (int i = 0, n = (int)p.size(); i < n; ++i)
17 | | ret += p[i] ^ p[(i + 1) % n];
18 | return ret;
19 }
20 LD polygon_area(cvp p) {
21 | return fabs1(polygon_signed_area2(p)) / 2;
22 }
23 LD polygon_perimeter(cvp p) {
24 | LD ret = 0;

```

```

27 | for (int i = 0, n = (int)p.size(); i < n; ++i)
28 | | ret += (p[i] - p[(i + 1) % n]).len();
29 | return ret;
30 }
31
32 P polygon_centroid(cvp p) { // 退化时返回边界按长度加权的质心
33 | LD area2 = polygon_signed_area2(p);
34 | if (!sgn(area2)) {
35 | | P ret;
36 | | LD perimeter = 0;
37 | | for (int i = 0, n = (int)p.size(); i < n; ++i) {
38 | | | LD length = (p[i] - p[(i + 1) % n]).len();
39 | | | ret = ret + (p[i] + p[(i + 1) % n]) * length;
40 | | | perimeter += length;
41 | | }
42 | | if (sgn(perimeter)) return ret / (2 * perimeter);
43 | | return p.empty() ? P{} : p[0];
44 | }
45 | P ret;
46 | for (int i = 0, n = (int)p.size(); i < n; ++i) {
47 | | LD cross = p[i] ^ p[(i + 1) % n];
48 | | ret = ret + (p[i] + p[(i + 1) % n]) * cross;
49 | }
50 | return ret / (3 * area2);
51 }
52
53 // -1: 外部, 0: 边界, 1: 内部; 与顶点顺时针无关。
54 int point_in_polygon(cp u, cvp p) {
55 | bool in = false;
56 | for (int i = 0, n = (int)p.size(); i < n; ++i) {
57 | | P a = p[i], b = p[(i + 1) % n];
58 | | if (point_on_segment(u, {a, b})) return 0;
59 | | if ((a.y > u.y) != (b.y > u.y)
60 | | | && sgn((b - a) ^ (u - a)) == (b.y > a.y ? 1 : -1))
61 | | | in = !in;
62 | }
63 | return in ? 1 : -1;
64 }

```

8.2.2 简单多边形内最长线段

```

1  struct PolygonChordSolver { // 逆时针简单多边形顶点间最长内含
2  | 线段
3  | vp p;
4  | int n;
5  | PolygonChordSolver(vp points) : p(move(points)),
6  | | n((int)p.size()) {}
7  |
8  | bool segment_inside(cp u, cp v) const {
9  | | for (int i = 0; i < n; ++i) {
10 | | | int j = (i + 1) % n, k = (i + 2) % n;
11 | | | cp a = p[i], b = p[j], c = p[k];
12 | | | if (intersection_judge_strict({u, v}, {a, b}))
13 | | | | return false;
14 | | | if (point_on_segment(b, {u, v})) {
15 | | | | bool good = true, convex = turn(a, b, c) >= 0;
16 | | | | for (cp x : {u, v}) {
17 | | | | | if (convex) good &= turn(a, b, x) >= 0 &&
18 | | | | | turn(b, c, x) >= 0;
19 | | | | | else good &= !(turn(b, x, c) > 0 && turn(b,
20 | | | | | a, x) > 0);
21 | | | | }
22 | | | | if (!good) return false;
23 | | | }
24 | | }
25 | | return true;
26 | }
27 |
28 | LD ray_exit_distance(int uid, int vid) const {
29 | | cp u = p[uid], v = p[vid];
30 | | LD far = INF;
31 | | for (int i = 0; i < n; ++i) {
32 | | | int j = (i + 1) % n, k = (i + 2) % n;
33 | | | cp a = p[i], b = p[j], c = p[k];
34 | | | if (two_side(a, b, {u, v})) {
35 | | | | LD s1 = (b - a) ^ (u - a), s2 = (b - a) ^ (v -
36 | | | | a);
37 | | | | if (sgn(s1 - s2) && sgn(s1) != sgn(s2 - s1))
38 | | | | | far = min(far, (u - ll_intersection({a, b},
39 | | | | | {u, v})).len());
40 | | | }
41 | | }
42 | | if (j != uid && point_on_ray(b, {u, v})) {
43 | | | bool good = turn(a, b, c) <= 0;
44 | | | for (cp x : {u - (b - u), b + (b - u)})

```

```

37 | | | | good &= !(turn(a, b, x) > 0 && turn(b, c, x)
    | | | | ↪ > 0);
38 | | | | if (!good) far = min(far, (u - b).len());
39 | | | }
40 | | }
41 | | return far;
42 | }
43
44 | LD solve() const {
45 | | LD ans = 0;
46 | | for (int i = 0; i < n; ++i)
47 | | | for (int j = i + 1; j < n; ++j) {
48 | | | | if (!segment_inside(p[i], p[j])) continue;
49 | | | | ans = max(ans, ray_exit_distance(i, j) +
    | | | | ↪ ray_exit_distance(j, i)
50 | | | | - (p[i] - p[j]).len());
51 | | | }
52 | | return ans;
53 | }
54 | };

```

8.2.3 $O(n^2)$ Ear Clipping 三角剖分

```

1 | vector<array<int, 3>> ear_clipping(cvp p) { // 简单多边形,
    | ↪  $O(n^2)$ 
2 | | int n = (int)p.size();
3 | | if (n < 3) return {};
4 | | vector<int> id(n);
5 | | iota(id.begin(), id.end(), 0);
6 | | if (polygon_signed_area2(p) < 0) reverse(id.begin(),
    | | ↪ id.end());
7 | | vector<array<int, 3>> ret;
8 | | for (bool changed = true; changed && id.size() > 3; ) {
9 | | | changed = false;
10 | | | for (int i = 0, m = (int)id.size(); i < m; ++i) {
11 | | | | int a = id[(i + m - 1) % m], b = id[i], c = id[(i +
    | | | | ↪ 1) % m];
12 | | | | if (!turn(p[a], p[b], p[c]) &&
    | | | | ↪ point_on_segment(p[b], {p[a], p[c]})) {
13 | | | | | id.erase(id.begin() + i), changed = true;
14 | | | | | break;
15 | | | | }
16 | | | }
17 | | }
18
19 | | auto in_triangle = [&](cp q, cp a, cp b, cp c) {
20 | | | return turn(a, b, q) >= 0 && turn(b, c, q) >= 0
21 | | | && turn(c, a, q) >= 0;
22 | | };
23 | | while (id.size() > 3) {
24 | | | bool found = false;
25 | | | for (int i = 0, m = (int)id.size(); i < m; ++i) {
26 | | | | int a = id[(i + m - 1) % m], b = id[i], c = id[(i +
    | | | | ↪ 1) % m];
27 | | | | if (turn(p[a], p[b], p[c]) <= 0) continue;
28 | | | | bool empty = true;
29 | | | | for (int x : id) if (x != a && x != b && x != c
    | | | | && in_triangle(p[x], p[a], p[b], p[c])) {
30 | | | | | empty = false;
31 | | | | | break;
32 | | | | }
33 | | | | if (!empty) continue;
34 | | | | ret.push_back({a, b, c});
35 | | | | id.erase(id.begin() + i);
36 | | | | found = true;
37 | | | | break;
38 | | | }
39 | | | }
40 | | | if (!found) return {}; // 自交、重复点或退化输入
41 | | }
42 | | ret.push_back({id[0], id[1], id[2]});
43 | | return ret;
44 | }

```

8.3 圆

8.3.1 圆基础

```

1 | struct C {
2 | | P c;
3 | | LD r;
4
5 | | C(cp c_ = P(), LD r_ = 0) : c(c_), r(r_) {}
6 | | C(cc a) : c(a.c), r(a.r) {}
7 | | C &operator=(cc a) { c = a.c, r = a.r; return *this; }
8 | | C(cp a, cp b) : c((a + b) / 2), r((a - c).len()) {}
9 | | C(cp x, cp y, cp z) { // 三点不能共线

```

```

10 | | P p(y - x), q(z - x);
11 | | P s(p.len2() / 2, q.len2() / 2);
12 | | LD d = p ^ q;
13 | | c = x + P(s ^ P(p.y, q.y), P(p.x, q.x) ^ s) / d;
14 | | r = (c - x).len();
15 | | }
16 | };
17
18 | bool in_circle(cp a, cc b) { return sgn((b.c - a).len() -
    | | ↪ b.r) <= 0; }
19
20 | C min_circle(vp p) { // 随机增量,  $O(n)$  期望
21 | | if (p.empty()) return C();
22 | | shuffle(p.begin(), p.end(), rnd);
23 | | C ret(p[0], 0);
24 | | for (int i = 1; i < (int)p.size(); ++i) if
    | | ↪ (!in_circle(p[i], ret)) {
25 | | | ret = C(p[i], 0);
26 | | | for (int j = 0; j < i; ++j) if (!in_circle(p[j], ret))
    | | | ↪ {
27 | | | | ret = C(p[i], p[j]);
28 | | | | for (int k = 0; k < j; ++k) if (!in_circle(p[k],
    | | | | ↪ ret))
29 | | | | | ret = C(p[i], p[j], p[k]);
30 | | | }
31 | | }
32 | | return ret;
33 | }
34
35 | vp lc_intersection(cl l, cc c) { // 直线与圆交点
36 | | LD d = point_to_line(c.c, l);
37 | | if (sgn(d - c.r) > 0) return {};
38 | | P p = project_to_line(c.c, l);
39 | | LD x = sqrtl(max(0.0L, c.r * c.r - d * d));
40 | | if (!sgn(x)) return {p};
41 | | P v = (l.t - l.s).unit() * x;
42 | | return {p - v, p + v};
43 | }
44
45 | LD circle_segment_factor(LD t) { //  $t \cdot \sin(t) \cos(t)$ , 小角时避
    | | ↪ 免灾难性消减
46 | | if (fabs1(t) < 0.1L) {
47 | | | LD t2 = t * t;
48 | | | return t * t2 * (2.0L / 3 + t2 * (-2.0L / 15
    | | | + t2 * (4.0L / 315 + t2 * (-2.0L / 2835
    | | | + t2 * (4.0L / 155925 - t2 * (4.0L / 6081075))))));
49 | | }
50 | | return t - sinl(t) * cosl(t);
51 | }
52
53 | LD cc_intersection_area(cc a, cc b) {
54 | | LD d = (a.c - b.c).len();
55 | | if (sgn(d - a.r - b.r) >= 0) return 0;
56 | | if (sgn(d - fabs1(a.r - b.r)) <= 0) {
57 | | | LD r = min(a.r, b.r);
58 | | | return pi * r * r;
59 | | }
60 | | // 用半角公式求两个弓形的半圆心角, 所有根号内的因子在相交时均
    | | ↪ 非负。
61 | | // 这样既不需要平方差, 也不会用两个大扇形减去一个大三角形。
62 | | LD u = max(0.0L, a.r + b.r - d);
63 | | LD v = max(0.0L, d + b.r - a.r);
64 | | LD w = max(0.0L, d + a.r - b.r);
65 | | LD z = max(0.0L, d + a.r + b.r);
66 | | LD t1 = 2 * atan2l(sqrtl(u * v), sqrtl(z * w));
67 | | LD t2 = 2 * atan2l(sqrtl(u * w), sqrtl(z * v));
68 | | return a.r * a.r * circle_segment_factor(t1)
    | | + b.r * b.r * circle_segment_factor(t2);
69 | }
70
71 | // 返回交点个数; -1 表示两圆重合, 有无穷多个交点。
72 | int cc_intersection_count(cc a, cc b) {
73 | | LD d = (a.c - b.c).len();
74 | | if (a.c == b.c) {
75 | | | if (sgn(a.r - b.r)) return 0;
76 | | | return !sgn(a.r) ? 1 : -1;
77 | | }
78 | | if (sgn(d - a.r - b.r) > 0)
79 | | | return 0;
80 | | if (sgn(d - fabs1(a.r - b.r)) < 0) return 0;
81 | | if (!sgn(d - a.r - b.r) || !sgn(d - fabs1(a.r - b.r)))
    | | ↪ return 1;
82 | | return 2;
83 | }
84
85 | }
86

```

```

87 vp cc_intersection(cc a, cc b) { // 仅返回有限交点; 重合圆返回
    ↪ 空集
88 | int cnt = cc_intersection_count(a, b);
89 | if (cnt <= 0) return {};
90 | if (a.c == b.c) return {a.c}; // 重合的零半径圆
91 | LD d = (a.c - b.c).len();
92 | P v = (b.c - a.c).unit();
93 | LD x = (a.r * a.r - b.r * b.r + d * d) / (2 * d);
94 | LD h = sqrt1(max(0.0L, a.r * a.r - x * x));
95 | P p = a.c + v * x;
96 | if (cnt == 1) return {p};
97 | return {p + v.rot90() * h, p - v.rot90() * h};
98 }
99
100 vp tangent(cp a, cc b) { // 点到圆的切点
101 | return cc_intersection(b, C(a, b.c));
102 }
103
104 vector<L> tangent(cc a, cc b, int f) { // f=1 外公切线, f=-1
    ↪ 内公切线
105 | P d = b.c - a.c;
106 | LD dr = b.r - f * a.r, d2 = d.len2();
107 | if (!sgn(d2) || sgn(d2 - dr * dr) < 0) return {};
108 | LD h = sqrt1(max(0.0L, d2 - dr * dr));
109 | vector<L> ret;
110 | for (int s : {-1, 1}) {
111 | | P v = (d * dr + d.rot90() * (h * s)) / d2;
112 | | ret.push_back({a.c + v * a.r, b.c + v * (f * b.r)});
113 | | if (!sgn(h)) break;
114 | }
115 | return ret;
116 }

```

8.3.2 圆反演与阿波罗尼茨圆

所有关于两点 A, B 满足 $PA/PB = k$ 且不等于 1 的点 P 的轨迹是一个圆。

圆幂: 半径为 R 的圆 O , 任意一点 P 到 O 的幂为 $h = OP^2 - R^2$

圆幂定理: 过 P 的直线交圆在 A 和 B 两点, 则 $PA \cdot PB = |h|$

根轴: 到两圆等幂点的轨迹是一条垂直于连心线的直线

反演: 已知一圆 C , 圆心为 O , 半径为 r , 如果 P 与 P' 在过圆心 O 的直线上, 且 $OP \cdot OP' = r^2$, 则称 P 与 P' 关于 O 互为反演。一般 C 取单位圆。

反演的性质:

不过反演中心的直线的反形是过反演中心的圆, 反之亦然。

不过反演中心的圆, 它的反形是一个不过反演中心的圆。

两条直线在交点 A 的夹角, 等于它们的反形在相应点 A' 的夹角, 但方向相反。

两个相交圆在交点 A 的夹角等于它们的反形在相应点 A' 的夹角, 但方向相反。

直线和圆在交点 A 的夹角等于它们的反演图形在相应点 A' 的夹角, 但方向相反。

正交圆反形也正交。相切圆反形也相切, 当切点为反演中心时, 反形为两条平行线。两

两相切的圆 r_1, r_2, r_3 , 求与他们都相切的圆 r_4 。分母取负号, 答案再取绝对值, 为外切圆半径。分母取正号为内切圆半径。 $r_4^{\pm} = \frac{r_1 r_2 r_3}{r_1 r_2 + r_1 r_3 + r_2 r_3 \pm 2\sqrt{r_1 r_2 r_3 (r_1 + r_2 + r_3)}}$

```

1 C inv_c2c(P O, LD R, C A) {
2 | LD OA = (A.c - O).len();
3 | LD RB = 0.5 * R * R * (1 / (OA - A.r) - 1 / (OA + A.r));
4 | LD OB = OA * RB / A.r;
5 | P B = O + (A.c - O) * (OB / OA);
6 | return {B, RB};
7 } // 点 O 在圆 A 外, 求圆 A 的反演圆 B, R 是反演半径
8 C inv_l2c(P O, LD R, L l) {
9 | P p = project_to_line(O, l);
10 | LD d = (O - p).len();
11 | LD RB = R * R / (2 * d);
12 | P VB = (p - O) / d * RB;
13 | return {O + VB, RB};
14 } // 不过 O 点的直线反演为过 O 点的圆, R 是反演半径
15 L inv_c2l(P O, LD R, C A) {
16 | LD t = R * R / (2 * A.r);
17 | P p = O + (A.c - O).unit() * t;
18 | return {p, p + (O - p).rot90()};
19 } // 过 O 点的圆反演为不过 O 点的直线, R 是反演半径

```

8.3.3 圆并

```

1 int circle_count;
2 C circles[MAXN];
3 LD coverage_area[MAXN]; // coverage_area[k]: 恰被 k 个圆覆盖
    ↪ 的面积
4
5 struct Event {
6 | LD ang;
7 | int delta;
8 | bool operator<(const Event &o) const { return ang <
    ↪ o.ang; }
9 };
10
11 bool same_circle(cc a, cc b) {

```

```

12 | return a.c == b.c && !sgn(a.r - b.r);
13 }
14 bool contains(cc a, cc b) { // 圆 a 包含圆 b
15 | return sgn(a.r - b.r - (a.c - b.c).len()) >= 0;
16 }
17
18 void add_interval(LD l, LD r, vector<Event> &e, int &base) {
19 | const LD tau = 2 * pi;
20 | while (l < 0) l += tau, r += tau;
21 | while (l >= tau) l -= tau, r -= tau;
22 | if (r < tau) e.push_back({l, 1}), e.push_back({r, -1});
23 | else {
24 | | ++base;
25 | | e.push_back({r - tau, -1});
26 | | e.push_back({l, 1});
27 | }
28 }
29
30 void circle_union() {
31 | fill(coverage_area, coverage_area + circle_count + 2, 0);
32 | vector<LD> delta(circle_count + 2);
33 | for (int i = 0; i < circle_count; ++i) {
34 | | bool duplicate = false;
35 | | for (int j = 0; j < i; ++j)
36 | | | if (same_circle(circles[i], circles[j])) duplicate
    ↪ = true;
37 | | if (duplicate) continue;
38 |
39 | | int multiplicity = 0;
40 | | for (int j = 0; j < circle_count; ++j)
41 | | | multiplicity += same_circle(circles[i],
    ↪ circles[j]);
42 | | int cnt = multiplicity;
43 | | vector<Event> e;
44 | | for (int j = 0; j < circle_count; ++j) if (i != j) {
45 | | | if (same_circle(circles[i], circles[j])) continue;
46 | | | if (contains(circles[j], circles[i])) { ++cnt;
    ↪ continue; }
47 | | | if (contains(circles[i], circles[j])) continue;
48 | | | LD d = (circles[j].c - circles[i].c).len();
49 | | | if (sgn(d - circles[i].r - circles[j].r) >= 0)
    ↪ continue;
50 | | | LD a = atan2(circles[j].c.y - circles[i].c.y,
51 | | | | circles[j].c.x - circles[i].c.x);
52 | | | LD b = acos1(clamp((circles[i].r * circles[i].r + d
    ↪ * d
53 | | | - circles[j].r * circles[j].r) / (2 *
    ↪ circles[i].r * d),
54 | | | -1.0L, 1.0L));
55 | | | add_interval(a - b, a + b, e, cnt);
56 | | }
57 | | e.push_back({0, 0});
58 | | e.push_back({2 * pi, 0});
59 | | sort(e.begin(), e.end());
60 | | for (int j = 0; j + 1 < (int)e.size(); ++j) {
61 | | | cnt += e[j].delta;
62 | | | LD l = e[j].ang, r = e[j + 1].ang, d = r - l;
63 | | | P p = circles[i].c + P(cos(l), sin(l)) *
    ↪ circles[i].r;
64 | | | P q = circles[i].c + P(cos(r), sin(r)) *
    ↪ circles[i].r;
65 | | | LD area = (p ^ q) / 2
66 | | | + circles[i].r * circles[i].r * (d - sin(d)) /
    ↪ 2;
67 | | | delta[cnt - multiplicity + 1] += area;
68 | | | delta[cnt + 1] -= area;
69 | | }
70 | }
71 | for (int k = 1; k <= circle_count; ++k)
72 | | coverage_area[k] = coverage_area[k - 1] + delta[k];
73 | for (int k = 1; k < circle_count; ++k)
74 | | coverage_area[k] -= coverage_area[k + 1];
75 }

```

8.3.4 多边形与圆交

```

1 LD angle (cp u, cp v) {
2 | return atan2(fabs1(u ^ v), u * v); }
3 LD circle_edge_area2(cp s, cp t, LD r) { // 2 * area
4 | if (!sgn(s ^ t)) return 0;
5 | LD theta = angle(s, t);
6 | LD d = point_to_segment({0, 0}, {s, t});
7 | if (sgn(d - r) >= 0) return theta * r * r;
8 | auto q = lc_intersection({s, t}, C({0, 0}, r));

```

```

9 | if (q.size() < 2) return theta * r * r;
10 | P lo = sgn(s ^ q[0]) >= 0 ? q[0] : s;
11 | P hi = sgn(q[1] ^ t) >= 0 ? q[1] : t;
12 | return (lo ^ hi) + (theta - angle(lo, hi)) * r * r; }
LD polygon_circle_intersection_area(cvp p, cc c) {
14 | LD ret = 0;
15 | for (int i = 0; i < (int) p.size(); ++i) {
16 | | auto u = p[i] - c.c;
17 | | auto v = p[(i + 1) % p.size()] - c.c;
18 | | int s = sgn(u ^ v);
19 | | if (s > 0) ret += circle_edge_area2(u, v, c.r);
20 | | else if (s < 0) ret -= circle_edge_area2(v, u, c.r);
21 | } return fabs1(ret) / 2; } // ret 在 p 逆时针时为正

```

8.3.5 圆上整点

```

1 | vector<LL> solve(LL r) {
2 | | auto isqrt = [](LL n) {
3 | | | LL x = (LL)sqrt1((LD)n);
4 | | | while ((__int128)(x + 1) * (x + 1) <= n) ++x;
5 | | | while ((__int128)x * x > n) --x;
6 | | | return x;
7 | | };
8 | vector<LL> ret; // non-negative Y pos
9 | ret.push_back(0);
10 | LL l = 2 * r, s = isqrt(l);
11 | for (LL d=1; d<=s; d++) if (l%d==0) {
12 | | LL lim = isqrt(l/(2*d));
13 | | for (LL a = 1; a <= lim; a++) {
14 | | | LL b = isqrt(l/d-a*a);
15 | | | if (a*a+b*b==l/d && __gcd(a,b)==1 && a!=b)
16 | | | | ret.push_back(d*a*b);
17 | | | } if (d*d==l) break;
18 | | lim = isqrt(d/2);
19 | | for (LL a=1; a<=lim; a++) {
20 | | | LL b = isqrt(d-a*a);
21 | | | if (a*a+b*b==d && __gcd(a,b)==1 && a!=b)
22 | | | | ret.push_back(l/d*a*b);
23 | | } }
24 | ret.push_back(r);
25 | sort(ret.begin(), ret.end());
26 | ret.erase(unique(ret.begin(), ret.end()), ret.end());
27 | return ret; }

```

8.4 凸几何

8.4.1 凸包

```

1 | vp convex_hull(vp a) {
2 | | sort(a.begin(), a.end()); // 小于号 (y, x) 字典序
3 | | a.erase(unique(a.begin(), a.end()), a.end()); // 必要时去重
4 | | int n = (int) a.size(), cnt = 0;
5 | | vp ret;
6 | | for (int i = 0; i < n; ++i) {
7 | | | while (cnt > 1
8 | | | && turn(ret[cnt - 2], ret[cnt - 1], a[i]) <= 0)
9 | | | | --cnt, ret.pop_back(); // 保留凸包边界上的点: < 0
10 | | | ++cnt, ret.push_back(a[i]); }
11 | | for (int i = n - 2, fixed = cnt; i >= 0; --i) {
12 | | | while (cnt > fixed
13 | | | && turn(ret[cnt - 2], ret[cnt - 1], a[i]) <= 0)
14 | | | | --cnt, ret.pop_back(); // 保留凸包边界上的点: < 0
15 | | | ++cnt, ret.push_back(a[i]); }
16 | | if (n > 1) ret.pop_back(); // n <= 2 是凸包吗?
17 | | return ret; } // 小于号为 (y, x) 时边 [0, 2pi) 逆时针

```

8.4.2 凸包询问: 凸包内、切点、交点、最近点

```

1 | struct ConvexQuery { // a 为逆时针严格凸包
2 | | vp a;
3 | | int n;
4 | | ConvexQuery(vp points = {}) : a(move(points)),
5 | | | n((int)a.size()) {}
6 | | P at(int i) const { // 空凸包没有合法下标
7 | | | if (!n) throw out_of_range("ConvexQuery::at() called
8 | | | | on an empty polygon");
9 | | | if (i % n;
10 | | | if (i < 0) i += n;
11 | | | return a[i];
12 | | }
13 | | pair<vp, vp> monotone_chains() const { // 返回下链 (最左到
14 | | | | 最右)、上链 (最右到最左)
15 | | | if (!n) return {};
16 | | | int left = 0, right = 0;
17 | | | for (int i = 1; i < n; ++i) {

```

```

16 | | if (a[i] < a[left]) left = i;
17 | | if (a[right] < a[i]) right = i;
18 | | }
19 | | vp lower{a[left]}, upper{a[right]};
20 | | for (int i = left; i != right; ) {
21 | | | i = (i + 1) % n;
22 | | | lower.push_back(a[i]);
23 | | }
24 | | for (int i = right; i != left; ) {
25 | | | i = (i + 1) % n;
26 | | | upper.push_back(a[i]);
27 | | }
28 | | return {lower, upper};
29 | }
30 | bool inside(cp u) const { // 含边界
31 | | if (!n) return false;
32 | | if (n == 1) return u == a[0];
33 | | if (n == 2) return point_on_segment(u, {a[0], a[1]});
34 | | if (turn(a[0], a[1], u) < 0 || turn(a[0], a[n - 1], u)
35 | | | < > 0)
36 | | | return false;
37 | | int l = 1, r = n - 1;
38 | | while (l + 1 < r) {
39 | | | int m = (l + r) / 2;
40 | | | if (turn(a[0], a[m], u) >= 0) l = m;
41 | | | else r = m;
42 | | }
43 | | return turn(a[l], a[(l + 1) % n], u) >= 0;
44 | }
45 | template<class F> int extreme(F better) const {
46 | | int l = 0, r = n - 1, d = 1;
47 | | if (better(a[r], a[l])) swap(l, r), d = -1;
48 | | while (d * (r - l) > 1) {
49 | | | int m = (l + r) / 2;
50 | | | if (better(a[m], a[l]) && better(a[m], a[m - d])) l
51 | | | | = m;
52 | | | else r = m;
53 | | }
54 | | return l;
55 | }
56 | pair<int, int> tangents(cp u) const { // u 严格在凸包外, n
57 | | | < >= 3
58 | | | return {
59 | | | | extreme([&](cp x, cp y) { return turn(u, y, x) > 0;
60 | | | | | < >)),
61 | | | | extreme([&](cp x, cp y) { return turn(u, x, y) > 0;
62 | | | | | < >))};
63 | | }
64 | int edge_crossing(cp u, cp v, int l, int r) const {
65 | | int side = turn(u, v, at(l));
66 | | while (l + 1 < r) {
67 | | | int m = (l + r) / 2;
68 | | | if (side == turn(u, v, at(m))) l = m;
69 | | | else r = m;
70 | | }
71 | | return l % n;
72 | }
73 | bool line_crossings(cp u, cp v, int &i, int &j) const {
74 | | int p0 = extreme([&](cp x, cp y) {
75 | | | return ((v - u) ^ (x - u)) < ((v - u) ^ (y - u));
76 | | | });
77 | | int p1 = extreme([&](cp x, cp y) {
78 | | | return ((v - u) ^ (x - u)) > ((v - u) ^ (y - u));
79 | | | });
80 | | if (turn(u, v, a[p0]) * turn(u, v, a[p1]) >= 0) return
81 | | | false;
82 | | if (p0 > p1) swap(p0, p1);
83 | | i = edge_crossing(u, v, p0, p1);
84 | | j = edge_crossing(u, v, p1, p0 + n);
85 | | return true;
86 | }
87 | LD chain_distance(cp u, int l, int r) const {
88 | | if (l > r) r += n;
89 | | int side = sgn((u - at(l)) * (at(l + 1) - at(l)));
90 | | LD ret = point_to_segment(u, {at(l), at(l + 1)});
91 | | while (l + 1 < r) {
92 | | | int m = (l + r) / 2;
93 | | | if (side == sgn((u - at(m)) * (at(m + 1) - at(m))))
94 | | | | l = m;
95 | | | else r = m;

```

```

93 | }
94 | return min(ret, point_to_segment(u, {at(1), at(1 +
    ↳ 1})));
95 | }
96 | LD distance(cvp u) const {
97 | if (inside(u)) return 0;
98 | if (n == 1) return (u - a[0]).len();
99 | if (n == 2) return point_to_segment(u, {a[0], a[1]});
100 | auto [x, y] = tangents(u);
101 | return min(chain_distance(u, x, y), chain_distance(u,
    ↳ y, x));
102 | }
103 | };

```

8.4.3 单插入动态凸包

```

1 struct hull { // upper hull, left to right
2 set<P> a; LD tot;
3 hull () {tot = 0;}
4 template<class It> LD calc(It it) {
5 auto u = it == a.begin() ? a.end() : prev(it);
6 auto v = next(it);
7 LD ret = 0;
8 if (u != a.end()) ret += *u ^ *it;
9 if (v != a.end()) ret += *it ^ *v;
10 if (u != a.end() && v != a.end()) ret -= *u ^ *v;
11 return ret; }
12 void insert (P p) {
13 auto same_x = a.lower_bound(P(p.x,
    ↳ numeric_limits<LD>::lowest()));
14 if (same_x != a.end() && same_x->x == p.x) {
15 if (same_x->y >= p.y) return;
16 erase(same_x);
17 }
18 if (!a.size()) { a.insert (p); return; }
19 auto it = a.lower_bound (p);
20 bool out;
21 if (it == a.begin()) out = (p < *it); // special case
22 else if (it == a.end()) out = true;
23 else out = turn(*prev(it), *it, p) > 0;
24 if (!out) return;
25 while (it != a.begin()) {
26 auto o = prev(it);
27 if (o == a.begin() || turn(*prev(o), *o, p) < 0)
    ↳ break;
28 else erase(o); }
29 while (it != a.end()) {
30 auto o = next(it);
31 if (o == a.end() || turn(p, *it, *o) < 0) break;
32 else erase(it), it = o; }
33 tot += calc(a.insert(p).first); }
34 template<class It> void erase(It it) { tot -= calc(it);
    ↳ a.erase(it); } };

```

8.4.4 旋转卡壳

```

1 LD convex_diameter(cvp p) { // p 为逆时针凸包, 无重复末点
2 int n = (int)p.size();
3 if (n < 2) return 0;
4 if (n == 2) return (p[0] - p[1]).len();
5 LD ans2 = 0;
6 for (int i = 0, j = 1; i < n; ++i) {
7 int ni = (i + 1) % n;
8 while (fabs1((p[ni] - p[i]) ^ (p[(j + 1) % n] - p[i]))
9 > fabs1((p[ni] - p[i]) ^ (p[j] - p[i])) + eps)
10 j = (j + 1) % n;
11 ans2 = max({ans2, (p[i] - p[j]).len2(), (p[ni] -
    ↳ p[j]).len2()});
12 }
13 return sqrt1(ans2);
14 }
15
16 vp remove_repeated_convex_vertices(cvp p) {
17 vp q;
18 for (cp x : p) if (q.empty() || x != q.back())
    ↳ q.push_back(x);
19 if (q.size() > 1 && q.front() == q.back()) q.pop_back();
20 return q;
21 }
22
23 LD convex_min_width(cvp input) { // input 为逆时针凸包, 允许
    ↳ 相邻重复点
24 vp p = remove_repeated_convex_vertices(input);
25 int n = (int)p.size();
26 if (n < 3) return 0;

```

```

27 LD ans = INF;
28 for (int i = 0, j = 1; i < n; ++i) {
29 int ni = (i + 1) % n;
30 while ((p[ni] - p[i]) ^ (p[(j + 1) % n] - p[i]))
31 > ((p[ni] - p[i]) ^ (p[j] - p[i])) + eps)
32 j = (j + 1) % n;
33 ans = min(ans, ((p[ni] - p[i]) ^ (p[j] - p[i]))
34 / (p[ni] - p[i]).len());
35 }
36 return ans;
37 }
38
39 struct MinRectangle {
40 LD area = 0;
41 array<P, 4> p{};
42 };
43
44 MinRectangle min_area_rectangle(cvp input) { // input 为逆时
    ↳ 针凸包, 允许相邻重复点
45 vp p = remove_repeated_convex_vertices(input);
46 int n = (int)p.size();
47 if (!n) return {};
48 if (n == 1) return {0, {p[0], p[0], p[0], p[0]}};
49 MinRectangle ans{INF, {}};
50 P first_u = (p[1] - p[0]).unit(), first_v =
    ↳ first_u.rot90();
51 int high = 0, right = 0, left = 0;
52 for (int j = 1; j < n; ++j) {
53 if (p[j] * first_v > p[high] * first_v) high = j;
54 if (p[j] * first_u > p[right] * first_u) right = j;
55 if (p[j] * first_u < p[left] * first_u) left = j;
56 }
57 for (int i = 0; i < n; ++i) {
58 P u = (p[(i + 1) % n] - p[i]).unit(), v = u.rot90();
59 while (p[(high + 1) % n] * v > p[high] * v + eps)
60 high = (high + 1) % n;
61 while (p[(right + 1) % n] * u > p[right] * u + eps)
62 right = (right + 1) % n;
63 while (p[(left + 1) % n] * u < p[left] * u - eps)
64 left = (left + 1) % n;
65 LD min_u = p[left] * u, max_u = p[right] * u;
66 LD min_v = p[i] * v, max_v = p[high] * v;
67 LD area = (max_u - min_u) * (max_v - min_v);
68 if (area < ans.area) ans = {area, {
69 u * min_u + v * min_v, u * max_u + v * min_v,
70 u * max_u + v * max_v, u * min_u + v * max_v}};
71 }
72 return ans;
73 }

```

8.4.5 闵可夫斯基和

```

1 vp minkowski_sum(vp a, vp b) {
2 // size > 0, counter-clockwise, no duplicate points
3 if (a.size() == 1 || b.size() == 1) {
4 vp ret;
5 for (auto i : a) for (auto j : b) ret.push_back(i+j);
6 return ret; }
7 auto rotate_to_lowest = [](vp &p) {
8 auto it = min_element(p.begin(), p.end(), [](cp u, cp
    ↳ v) {
9 return u.y == v.y ? u.x < v.x : u.y < v.y;
10 });
11 rotate(p.begin(), it, p.end());
12 };
13 rotate_to_lowest(a);
14 rotate_to_lowest(b);
15 vp x, y;
16 for (size_t i = 0; i < a.size(); ++i)
17 x.push_back(a[(i + 1) % a.size()] - a[i]);
18 for (size_t i = 0; i < b.size(); ++i)
19 y.push_back(b[(i + 1) % b.size()] - b[i]);
20 vp ret (x.size() + y.size());
21 merge(x.begin(), x.end(), y.begin(), y.end(),
22 ret.begin(), [](cp u, cp v) {
23 return half(u) != half(v) ? half(u) > half(v) : (u ^
    ↳ v) > 0;});
24 P cur = a[0] + b[0];
25 for (auto &i : ret) swap(i, cur), cur = cur + i;
26 return ret; } // ret 可能共线, 但没有重点

```

8.4.6 直线半平面交

```

1 bool turn_left(cl a, cl b, cl c) {

```

```

2   return turn(a.s, a.t, ll_intersection(b, c)) > 0; }
3 bool is_para(cl a, cl b){return !sgn((a.t-a.s) ^
  ↪ (b.t-b.s));}
4 bool cmp(cl a, cl b) {
5   int sign = half(a.t - a.s) - half(b.t - b.s);
6   int dir = sgn((a.t - a.s) ^ (b.t - b.s));
7   if (!dir && !sign) return turn(a.s, a.t, b.t) < 0;
8   else return sign ? sign > 0 : dir > 0; }
9 vp hpi(vector<L> h) { // 半平面交
10  sort(h.begin(), h.end(), cmp);
11  vector<L> q(h.size()); int l = 0, r = -1;
12  for(auto &i : h) {
13    while (l < r && !turn_left(i, q[r-1], q[r])) --r;
14    while (l < r && !turn_left(i, q[l], q[l+1])) ++l;
15    if (l <= r && is_para(i, q[r])) continue;
16    q[++r] = i; }
17  while (r - l > 1 && !turn_left(q[l], q[r-1], q[r])) --r;
18  while (r - l > 1 && !turn_left(q[r], q[l], q[l+1])) ++l;
19  if(r - l < 2) return {};
20  vp ret(r - l + 1);
21  for(int i = l; i <= r; i++)
22  | ret[i - l] = ll_intersection(q[i], q[i == r ? l : i +
  ↪ 1]);
23  return ret; }
24 // 空集会在队列里留下一个开区域; 开区域会被判定为空集。
25 // 为了保证正确性, 一定要加足够大的框, 尽可能避免零面积区域。
26 // 实在需要零面积区域边缘, 需要仔细考虑 turn_left 的实现。

```

8.4.7 整数半平面交

```

1 struct IL : P { // ax + by + c >= 0, 适合整数半平面
2   | LD z;
3   | IL(LD a = 0, LD b = 0, LD c = 0) : P(a, b), z(c) {}
4   | IL(cp a, cp b) : P((b - a).rot90()), z(a ^ b) {}
5   | LD operator()(cp a) const { return a * static_cast<const
  ↪ P &>(*this) + z; }
6 };
7 using cil = const IL &;
8
9 P il_intersection(cil u, cil v) {
10  | return P(P(u.z, u.y) ^ P(v.z, v.y),
11  | | P(u.x, u.z) ^ P(v.x, v.z)) / -(static_cast<const P
  ↪ &>(u)
12  | | ^ static_cast<const P &>(v));
13 }
14 LD signed_distance(cil l, cp x = P()) { return l(x) /
  ↪ l.len(); }
15 bool il_parallel(cil a, cil b) { return !sgn(a ^ b); }
16 LD det3(cil a, cil b, cil c) {
17  | return (a ^ b) * c.z + (b ^ c) * a.z + (c ^ a) * b.z;
18 }
19 int check(cil a, cil b, cil c) {
20  | return sgn(det3(b, c, a)) * sgn(b ^ c);
21 }
22 bool il_turn_left(cil a, cil b, cil c) { return check(a, b,
  ↪ c) > 0; }
23 bool il_cmp(cil a, cil b) {
24  | if (il_parallel(a, b) && a * b > 0)
25  | | return signed_distance(a) < signed_distance(b);
26  | return half(a) == half(b) ? sgn(a ^ b) > 0 : half(b) > 0;
27 }
28
29 IL perpendicular(cil l) { return {l.y, -l.x, 0}; }
30 IL parallel_through(cil l, cp o) { return {l.x, l.y, l.z -
  ↪ l(o)}; }
31 P project(cp x, cil l) { return x - static_cast<const P
  ↪ &>(l) * (l(x) / l.len2()); }
32 P reflect(cp x, cil l) { return x - static_cast<const P
  ↪ &>(l) * (2 * l(x) / l.len2()); }
33 bool il_perpendicular(cil a, cil b) { return !sgn(a * b); }
34 LD triangle_area(cil a, cil b, cil c) {
35  | LD d = det3(a, b, c);
36  | return d * d / (2 * (a ^ b) * (b ^ c) * (c ^ a));
37 }
38
39 vector<IL> cut_integral_hpi(const vector<IL> &o, IL l) {
40  | vector<IL> ret;
41  | int n = (int)o.size();
42  | for (int i = 0; i < n; ++i) {
43  | | cil u = o[i], v = o[(i + 1) % n], w = o[(i + 2) % n];
44  | | int va = check(l, u, v), vb = check(l, v, w);
45  | | if (va > 0 || vb > 0 || (!va && !vb))
  ↪ ret.push_back(v);
46  | | if (va >= 0 && vb < 0) ret.push_back(l);
47  | }

```

```

48 | return ret;
49 }

```

8.5 点集算法

8.5.1 最近点对

```

1 LD closest_pair(vp p) { // O(n log n), 返回欧氏距离
2   | int n = (int)p.size();
3   | if (n < 2) return INF;
4   | sort(p.begin(), p.end());
5   | vector<P> tmp(n);
6   | function<LD(int, int)> solve = [&](int l, int r) -> LD {
7   | | if (r - l <= 3) {
8   | | | LD ret = INF;
9   | | | for (int i = l; i < r; ++i)
10  | | | | for (int j = i + 1; j < r; ++j)
11  | | | | | ret = min(ret, (p[i] - p[j]).len2());
12  | | | sort(p.begin() + l, p.begin() + r,
13  | | | | [](cp a, cp b) { return a.y == b.y ? a.x < b.x :
  ↪ a.y < b.y; });
14  | | | return ret;
15  | | }
16  | | int m = (l + r) / 2;
17  | | LD mid_x = p[m].x;
18  | | LD ret = min(solve(l, m), solve(m, r));
19  | | merge(p.begin() + l, p.begin() + m, p.begin() + m,
  ↪ p.begin() + r,
20  | | | tmp.begin(), [](cp a, cp b) {
21  | | | | return a.y == b.y ? a.x < b.x : a.y < b.y;
22  | | | });
23  | | copy(tmp.begin(), tmp.begin() + r - l, p.begin() + l);
24  | | vector<P> strip;
25  | | for (int i = l; i < r; ++i) if ((p[i].x - mid_x) *
  ↪ (p[i].x - mid_x) < ret) {
26  | | | for (int j = (int)strip.size() - 1; j >= 0
27  | | | | && (p[i].y - strip[j].y) * (p[i].y - strip[j].y)
  ↪ < ret; --j)
28  | | | | ret = min(ret, (p[i] - strip[j]).len2());
29  | | | strip.push_back(p[i]);
30  | | }
31  | | return ret;
32  | }
33 | return sqrtl(solve(0, n));
34 }

```

8.5.2 最小覆盖球

```

1 struct Sphere {
2   | p3 c; LD r;
3 };
4 bool in_sphere(p3 p, const Sphere &s) {return
  ↪ sgn((p-s.c).len()-s.r)<=0;}
5 Sphere sphere(p3 a) {return {a,0};}
6 Sphere sphere(p3 a,p3 b) {p3 c=(a+b)/2;return {c,
  ↪ (a-c).len()};}
7 Sphere sphere3(p3 a,p3 b,p3 c) {
8   | p3 u=b-a,v=c-a,n=cross(u,v);
9   | if (!sgn(n.len2())) {
10  | | Sphere s=sphere(a,b);
11  | | array<pair<p3,p3>,2> candidate{{{a,c},{b,c}}};
12  | | for (auto [x,y]:candidate) {
13  | | | Sphere t=sphere(x,y);
14  | | | if (t.r>s.r) s=t;
15  | | }
16  | | return s;
17  | }
18  | p3 o=a+(cross(n,u)*v.len2()+cross(v,n)*u.len2())
19  | | /(2*n.len2());
20  | return {o,(a-o).len()};
21 }
22 Sphere sphere4(p3 a,p3 b,p3 c,p3 d) {
23  | p3 u=b-a,v=c-a,w=d-a;
24  | LD det=mix(u,v,w);
25  | if (!sgn(det)) {
26  | | array<p3,4> q{a,b,c,d}; Sphere best{{{,INF}};
27  | | for (int i=0;i<4;++i) for (int j=i+1;j<4;++j) {
28  | | | Sphere s=sphere(q[i],q[j]); bool ok=true;
29  | | | for (p3 x:q) ok&=in_sphere(x,s);
30  | | | if (ok&&s.r<best.r) best=s;
31  | | }
32  | | for (int i=0;i<4;++i) {
33  | | | Sphere s=sphere3(q[(i+1)%4],q[(i+2)%4],q[(i+3)%4]);
  ↪ bool ok=true;
34  | | | for (p3 x:q) ok&=in_sphere(x,s);

```

```

35 | | | if (ok&&s.r<best.r) best=s;
36 | | }
37 | | return best;
38 | }
39 | p3 o=a+(cross(v,w)*(u.len2()/2)+cross(w,u)*(v.len2()/2)
40 | | +cross(u,v)*(w.len2()/2))/det;
41 | return {o,(a-o).len()};
42 | }
43 | Sphere min_sphere(vector<p3> p) { // 随机增量, O(n) 期望
44 | | if (p.empty()) return {{},0};
45 | | shuffle(p.begin(),p.end(),rnd); Sphere ret=sphere(p[0]);
46 | | for (int i=1;i<(int)p.size();++i) if
47 | | | (!in_sphere(p[i],ret)) {
48 | | | ret=sphere(p[i]);
49 | | | for (int j=0;j<i;++j) if (!in_sphere(p[j],ret)) {
50 | | | | ret=sphere(p[i],p[j]);
51 | | | | for (int k=0;k<j;++k) if (!in_sphere(p[k],ret)) {
52 | | | | | ret=sphere3(p[i],p[j],p[k]);
53 | | | | | for (int l=0;l<k;++l) if (!in_sphere(p[l],ret))
54 | | | | | | ret=sphere4(p[i],p[j],p[k],p[l]);
55 | | | | }
56 | | | }
57 | | }
58 | return ret;

```

8.5.3 Delaunay 三角剖分

```

1 /* Delaunay Triangulation 随机增量算法 :
2 节点数至少为点数的 6 倍, 空间消耗较大注意计算内存使用
3 建图的过程在 build 中, 会根据输入范围初始化内存池和初始三角形
4 Triangulation::find 返回包含某点的三角形
5 Triangulation::add_point 将某点加入三角剖分, 点在初始三角形外
6 时返回 false
7 某个 Tri 在三角剖分中当且仅当它的 has_ch 为 0
8 如果要找到三角形 u 的邻域, 则枚举它的所有 u.edge[i].tri, 该条
9 边的两个点为 u.p[(i+1)%3], u.p[(i+2)%3] */
10 const int N = 100000 + 5, MAX_TRIS = N * 6;
11 bool in_circum(cp p1, cp p2, cp p3, cp p4) {
12 | LD u11 = p1.x-p4.x, u21 = p2.x-p4.x, u31 = p3.x-p4.x;
13 | LD u12 = p1.y-p4.y, u22 = p2.y-p4.y, u32 = p3.y-p4.y;
14 | LD u13 = p1.len2()-p4.len2(), u23 = p2.len2()-p4.len2();
15 | LD u33 = p3.len2()-p4.len2();
16 | LD d = -u13*u22*u31 + u12*u23*u31 + u13*u21*u32
17 | | - u11*u23*u32 - u12*u21*u33 + u11*u22*u33;
18 | return sgn(d) > 0; }
19 LD dir(cp a, cp b, cp p) { return (b-a) ^ (p-a);}
20 typedef int SideRef; struct Tri; typedef Tri* TriRef;
21 struct Edge {
22 | TriRef tri; SideRef side; Edge() : tri(0), side(0) {}
23 | Edge(TriRef tri_, SideRef side_) : tri(tri_), side(side_)
24 | | { } };
25 struct Tri { // Triangle
26 | | P p[3];Edge edge[3];TriRef ch[3]; Tri(){ }
27 | | Tri(cp p0,cp p1,cp p2){
28 | | | p[0] = p0; p[1] = p1; p[2] = p2;
29 | | | | ch[0] = ch[1] = ch[2] = 0; }
30 | | | bool has_ch() const { return ch[0] != 0; }
31 | | | int num_ch() const {
32 | | | | return ch[0] == 0 ? 0
33 | | | | : ch[1] == 0 ? 1
34 | | | | : ch[2] == 0 ? 2 : 3; }
35 | | | bool contains(cp q) const {
36 | | | | LD a=dir(p[0],p[1],q), b=dir(p[1],p[2],q),
37 | | | | c=dir(p[2],p[0],q);
38 | | | | return sgn(a) >= 0 && sgn(b) >= 0 && sgn(c) >= 0; }
39 | | } triange_pool[MAX_TRIS], *tot_tri;
40 void set_edge(Edge a, Edge b) {
41 | if (a.tri) a.tri->edge[a.side] = b;
42 | if (b.tri) b.tri->edge[b.side] = a; }
43 class Triangulation {
44 | public:
45 | | explicit Triangulation(LD lots = 1e6L) { init(lots); }
46 | | TriRef find(P p) const { return find(the_root,p); }
47 | | bool add_point(cp p) {
48 | | | TriRef root = find(the_root,p);
49 | | | if (!root) return false;
50 | | | add_point(root,p);
51 | | | return true;
52 | | }
53 | | bool build(vp p) {
54 | | | LD bound = 1;
55 | | | for (cp x : p)
56 | | | | bound = max({bound, fabs1(x.x), fabs1(x.y)});
57 | | | init(4*bound+1);

```

```

54 | | | shuffle(p.begin(), p.end(), rnd);
55 | | | for (cp x : p) if (!add_point(x)) return false;
56 | | | return true;
57 | | }
58 private:
59 | | TriRef the_root;
60 | | void init(LD lots) {
61 | | | tot_tri = triange_pool;
62 | | | the_root = new(tot_tri++) Tri(
63 | | | | P(-lots,-lots), P(+lots,-lots), P(0,+lots));
64 | | | }
65 | | static TriRef find(TriRef root,cp p){
66 | | | for( ; ; ) {
67 | | | | if (!root->has_ch()) return root->contains(p) ?
68 | | | | | root : nullptr;
69 | | | | TriRef next = nullptr;
70 | | | | for (int i = 0; i < 3 && root->ch[i]; ++i)
71 | | | | | if (root->ch[i]->contains(p)) {
72 | | | | | | next = root->ch[i]; break;
73 | | | | | }
74 | | | | if (!next) return nullptr;
75 | | | | root = next;
76 | | | }
77 | | void add_point(TriRef root, cp p) {
78 | | | TriRef tab,tbc,tca;
79 | | | tab = new(tot_tri++) Tri(root->p[0], root->p[1],
80 | | | | p);
81 | | | tbc = new(tot_tri++) Tri(root->p[1], root->p[2],
82 | | | | p);
83 | | | tca = new(tot_tri++) Tri(root->p[2], root->p[0],
84 | | | | p);
85 | | | set_edge(Edge(tab,0),Edge(tbc,1));
86 | | | set_edge(Edge(tbc,0),Edge(tca,1));
87 | | | set_edge(Edge(tca,0),Edge(tab,1));
88 | | | set_edge(Edge(tab,2),root->edge[2]);
89 | | | set_edge(Edge(tbc,2),root->edge[0]);
90 | | | set_edge(Edge(tca,2),root->edge[1]);
91 | | | root->ch[0]=tab;root->ch[1]=tbc;root->ch[2]=tca;
92 | | | flip(tab,2); flip(tbc,2); flip(tca,2); }
93 | | void flip(TriRef tri, SideRef side_index) {
94 | | | TriRef trj = tri->edge[side_index].tri;
95 | | | int pj = tri->edge[side_index].side;
96 | | | if(!trj) return;
97 | | | if(!in_circum(tri->p[0],tri->p[1],
98 | | | | tri->p[2],trj->p[pj])) return;
99 | | | TriRef trk = new(tot_tri++)
100 | | | | Tri(tri->p[(side_index+1)%3], trj->p[pj],
101 | | | | | tri->p[side_index]);
102 | | | TriRef trl = new(tot_tri++) Tri(trj->p[(pj+1)%3],
103 | | | | tri->p[side_index], trj->p[pj]);
104 | | | set_edge(Edge(trk,0), Edge(trl,0));
105 | | | set_edge(Edge(trk,1), tri->edge[(side_index+2)%3]);
106 | | | set_edge(Edge(trk,2), trj->edge[(pj+1)%3]);
107 | | | set_edge(Edge(trl,1), trj->edge[(pj+2)%3]);
108 | | | set_edge(Edge(trl,2), tri->edge[(side_index+1)%3]);
109 | | | tri->ch[0]=trk; tri->ch[1]=trl; tri->ch[2]=0;
110 | | | trj->ch[0]=trk; trj->ch[1]=trl; trj->ch[2]=0;
111 | | | flip(trk,1); flip(trk,2); flip(trl,1); flip(trl,2);
112 | | | }
113 };
114 //The Euclidean minimum spanning tree of a set of points is
115 | | a subset of the Delaunay triangulation of the same
116 | | points.
117 //Connecting the centers of the circumcircles produces the
118 | | Voronoi diagram.
119 //No point in P is inside the circumcircle of any triangle.
120 //Maximize the minimum angle of all the angles of the
121 | | triangles.

```

8.6 球面与三维几何

8.6.1 球面基础与经纬度球面距离

球面距离: 连接球面两点的大圆劣弧 (所有曲线中最短)

球面角: 球面两个大圆弧所在半平面形成的二面角

球面凸多边形: 把一个球面多边形任意一边向两方无限延长成大圆, 其余边都在此大圆的同旁。

球面角盈 E: 球面凸n边形的内角和与 $(n-2)\pi$ 的差

离北极夹角 θ , 距离 h 的球冠: $S = 2\pi Rh = 2\pi R^2(1 - \cos \theta)$, $V = \frac{\pi h^2}{3}(3R - h)$

球面凸n边形面积: $S = ER^2$

```

1 // longitude 经度范围:  $\pm\pi$ , latitude 纬度范围:  $\pm\pi/2$ 
2 LD sphereDis(LD lon1, LD lat1, LD lon2, LD lat2, LD R) {

```

```

3 | LD d = cosl(lat1) * cosl(lat2) * cosl(lon1 - lon2) +
  |   sinl(lat1) * sinl(lat2);
4 | return R * acosl(clamp(d, -1.0L, 1.0L)); }

```

8.6.2 三维几何基础操作

```

1 struct p3 {
2 | LD x, y, z;
3 | p3(LD x_ = 0, LD y_ = 0, LD z_ = 0) : x(x_), y(y_), z(z_)
  | {}
4 | LD &operator[](int i) { return i == 0 ? x : (i == 1 ? y :
  |   z); }
5 | LD operator[](int i) const { return i == 0 ? x : (i == 1
  |   ? y : z); }
6 | LD len2() const { return x*x + y*y + z*z; }
7 | LD len() const { return sqrtl(len2()); }
8 | p3 unit() const { LD d = len(); return {x/d, y/d, z/d}; }
9 | void read() { if (scanf("%Lf%Lf%Lf", &x, &y, &z) != 3) x
  |   = y = z = 0; }
10 };
11 p3 operator+(p3 a, p3 b) { return {a.x+b.x, a.y+b.y,
  |   a.z+b.z}; }
12 p3 operator-(p3 a, p3 b) { return {a.x-b.x, a.y-b.y,
  |   a.z-b.z}; }
13 p3 operator*(p3 a, LD k) { return {a.x*k, a.y*k, a.z*k}; }
14 p3 operator*(LD k, p3 a) { return a*k; }
15 p3 operator/(p3 a, LD k) { return {a.x/k, a.y/k, a.z/k}; }
16 bool operator==(p3 a, p3 b) {
17 | return !sgn(a.x-b.x) && !sgn(a.y-b.y) && !sgn(a.z-b.z);
18 }
19 LD dot(p3 a, p3 b) { return a.x*b.x + a.y*b.y + a.z*b.z; }
20 p3 cross(p3 a, p3 b) {
21 | return {a.y*b.z-a.z*b.y, a.z*b.x-a.x*b.z,
22 |   a.x*b.y-a.y*b.x};
23 }
24 LD mix(p3 a, p3 b, p3 c) { return dot(cross(a, b), c); }
25
26 LD rotation[3][3]; // 右手系, 绕单位向量 axis 旋转
27 void rotation_matrix(p3 axis, LD angle) {
28 | axis = axis.unit();
29 | LD c = cosl(angle), s = sinl(angle);
30 | for (int i = 0; i < 3; ++i) {
31 | | int j = (i + 1) % 3, k = (i + 2) % 3;
32 | | LD x = axis[i], y = axis[j], z = axis[k];
33 | | rotation[i][i] = (y*y + z*z)*c + x*x;
34 | | rotation[i][j] = x*y*(1-c) + z*s;
35 | | rotation[i][k] = x*z*(1-c) - y*s;
36 | }
37 }
38
39 struct l3 { p3 s, t; };
40 struct plane {
41 | p3 normal; LD offset; // dot(normal, x) = offset, normal
  |   为 单位向量
42 | plane(p3 normal_ = {1,0,0}, p3 point = {})
43 | : normal(normal_.unit()), offset(dot(normal, point))
  | {}
44 };
45 p3 project_to_plane(p3 a, const plane &b) {
46 | return a + b.normal * (b.offset - dot(a, b.normal));
47 }
48 pair<p3, p3> closest_points(l3 x, l3 y) { // 两直线需异面且不
  |   平行
49 | p3 u = x.t-x.s, v = y.t-y.s, r = x.s-y.s;
50 | LD a = dot(u,u), b = dot(u,v), e = dot(v,v), d = a*e-b*b;
51 | LD c = dot(u,r), f = dot(v,r);
52 | LD s = (b*f-c*e)/d, t = (a*f-b*c)/d;
53 | return {x.s+u*s, y.s+v*t};
54 }
55 p3 line_plane_intersection(const plane &a, l3 b) {
56 | LD t =
  |   (a.offset-dot(a.normal,b.s))/dot(a.normal,b.t-b.s);
57 | return b.s+(b.t-b.s)*t;
58 }
59 l3 plane_plane_intersection(const plane &a, const plane &b)
  | {}
60 | p3 d = cross(a.normal,b.normal);
61 | LD d2 = d.len2();
62 | p3 s = cross(b.normal*a.offset-a.normal*b.offset,d)/d2;
63 | return {s,s+d};
64 }
65 p3 three_plane_intersection(const plane &a, const plane &b,
  |   const plane &c) {
66 | LD d = mix(a.normal,b.normal,c.normal);

```

```

67 | return (cross(b.normal,c.normal)*a.offset
68 |   + cross(c.normal,a.normal)*b.offset
69 |   + cross(a.normal,b.normal)*c.offset)/d;
70 }

```

8.6.3 三维凸包

```

1 // 随机增量; 输入需不存在重复点且仿射维数为 3。
2 // 点数不足、共线或全部共面时 build 返回 false, face 为空。
3 struct ConvexHull3D {
4 | using Face = array<int,3>;
5 | vector<p3> p;
6 | vector<Face> face;
7
8 | LD volume(Face f, int x) const {
9 | | return
  |   mix(p[f[1]]-p[f[0]],p[f[2]]-p[f[0]],p[x]-p[f[0]]);
10 | }
11 | Face outward(int a,int b,int c,int inside) const {
12 | | Face f{a,b,c};
13 | | if (sgn(volume(f,inside))>0) swap(f[1],f[2]);
14 | | return f;
15 | }
16 | bool build(vector<p3> points) {
17 | | p=move(points), face.clear();
18 | | if (p.size(<4) return false;
19 | | shuffle(p.begin(),p.end(),rnd);
20 | | int n=(int)p.size(),b=1,c=-1,d=-1;
21 | | for (int i=2;i<n;++i) if
  |   (cross(p[b]-p[0],p[i]-p[0]).len(>eps)
  |   < {c=i;break;})
22 | | if (c<0) return false;
23 | | swap(p[2],p[c]);
24 | | for (int i=3;i<n;++i) if
  |   (fabsl(mix(p[1]-p[0],p[2]-p[0],p[i]-p[0]))>eps)
  |   < {d=i;break;})
25 | | if (d<0) return false;
26 | | swap(p[3],p[d]);
27 | | face={outward(0,1,2,3),outward(0,3,1,2),
28 | |   outward(0,2,3,1),outward(1,3,2,0)};
29 | | for (int x=4;x<n;++x) {
30 | | | set<pair<int,int>> horizon;
31 | | | vector<Face> keep;
32 | | | for (Face f:face) if (sgn(volume(f,x))>0) {
33 | | | | for (int k=0;k<3;++k) {
34 | | | | | pair<int,int> e={f[k],f[(k+1)%3]};
35 | | | | | pair<int,int> rev={e.second,e.first};
36 | | | | | if (horizon.count(rev)) horizon.erase(rev);
  |   else horizon.insert(e);
37 | | | }
38 | | | } else keep.push_back(f);
39 | | | for (auto [u,v]:horizon) keep.push_back({u,v,x});
40 | | | face.swap(keep);
41 | | }
42 | | return true;
43 | }
44 };

```

8.7 数值工具

8.7.1 黄金三分

```

1 constexpr LD R = (sqrt(5) - 1) / 2;
2 auto split = [](LD l, LD r) { return l + (r - l) * R; };
3 LD solve(LD a, LD c, auto f) {
4 | LD b = split(a, c), bv = f(b);
5 | for (int _ = T; _; --) {
6 | | LD x = split(a, b), xv = f(x);
7 | | if (xv < bv) c = b, b = x, bv = xv; // 最小化, 注意符号
8 | | else a = c, c = x;
9 | } return bv; }

```

8.7.2 自适应 Simpson

```

1 // Adaptive Simpson's method : LD simpson::solve (LD (*f)
  |   (LD), LD l, LD r, LD eps) : integrates f over (l, r)
  |   with error eps.
2 struct simpson {
3 | LD area (LD (*f) (LD), LD l, LD r) {
4 | | LD m = l + (r - l) / 2;
5 | | return (f(l) + 4 * f(m) + f(r)) * (r - l) / 6;
6 | }
7 LD solve (LD (*f) (LD), LD l, LD r, LD eps, LD a) {
8 | LD m = l + (r - l) / 2;
9 | LD left = area (f, l, m), right = area (f, m, r);

```

```

10 | if (abs (left + right - a) <= 15 * eps) // TLE: || eps <
    |   EPS ** 2
11 | | return left + right + (left + right - a) / 15.0;
12 | return solve (f, l, m, eps / 2, left) + solve (f, m, r,
    |   eps / 2, right);
13 }
14 LD solve (LD (*f) (LD), LD l, LD r, LD eps) {
15 | return solve (f, l, r, eps, area (f, l, r));
16 };
```

9. 图论

9.1 Hopcroft-Karp $O(\sqrt{VE})$

```

1 vector<int> E[N];
2 vector<int> ml, mr, a, p;
3 void match (int nl, int nr) { // 1-based
4 | ml.assign(nl + 1, 0);
5 | mr.assign(nr + 1, 0); // nr
6 | while (true) {
7 | | bool ok = 0;
8 | | a.assign(nl + 1, 0);
9 | | p.assign(nl + 1, 0); // nl
10 | | static queue<int> q;
11 | | for (int i = 1; i <= nl; i++)
12 | | | if (!ml[i]) a[i] = p[i] = i, q.push(i);
13 | | while (!q.empty()) {
14 | | | int x = q.front(); q.pop();
15 | | | if (ml[a[x]]) continue;
16 | | | for (auto y : E[x]) {
17 | | | | if (!mr[y]) {
18 | | | | | for (ok = 1; y; x = p[x])
19 | | | | | | mr[y] = x, swap(ml[x], y);
20 | | | | | break;
21 | | | | } else if (!p[mr[y]])
22 | | | | | q.push(y = mr[y]), p[y] = x, a[y] = a[x];
23 | | | } } // while (!q.empty())
24 | | if (!ok) break; } }
25 array<vector<int>, 2> min_vertex_cover(int nl, int nr) { //
    |   Konig
26 match(nl, nr); vector<int> l, r;
27 for (int i = 1; i <= nl; i++) if (!a[i]) l.push_back(i);
28 for (int i = 1; i <= nr; i++) if (a[mr[i]]) r.push_back(i);
29 return {l, r}; }
30 vector<pair<int, int>> min_edge_cover(int nl, int nr) { // |
    |   V|-matching
31 match(nl, nr); vector<pair<int, int>> es;
32 vector<int> uL(nl+1), uR(nr+1);
33 for (int i = 1; i <= nl; i++) if (ml[i])
34 | es.push_back({i, ml[i]}), uL[i] = uR[ml[i]] = 1;
35 for (int i = 1; i <= nl; i++) if (!uL[i])
36 | for (int y : E[i]) { es.push_back({i, y}); uL[i] = uR[y] =
    |   1; break; }
37 for (int j = 1; j <= nr; j++) if (!uR[j])
38 | for (int i = 1; i <= nl && !uR[j]; i++)
39 | | for (int y : E[i]) if (y == j) { es.push_back({i, j});
    |   uR[j] = 1; break; }
40 return es; }
```

9.2 Hungarian $O(VE/w)$

```

1 using B = bitset<N>; B edge[N];
2 bool dfs(int x, B& unvis, vector<int>& match) {
3 | for(B z = edge[x];;) {
4 | | z &= unvis;
5 | | int y = z._Find_first();
6 | | if(y == N) return 0;
7 | | unvis.reset(y);
8 | | if(!match[y] || dfs(match[y], unvis, match))
9 | | | return match[y] = x, 1; } }
10 vector<int> match(int nl, int nr) {
11 | B unvis; unvis.set();
12 | vector<int> match(nr + 1), ret(nl + 1);
13 | for(int i = 1; i <= nl; i++)
14 | | if(dfs(i, unvis, match)) unvis.set();
15 | for(int i = 1; i <= nr; i++) ret[match[i]] = i;
16 | return ret[0] = 0, ret; }
```

9.3 Shuffle 一般图最大匹配 $O(VE)$

```

1 // 该算法的正确性无法保证
2 int n, m, mat[N], vis[N]; vector<int> E[N];
3 bool dfs(int tim, int x) {
4 | shuffle(E[x].begin(), E[x].end(), rng);
```

```

5 | vis[x] = tim;
6 | for (auto y : E[x]) {
7 | | int z = mat[y]; if (vis[z] == tim) continue;
8 | | mat[x] = y, mat[y] = x, mat[z] = 0;
9 | | if (!z || dfs(tim, z)) return true;
10 | | mat[x] = 0, mat[y] = z, mat[z] = y; }
11 | return false; }
12 int main() { // 暗含二分图性质跑一次即可
13 | for (int _ = 0; _ < 10; _++) {
14 | | fill(vis + 1, vis + n + 1, 0);
15 | | for (int i = 1; i <= n; ++i) if (!mat[i]) dfs(i, i);}}
```

9.4 极大团计数

```

1 // 0下标, 需删除自环 (即确保  $E_{ii} = 0$ , 补图要特别注意)
2 // 极大团计数, 最坏情况 $O(3^{n/3})$ 
3 ll ans; ull E[64];
4 #define bit(i) (1ULL << (i))
5 void dfs(ull P, ull X, ull R) { // 不要方案时可去掉R
6 | if (!P && !X) { ++ans; sol.pb(R); return; }
7 | ull Q = P & ~E[__builtin_ctzll(P | X)];
8 | for (int i; Q; Q &= ~bit(i)) {
9 | | i = __builtin_ctzll(Q);
10 | | dfs(P & E[i], X & E[i], R | bit(i));
11 | | P &= ~bit(i), X |= bit(i); }
12 ans = 0; dfs(n == 64 ? ~0ULL : bit(n) - 1, 0, 0);
```

9.5 KM 最大权匹配 $O(V^3)$

```

1 ll e[N][M]; //  $O(1^2 * r)$  最大权完备匹配; 要求  $n \leq m$  (不足则右侧
    |   补 0 列)
2 vector<int> KM(int n, int m) {
3 | if (n > m) m = n; // 右侧补虚点, e[][] 缺省为 0
4 | vector<ll> l(n + 1), r(m + 1);
5 | vector<int> p(m + 1), ans(n + 1);
6 | for (int i = 1; i <= n; ++i) {
7 | | l[i] = e[i][1];
8 | | for (int j = 2; j <= m; ++j) l[i] = max(l[i], e[i]
    |   [j]); }
9 | for (int i = 1; i <= n; ++i) {
10 | | vector<ll> d(m + 1, 1e18);
11 | | vector<int> pre(m + 1), done(m + 1);
12 | | int x, v = 0, u = 0;
13 | | for (p[0] = i; x = p[u]; u = v) {
14 | | | done[u] = 1;
15 | | | ll mn = 1e18; v = 0;
16 | | | for (int j = 1; j <= m; ++j) if (!done[j]) {
17 | | | | auto w = l[x] + r[j] - e[x][j];
18 | | | | if (w < d[j]) d[j] = w, pre[j] = u;
19 | | | | if (d[j] < mn) mn = d[j], v = j;
20 | | | }
21 | | | if (!v) break; // 无完备匹配
22 | | | for (int j = 0; j <= m; ++j) {
23 | | | | if (done[j]) l[p[j]] -= mn, r[j] += mn;
24 | | | | else d[j] -= mn;
25 | | | }
26 | | }
27 | | if (v) for (int vv; u; u = vv) vv = pre[u], p[u] =
    |   v;
28 | }
29 | for (int j = 1; j <= m; ++j) if (p[j] && p[j] <= n)
    |   ans[p[j]] = j;
30 | return ans; }
```

9.6 欧拉回路

```

1 /* comment : directed */
2 int e, cur[N]/*, deg[N]*/;
3 vector<int> E[N];
4 int id[M]; bool vis[M];
5 stack<int> stk;
6 void dfs(int u) {
7 | for (cur[u]; cur[u] < E[u].size(); cur[u]++) {
8 | | int i = cur[u];
9 | | if (vis[abs(E[u][i])]) continue;
10 | | int v = id[abs(E[u][i])] ^ u;
11 | | vis[abs(E[u][i])] = 1; dfs(v);
12 | | stk.push(E[u][i]); }
13 } // dfs for all when disconnect
14 void add(int u, int v) {
15 | id[++e] = u ^ v; // s = u
16 | E[u].push_back(e); E[v].push_back(-e);
17 | /* | E[u].push_back(e); deg[v]++; */
```

```

18 } bool valid() {
19     for (int i = 1; i <= n; i++)
20         if (E[i].size() & 1) return 0;
21     /* if (E[i].size() != deg[i]) return 0; */
22     return 1;

```

```

12 vis.set();
13 auto s = sta;
14 vector<vector<int>> ret;
15 for (int i = n - 1; i >= 0; --i) if (vis.test(s[i])) {
16     sta.clear(), dfs0(s[i], re), ret.push_back(sta);
17     return ret;

```

9.7 Blossom 一般图最大匹配 $O(|V||E|)$

```

1 int mch[MAXN], pre[MAXN], fa[MAXN], col[MAXN];
2 int hscnt = 0, hs[MAXN]; queue<int> q;
3 int find(int x) {return fa[x]==x ? x : fa[x]=find(fa[x]);}
4 int lca(int x, int y) {
5     for(++hscnt; x; y ? swap(x, y) : (void)0) {
6         x = find(x);
7         if (hs[x] == hscnt) return x;
8         hs[x] = hscnt, x = pre[mch[x]]; } }
9 void blossom(int x, int y, int rt) {
10     for (int z; find(x) != rt; y = z, x = pre[y]) {
11         pre[x] = y, z = mch[x];
12         if (col[z] == 1) q.push(z), col[z] = 0;
13         if (find(x) == x) fa[x] = rt;
14         if (find(z) == z) fa[z] = rt; } }
15 bool bfs(int st) {
16     while (!q.empty()) q.pop();
17     q.push(st), col[st] = 0;
18     while (!q.empty()) {
19         int x = q.front(); q.pop();
20         for (auto v : e[x]) {
21             if (col[v] == -1) {
22                 pre[v] = x, col[v] = 1;
23                 if (!mch[v]) {
24                     for (int y; v; swap(mch[y], v))
25                         mch[v] = y = pre[v];
26                     return true; }
27                 q.push(mch[v]); col[mch[v]] = 0;
28             } else if (col[v] == 0 && find(x) != find(v)) {
29                 int f = lca(x, v); blossom(x, v, f);
30                 blossom(v, x, f); } } }
31     return false;
32 int solve() {
33     int ans = 0;
34     for (int i = 1; i <= n; i++) {
35         for (int j = 1; j <= n; j++) col[j] = -1, fa[j] = j;
36         if (!mch[i] && bfs(i)) ans++;
37     }
    return ans;

```

9.8 2-SAT, 强连通分量 / Bitset Kosaraju

```

1 int stp, sccs, top; // N 开 **两倍**
2 int dfn[N], low[N], scc[N], stk[N], ans[N];
3 void add(int x, int a, int y, int b) { // 注意连边对称
4     E[x << 1 | a].push_back(y << 1 | b); }
5 void tarjan(int x) {
6     dfn[x] = low[x] = ++stp;
7     stk[top++] = x;
8     for (auto y : E[x]) {
9         if (!dfn[y])
10             tarjan(y), low[x] = min(low[x], low[y]);
11         else if (!scc[y])
12             low[x] = min(low[x], dfn[y]); }
13     if (low[x] == dfn[x]) {
14         sccs++;
15         do scc[stk[--top]] = sccs;
16         while (stk[top] != x); } }
17 bool solve() {
18     int cnt = n + n; stp = top = sccs = 0;
19     fill(dfn, dfn + cnt, 0); fill(scc, scc + cnt, 0);
20     for (int i = 0; i < cnt; ++i) if (!dfn[i]) tarjan(i);
21     for (int i = 0; i < n; ++i) {
22         if (scc[i << 1] == scc[i << 1 | 1]) return false;
23         ans[i] = (scc[i << 1 | 1] < scc[i << 1]); }
24     return true; }

```

```

1 bitset<N> e[N], re[N], vis; vector<int> sta;
2 void dfs0(int x, bitset<N> e[]) {
3     vis.reset(x);
4     while (true) {
5         int go = (e[x] & vis)._Find_first();
6         if (go == N) break;
7         dfs0(go, e); }
8     sta.push_back(x); }
9 vector<vector<int>> solve() { // re 需要连好反向边
10     vis.set();
11     for (int i = 1; i <= n; ++i) if (vis.test(i)) dfs0(i, e);

```

9.9 Tarjan 点双, 边双

```

1 // 强联通分量
2 void tar(int x) {
3     dfn[x] = low[x] = ++tot;
4     S[++top] = x, vis[x] = 1;
5     for (auto v : e[x]) {
6         if (!dfn[v]) tar(v), low[x] = min(low[x], low[v]);
7         else if (vis[v]) low[x] = min(low[x], dfn[v]);
8         if (dfn[x] == low[x]) {
9             ++bcnt;
10            while (S[top] != x) {
11                int v = S[top--];
12                vis[v] = 0;
13                bel[v] = bcnt; }
14            top--, vis[x] = 0, bel[x] = bcnt; } }
15 /*边双*/
16 // 注意! 建立边双树或者圆方树后【边表大小】是否开够
17 // 求边双: 无视掉 bri 后搜出每个连通块, 记得多测
18 int dfn[N], low[N], dfscnt; // clear dfn low bri dfscnt
19 bool bri[M << 1]; // 注意此处是边数
20 void tarjan(int x, int last) { // last 是边
21     dfn[x] = low[x] = ++dfscnt;
22     for (int i = head[x]; i; i = e[i].next) {
23         int v = e[i].to;
24         if (!dfn[v]) {
25             tarjan(v, i);
26             low[x] = min(low[x], low[v]);
27             if (low[v] > dfn[x]) bri[i] = bri[i ^ 1] = 1;
28             } else if (dfn[v] < dfn[x] && ((i ^ 1) != last))
29                 low[x] = min(low[x], dfn[v]); } }
30 /** 建立圆方树+求割点 **/
31 int is_cut[N], dfn[N], low[N], dfscnt, pcnt;
32 int stk[N], dep; // clear dfn low is_cut dfscnt, let pcnt=n
33 void tarjan(int x, int fa) {
34     int child = 0;
35     dfn[x] = low[x] = ++dfscnt; stk[++dep] = x;
36     for (auto v : e[x]) {
37         if (!dfn[v]) {
38             ++child; tarjan(v, x);
39             low[x] = min(low[x], low[v]);
40             if (low[v] >= dfn[x]) {
41                 is_cut[x] = true;
42                 ++pcnt; // square node index
43                 int j = 0, sz = 1;
44                 do {
45                     j = stk[dep--];
46                     addedge(pcnt, j);
47                     ++sz;
48                 } while (j != v);
49                 addedge(pcnt, x); }
50             } else if (low[x] == min(low[x], dfn[v]))
51                 if (!fa && child == 1) is_cut[x] = false; } }

```

9.10 Dominator Tree 支配树

```

1 vector<int> G[MAXN], R[MAXN], son[MAXN];
2 int ufs[MAXN]; // R 是反图, son 存的是 sdom 树上的儿子
3 int idom[MAXN], sdom[MAXN], anc[MAXN];
4 // anc: sdom的dfn最小的祖先
5 int pr[MAXN], dfn[MAXN], id[MAXN], stamp;
6 int findufs(int x) { if (ufs[x] == x) return x;
7     int t = ufs[x]; ufs[x] = findufs(ufs[x]);
8     if (dfn[sdom[anc[x]]] > dfn[sdom[anc[t]]])
9         anc[x] = anc[t];
10    return ufs[x]; }
11 void dfs(int x) {
12     dfn[x] = ++stamp; id[stamp] = x; sdom[x] = x;
13     for (int y : G[x]) if (!dfn[y]) { pr[y] = x; dfs(y); } }
14 void get_dominator(int n) {
15     for (int i = 1; i <= n; i++) ufs[i] = anc[i] = i;
16     dfs(1);
17     for (int i = stamp; i > 1; i--) { int x = id[i];
18         for (int y : R[x]) if (dfn[y]) { findufs(y);
19             if (dfn[sdom[x]] > dfn[sdom[anc[y]]])
20                 sdom[x] = sdom[anc[y]]; }
21         son[sdom[x]].push_back(x); ufs[x] = pr[x];

```

```

22 |   for (int u : son[pr[x]]) { findufs(u);
23 |   |   idom[u] = (sdom[u] == sdom[anc[u]] ?
24 |   |   |   pr[x] : anc[u]); }
25 |   |   son[pr[x]].clear(); }
26 |   for (int i = 2; i <= stamp; i++) { int x = id[i];
27 |   |   if (idom[x] != sdom[x]) idom[x] = idom[idom[x]];
28 |   |   son[idom[x]].push_back(x); } }

```

9.11 环计数

```

1 // Time complexity:  $O(m\sqrt{m})$ 
2 for(int i=1;i<=m;i++) {
3 |   g[x[i]].push_back(y[i]); g[y[i]].push_back(x[i]);
4 |   if(deg[x[i]] > deg[y[i]] || (deg[x[i]] == deg[y[i]] &&
5 |   |   x[i] > y[i])) swap(x[i], y[i]);
6 |   e[x[i]].push_back(y[i]);}
7 // 四元环计数: 第一跳走向边, 第二跳走向邻域, 终点严格在 i
8 // 之上
9 for(int i=1;i<=n;i++) {
10 |   for(auto x : e[i]) for(auto y : g[x])
11 |   |   if(y != i && (deg[i] < deg[y] || (deg[i] == deg[y] &&
12 |   |   x[i] < y)))
13 |   |   |   ans += cnt[y]++;
14 |   for(auto x : e[i]) for(auto y : g[x]) cnt[y] = 0;}

```

9.12 Dinic 最大流

单位流量网络 在 0-1 流量图上 Dinic 有更好的性质.

- 复杂度为 $O(\min\{V^{2/3}, E^{1/2}\}E)$.
- $\text{dist}(s, t) = d$, 残余网络上至多还存在 E/d 的流.
- 每个点只有一个入/出度时复杂度 $O(V^{1/2}E)$, 例如 Hopcroft-Karp.

```

1 bool bfs() { // 约定s=n+1, t=s+1
2 |   for(int i=1;i<=t;i++) dep[i] = 0, cur[i] = head[i];
3 |   queue<int> q; q.push(s); dep[s] = 1;
4 |   while(!q.empty()) {
5 |   |   int x = q.front(); q.pop();
6 |   |   for(int i=head[x];i;i=e[i].next) {
7 |   |   |   int v = e[i].to;
8 |   |   |   if(dep[v] || !e[i].w) continue;
9 |   |   |   dep[v] = dep[x] + 1; q.push(v);}
10 |   return dep[t];}
11 int dfs(int x, int in) {
12 |   if(x == t) return in;
13 |   int out = 0;
14 |   for(int &i=cur[x];i;i=e[i].next) {
15 |   |   int v = e[i].to;
16 |   |   if(dep[v] != dep[x] + 1 || !e[i].w) continue;
17 |   |   int res = dfs(v, min(in, e[i].w));
18 |   |   in -= res, out += res;
19 |   |   e[i].w -= res, e[i^1].w += res;
20 |   |   if(!in) break;}
21 |   if(!out) dep[x] = 0;
22 |   return out;}

```

9.13 原始对偶费用流

```

1 void dij() { // reversed: from t to s
2 |   for (int i = 1; i <= t; i++) dis[i] = inf;
3 |   for (int i = 1; i <= t; i++) vis[i] = 0;
4 |   dis[t] = 0;
5 |   priority_queue<Node> q; q.push({t, 0});
6 |   while(!q.empty()){
7 |   |   auto [x, t] = q.top(); q.pop();
8 |   |   if(vis[x]) continue;
9 |   |   vis[x] = 1;
10 |   |   for(int i=head[x];i;i=e[i].next){
11 |   |   |   int v = e[i].to;
12 |   |   |   if(vis[v] || !e[i^1].w) continue;
13 |   |   |   ll cost = -e[i].cost + h[x] - h[v];
14 |   |   |   if(dis[v] > dis[x] + cost) {
15 |   |   |   |   dis[v] = dis[x] + cost;
16 |   |   |   |   q.push({v, dis[v]});
17 |   |   } } }
18 bool bfs(){
19 |   for(int i=1;i<=t;i++) dep[i] = 0, cur[i] = head[i];
20 |   queue<int> q; q.push(s); dep[s] = 1;
21 |   while(!q.empty()){
22 |   |   int x = q.front(); q.pop();
23 |   |   for(int i=head[x];i;i=e[i].next){
24 |   |   |   int v = e[i].to;
25 |   |   |   ll cost = e[i].cost + h[v] - h[x];
26 |   |   |   if(dep[v] || !e[i].w || dis[v] != dis[x] - cost)
27 |   |   |   |   continue;

```

```

27 |   |   dep[v] = dep[x] + 1;
28 |   |   q.push(v);
29 |   } } return dep[t]; }
30 ll dfs(int x, ll in){
31 |   if(x == t) return in;
32 |   ll out = 0;
33 |   for(int &i=cur[x];i;i=e[i].next){
34 |   |   int v = e[i].to;
35 |   |   ll cost = e[i].cost + h[v] - h[x];
36 |   |   if(dep[v] != dep[x] + 1 ||
37 |   |   |   dis[v] != dis[x] - cost || !e[i].w) continue;
38 |   |   ll res = dfs(v, min(in, e[i].w));
39 |   |   in -= res, out += res;
40 |   |   e[i].w -= res, e[i^1].w += res;
41 |   |   if(!in) break; }
42 |   if(!out) dep[x] = 0;
43 |   return out; }
44 pair<ll, ll> solve(){ // first_sssp: DAG最短路或SPFA
45 |   ll flow = 0, cost = 0;
46 |   for (first_sssp(); dis[s] != inf; dij()) {
47 |   |   while (bfs()) {
48 |   |   |   ll tmp = dfs(s, inf);
49 |   |   |   flow += tmp, cost += tmp * (dis[s] + h[s]); }
50 |   |   for(int i=1;i<=t;i++) h[i] += min(dis[i], dis[s]);
51 |   } return {flow, cost}; }

```

9.14 网络流总结

9.15 网络流总结

最小割集, 最小割必须边以及可行边

最小割集 从 S 出发, 在残余网络中BFS所有权值非 0 的边 (包括反向边), 得到点集 $\{S\}$, 另一集为 $\{V\} - \{S\}$.

最小割集必须点 残余网络中 S 直接连向的点必在 S 的割集中, 直接连向 T 的点必在 T 的割集中; 若这些点的并集为全集, 则最小割方案唯一.

最小割可行边 在残余网络中求强联通分量, 将强联通分量缩点后, 剩余的边即为最小割可行边, 同时这些边也必然满流.

最小割必须边 在残余网络中求强联通分量, 若 S 出发可到 u , T 出发可到 v , 等价于 $\text{scc}_S = \text{scc}_u$ 且 $\text{scc}_T = \text{scc}_v$, 则该边为必须边.

常见问题

最大权闭合子图 点权, 限制条件形如: 选择 A 则必须选择 B , 选择 B 则必须选择 C, D . 建图方式: B 向 A 连边, CD 向 B 连边. 求解: S 向正权点连边, 负权点向 T 连边, 其余边容量 ∞ , 求最小割, 答案为 S 所在最小割集.

二次布尔型 (文理分科) n 个点分为两类, i 号点有 l_i 或 r_i 的代价, i, j 同属一侧分别获得 l_{ij} 或 r_{ij} 的代价, 问最小代价. $L \rightarrow i: (l_i + 1/2 \sum_j l_{ij})$, $i \rightarrow R: (r_i + 1/2 \sum_j r_{ij})$, $i \leftrightarrow j: 1/2(l_{ij} + r_{ij})$. 实现时边权乘 2 为整数, 求解后答案除 2 为整数. 图拆点可以看作二分图.

如果是二元限制是分类不同时有一个代价 d_{ij} , 建图可以简化为 $L \rightarrow i: l_i$, $i \rightarrow R: r_i$, $i \leftrightarrow j: d_{ij}$. 经典例子: xor 最小值, 按位拆开建图.

混合图欧拉回路 把无向边随便定向, 计算每个点的入度和出度, 如果有某个点出入度之差 $\text{deg}_i = \text{in}_i - \text{out}_i$ 为奇数, 肯定不存在欧拉回路. 对于 $\text{deg}_i > 0$ 的点, 连接边 $(i, T, \text{deg}_i/2)$; 对于 $\text{deg}_i < 0$ 的点, 连接边 $(S, i, -\text{deg}_i/2)$. 最后检查是否满流即可.

二物流 水源 S_1 , 水汇 T_1 , 油源 S_2 , 油汇 T_2 , 每根管道流量共用. 求流量和最大. 建超级源 SS_1 汇 TT_1 , 连边 $SS_1 \rightarrow S_1, SS_1 \rightarrow S_2, T_1 \rightarrow TT_1, T_2 \rightarrow TT_1$, 设最大流为 x_1 . 建超级源 SS_2 汇 TT_2 , 连边 $SS_2 \rightarrow S_1, SS_2 \rightarrow T_2, T_1 \rightarrow TT_2, S_2 \rightarrow TT_2$, 设最大流为 x_2 . 则最大流中水流量 $\frac{x_1+x_2}{2}$, 油流量 $\frac{x_1-x_2}{2}$.

无源汇有上下界可行流 每条边 (u, v) 有一个上界容量 $C_{u,v}$ 和下界容量 $B_{u,v}$, 我们让下界变为 0, 上界变为 $C_{u,v} - B_{u,v}$, 但这样做流量不守恒. 建立超级源点 SS 和超级汇点 TT , 用 du_i 来记录每个节点的流量情况, $du_i = \sum B_{j,i} - \sum B_{i,j}$. 添加一些附加弧. 当 $du_i > 0$ 时, 连边 (SS, i, du_i) ; 当 $du_i < 0$ 时, 连边 $(i, TT, -du_i)$. 最后对 (SS, TT) 求一次最大流即可, 当所有附加边全部满流时 (即 $\text{maxflow} ==$ 所有 $du_i > 0$ 之和) 时有可行解.

有源汇有上下界最大可行流 建立超级源点 SS 和超级汇点 TT , 首先判断是否存在可行流, 用无源汇有上下界可行流的方法判断. 增设一条从 T 到 S 没有下界容量为无穷的边, 那么原图就变成了一个无源汇有上下界可行流问题. 同样地建图后, 对 (SS, TT) 进行一次最大流, 判断是否有可行解. 如果有可行解, 删除超级源点 SS 和超级汇点 TT , 并删去 T 到 S 的这条边, 再对 (S, T) 进行一次最大流, 此时得到的 maxflow 即为有源汇有上下界最大可行流.

有源汇有上下界最小可行流 建立超级源点 SS 和超级汇点 TT , 和无源汇有上下界可行流一样新增一些边, 然后从 SS 到 TT 跑最大流. 接着加上边 (T, S, ∞) , 再从 SS 到 TT 跑一遍最大流. 如果所有新增边都是满的, 则存在可行流, 此时 T 到 S 这条边的流量即为最小可行流.

有上下界费用流 如果求无源汇有上下界最小费用可行流或有源汇有上下界最小费用最大可行流, 用 1.6.3.1/1.6.3.2 的构图方法, 给边加上费用即可. 求有源汇有上下界最小费用可行流, 要先用 1.6.3.3 的方法建图, 先求出一个保证必要边满流情况下的最小费用. 如果费用全部非负, 那么这时的费用就是答案. 如果费用有负数, 那么流多了可能更好, 继续做从 S 到 T 的流量任意的最小费用流, 加上原来的费用就是答案.

费用流消负环 新建超级源SS汇TT, 对于所有流量非空的负权边 e , 先流满 ($ans += e.f * e.c$, $e.rev.f += e.f$, $e.f = 0$), 再连边 $SS \rightarrow e.to$, $e.from \rightarrow TT$, 流量均为 $e.f(> 0)$, 费用均为0. 再连边 $T \rightarrow S$ 流量 ∞ 费用0. 此时没有负环了. 做一遍SS到TT的最小费用最大流, 将费用累加ans, 拆掉 $T \rightarrow S$ 的那条边 (此边的流量为残量网络中 $S \rightarrow T$ 的流量). 此时负环已消, 再继续跑最小费用最大流.

整数线性规划转费用流

首先将约束关系转化为所有变量下界为 0, 上界没有要求, 并满足一些等式, 每个变量在均在等式左边且出现恰好两次, 系数为 +1 和 -1, 优化目标为 $\max \sum v_i x_i$ 的形式. 将等式看做点, 等式右边的值 b_i 若为正, 则 S 向 i 连边 $(b_i, 0)$, 否则向 T 连边 $(-b_i, 0)$. 将变量看做边, 记变量 x_i 的上界为 m_i (无上界则 $m_i = \infty$), 将 x_i 系数为 +1 的那个等式 u 向系数为 -1 的等式 v 连边 (m_i, v_i) .

9.16 Gomory-Hu 无向图最小割树 $O(V^3 E)$

每次随便找两个点 s, t 求在原图的最小割, 在最小割树上连 (s, t, w_{cut}) , 递归对由割集隔开的部分继续做. 在得到的树上, 两点最小割即为树上瓶颈路. 实现时, 由于是随意找点, 可以写为分治的形式.

9.17 Stoer-Wagner 无向图最小割 $O(VE + V^2 \log V)$

```
1 const int N = 601;
2 int f[N], siz[N], G[N][N];
3 int getf(int x) {return f[x] == x ? x : f[x] = getf(f[x]);}
4 int dis[N], vis[N], bin[N];
5 int n, m;
6 int contract(int &s, int &t) { // Find s,t
7     memset(vis, 0, sizeof(vis));
8     memset(dis, 0, sizeof(dis));
9     int i, j, k, mincut, maxc;
10    for (i = 1; i <= n; i++) {
11        k = -1; maxc = -1;
12        for (j = 1; j <= n; j++)
13            if (!bin[j] && !vis[j] && dis[j] > maxc) {
14                k = j;
15                maxc = dis[j];
16            }
17        if (k == -1) return mincut;
18        s = t; t = k; mincut = maxc; vis[k] = true;
19        for (j = 1; j <= n; j++)
20            if (!bin[j] && !vis[j]) dis[j] += G[k][j];
21    } return mincut;
22 }
23 const int inf = 0x3f3f3f3f;
24 int solve() {
25     int mincut, i, j, s, t, ans;
26     if (n <= 1) return 0;
27     memset(bin, 0, sizeof(bin));
28     for (mincut = inf, i = 1; i < n; i++) {
29         ans = contract(s, t);
30         bin[t] = true;
31         if (mincut > ans) mincut = ans;
32         if (mincut == 0) return 0;
33         for (j = 1; j <= n; j++)
34             if (!bin[j]) G[s][j] = (G[j][s] += G[j][t]);
35     } return mincut;
36 }
37 int main() {
38     cin >> n >> m;
39     for (int i = 1; i <= n; ++i) f[i] = i, siz[i] = 1;
40     for (int i = 1, u, v, w; i <= m; ++i) {
41         cin >> u >> v >> w;
42         int fu = getf(u), fv = getf(v);
43         if (fu != fv) {
44             if (siz[fu] > siz[fv]) swap(fu, fv);
45             f[fu] = fv, siz[fv] += siz[fu];
46             G[u][v] += w, G[v][u] += w;
47         }
48     }
49     cout << (siz[getf(1)] != n ? 0 : solve());
50 }
```

9.18 弦图

弦图的定义 连接环中不相邻的两个点的边称为弦. 一个无向图称为弦图, 当图中任意长度都大于 3 的环都至少有一个弦.

单纯点 一个点称为单纯点当 $\{v\} \cup A(v)$ 的导出子图为一个团. 任何一个弦图都至少有一个单纯点, 不是完全图的弦图至少有两个不相邻的单纯点.

完美消除序列 一个序列 $v_1, v_2, \dots, v_n$ 满足 v_i 在 $v_i, \dots, v_n$ 的诱导子图中为一个单纯点. 一个无向图是弦图当且仅当它有一个完美消除序列.

最大势算法 从 n 到 1 的顺序依次给点标号. 设 $label_i$ 表示第 i 个点与多少个已标号的点相邻, 每次选择 $label$ 最大的未标号的点进行标号. 用桶维护优先队列可以做到 $O(n + m)$.

弦图的判定 判定最大势算法输出是否合法即可. 如果依次判断是否构成团, 时间复杂度为 $O(nm)$. 考虑优化, 设 $v_{i+1}, \dots, v_n$ 中所有与 v_i 相邻的点依次为 $N(v_i) = \{v_{j_1}, \dots, v_{j_k}\}$. 只需判断 v_{j_1} 是否与 $v_{j_2}, \dots, v_{j_k}$ 相邻即可. 时间复杂度 $O(n + m)$.

弦图的染色 完美消除序列从后往前染色, 染上出度的 mex .

最大独立集 完美消除序列从前往后能选就选.

团数 最大团的点数. 一般团团数 $\leq$ 色数, 弦图团数 = 色数.

极大团 弦图的极大团一定为 $\{x\} \cup N(x)$.

最小团覆盖 用最少的团覆盖所有的点. 设最大独立集为 $\{p_1, \dots, p_t\}$, 则 $\{p_1 \cup N(p_1), \dots, p_t \cup N(p_t)\}$ 为最小团覆盖.

弦图 k 染色计数 $\prod_{v \in V} k - N(v) + 1$.

区间图 每个顶点代表一个区间, 有边当且仅当区间有交. 区间图是弦图, 一个完美消除序列是右端点排序.

```
1 vector<int> L[N];
2 int seq[N], lab[N], col[N], id[N], vis[N];
3 void mcs() {
4     for (int i = 0; i < n; i++) L[i].clear();
5     fill(lab + 1, lab + n + 1, 0);
6     fill(id + 1, id + n + 1, 0);
7     for (int i = 1; i <= n; i++) L[0].push_back(i);
8     int top = 0;
9     for (int k = n; k; k--) {
10         int x = -1;
11         for ( ; ; ) {
12             if (L[top].empty()) top--;
13             else {
14                 x = L[top].back(), L[top].pop_back();
15                 if (lab[x] == top) break;
16             }
17         }
18         seq[k] = x; id[x] = k;
19         for (auto v : E[x]) {
20             if (!id[v]) {
21                 L[++lab[v]].push_back(v);
22                 top = max(top, lab[v]);
23             }
24         }
25     }
26 }
27 bool check() {
28     fill(vis + 1, vis + n + 1, 0);
29     for (int i = n; i; i--) {
30         int x = seq[i];
31         vector<int> to;
32         for (auto v : E[x])
33             if (id[v] > i) to.push_back(v);
34         if (to.empty()) continue;
35         int w = to.front();
36         for (auto v : to) if (id[v] < id[w]) w = v;
37         for (auto v : E[w]) vis[v] = i;
38         for (auto v : to)
39             if (v != w && vis[v] != i) return false;
40     } return true;
41 }
42 void color() {
43     fill(vis + 1, vis + n + 1, 0);
44     for (int i = n; i; i--) {
45         int x = seq[i];
46         for (auto v : E[x]) vis[col[v]] = x;
47         for (int c = 1; !col[x]; c++)
48             if (vis[c] != x) col[x] = c;
49     }
50 }
```

9.19 Minimum Mean Cycle 最小平均值环 $O(n^2)$

```
1 // 点标号为 1, 2, ..., n. f[k][v] : 恰好 k 条边走到 v 的最短路
2 ll f[N][N]; int u[M], v[M], w[M];
3 for (int i = 0; i < N; i++) for (int j = 0; j < N; j++) f[i][j] = Inf;
4 for (int i = 1; i <= n; i++) f[0][i] = 0;
5 for (int i = 1; i <= n; i++)
6     for (int j = 0; j < m; j++)
7         f[i][v[j]] = min(f[i][v[j]], f[i - 1][u[j]] + w[j]);
8 double ans = Inf;
9 for (int i = 1; i <= n; i++) if (f[n][i] < Inf / 2) {
10     double t = -Inf; bool ok = 0;
11     for (int j = 0; j < n; j++) if (f[j][i] < Inf / 2) {
12         t = max(t, (f[n][i] - f[j][i]) / (double)(n - j)); ok = 1;
13     }
14     if (ok) ans = min(t, ans);
15 }
```

9.20 斯坦纳树

```
1 LL d[1 << 10][N]; int c[15];
2 priority_queue<pair<LL, int>> q;
3 void dij(int S) {
4     for (int i = 1; i <= n; i++) q.push(mp(-d[S][i], i));
5     while (!q.empty()) {
6         pair<LL, int> o = q.top(); q.pop();
7         if (-o.x != d[S][o.y]) continue;
8         int x = o.y;
9         for (auto v : E[x]) if (d[S][v.v] > d[S][x] + v.w) {
10             d[S][v.v] = d[S][x] + v.w;
11             q.push(mp(-d[S][v.v], v.v));
12         }
13     }
14 }
```

```

13 | for (int i = 1; i < (1 << K); i++)
14 | | for (int j = 1; j <= n; j++) d[i][j] = INF;
15 | for (int i = 0; i < K; i++) read(c[i]), d[1 << i][c[i]] =
    | ↪ 0;
16 | for (int S = 1; S < (1 << K); S++) {
17 | | for (int k = S; k > (S >> 1); k = (k - 1) & S) {
18 | | | for (int i = 1; i <= n; i++) {
19 | | | | d[S][i] = min(d[S][i], d[k][i] + d[S ^ k][i]);
20 | | | } } } dij(S);}

```

9.21 图论结论

9.21.1 最小乘积问题原理

每个元素有两个权值 $\{x_i\}$ 和 $\{y_i\}$, 要求在某个限制下 (例如生成树, 二分图匹配) 使得 $\sum x \sum y$ 最小. 对于任意一种符合限制的选取方法, 记 $X = \sum x_i, Y = \sum y_i$, 可看做平面内一点 (X, Y) . 答案必在下凸壳上, 找出该下凸壳所有点, 即可枚举获得最优答案. 可以递归求出此下凸壳所有点, 分别找出距 x, y 轴最近的两点 A, B , 分别对应于 $\sum y_i, \sum x_i$ 最小. 找出距离线段最远的点 C , 则 C 也在下凸壳上, C 点满足 $AB \times AC$ 最小, 也即

$$(X_B - X_A)Y_C + (Y_A - Y_B)X_C - (X_B - X_A)Y_A - (Y_B - Y_A)X_A$$

最小, 后两项均为常数, 因此将所以权值改成 $(X_B - X_A)y_i + (Y_B - Y_A)x_i$, 求同样问题 (例如最小生成树, 最小权匹配) 即可. 求出 C 点以后, 递归 AC, BC.

9.21.2 最小环

无向图最小环: 每次floyd到 k 时, 判断 1 到 $k-1$ 的每一个 i, j :

$$\text{ans} = \min\{\text{ans}, d(i, j) + G(i, k) + G(k, j)\}.$$

有向图最小环: 做完floyd后, $d(i, i)$ 即为经过 i 的最小环.

9.21.3 度序列的可图性

判断一个度序列是否可转化为简单图, 除了一种贪心构造的方法外, 下列方法更快速. EG定理: 将度序列从大到小排序得到 $\{d_i\}$, 此序列可转化为简单图当且仅当 $\sum d_i$ 为偶数, 且对于任意的 $1 \leq k \leq n-1$ 满足 $\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(k, d_i)$.

9.21.4 切比雪夫距离与曼哈顿距离转化

曼哈顿转切比雪夫: $(x+y, x-y)$, 适用于一些每次只能向四联通的格子走一格的问题. 切比雪夫转曼哈顿: $(\frac{x+y}{2}, \frac{x-y}{2})$, 适用于统计距离.

9.21.5 树链的交

```

1 | bool cmp(int a, int b){return dep[a]<dep[b];}
2 | path merge(path u, path v){
3 | | int d[4], c[2];
4 | | if (!u.x || !v.x) return path(0, 0);
5 | | d[0]=lca(u.x, v.x); d[1]=lca(u.x, v.y);
6 | | d[2]=lca(u.y, v.x); d[3]=lca(u.y, v.y);
7 | | c[0]=lca(u.x, u.y); c[1]=lca(v.x, v.y);
8 | | sort(d, d+4, cmp); sort(c, c+2, cmp);
9 | | if (dep[c[0]] <= dep[d[0]] && dep[c[1]] <= dep[d[2]])
10 | | | return path(d[2], d[3]);
11 | | else return path(0, 0); }

```

9.21.6 带修改MST

维护少量修改的最小生成树, 可以缩点缩边使暴力复杂度变低. (银川 21: 求有 16 个 '某两条边中至少选一条' 的限制条件的最小生成树)

找出必须边 将修改边标 $-\infty$, 在MST上的其余边为必须边, 以此缩点.

找出无用边 将修改边标 ∞ , 不在MST上的其余边为无用边, 删除之.

假设修改边数为 k , 操作后图中最多剩下 $k+1$ 个点和 $2k$ 条边.

9.21.7 差分约束

$x_r - x_l \leq c : \text{add}(l, r, c)$ $x_r - x_l \geq c : \text{add}(r, l, -c)$

9.21.8 李超线段树

添加若干条线段或直线 $(a_i, b_i) \rightarrow (a_j, b_j)$, 每次求 $[l, r]$ 上最上面的那条线段的值. 思想是让线段树中一个节点只对应一条直线, 如果在这个区间加入一条直线, 如果一段比原来的优, 一段比原来的劣, 那么判断一下两条线的交点, 判断哪条直线可以完全覆盖一段一半的区间, 把它保留, 另一条直线下传到另一半区间. 时间复杂度 $O(n \log n)$.

9.21.9 Segment Tree Beats

区间 $\min, \max$, 区间求和. 以区间取 $\min$ 为例, 额外维护最大值 m , 严格次大值 s 以及最大值个数 t . 现在假设我们要让区间 $[L, R]$ 对 x 取 $\min$, 先在线段树中定位若干个节点, 对于每个节点分三种情况讨论: 1, 当 $m \leq x$ 时, 显然这一次修改不会对这个节点产生影响, 直接退出; 2, 当 $se < x < ma$ 时, 显然这一次修改只会影响到所有最大值, 所以把 num 加上 $t * (x - ma)$, 把 ma 更新为 x , 打上标记退出; 3, 当 $se \geq x$ 时, 无法直接更新着一个节点的信息, 对当前节点的左儿子和右儿子递归处理. 单次操作均摊复杂度 $O(\log^2 n)$.

9.21.10 二分图

最小点覆盖=最大匹配数. 独立集与覆盖集互补. 最小点覆盖构造方法: 对二分图流图求割集, 跨过的边指示最小点覆盖. Hall定理 $G = (X, Y, E), |M| = |X| \Leftrightarrow \forall S \subseteq X, |S| \leq |A(S)|$.

9.21.11 稳定婚姻问题

男士按自己喜欢程度从高到底依次向每位女士求婚, 女士遇到更喜欢的男士时就接受他, 并抛弃以前的配偶. 被抛弃的男士继续按照列表向剩下的女士依次求婚, 直到所有人都有配偶. 算法一定能得到一个匹配, 而且这个匹配一定是稳定的. 时间复杂度 $O(n^2)$.

9.21.12 三元环

对于无向边 (u, v) , 如果 $\deg_u < \deg_v$, 那么连有向边 (u, v) (以点标号为第二关键字). 枚举 x 暴力即可. 时间复杂度 $O(m\sqrt{m})$.

9.21.13 图同构

令 $F_i(i) = (F_{i-1}(i) * A + \sum_{i \rightarrow j} F_{i-1}(j) * B + \sum_{j \rightarrow i} F_{i-1}(j) * C + D * (i - a)) \bmod P$, 枚举点 a , 迭代 K 次后求得的就是 a 点所对应的 $hash$ 值, 其中 K, A, B, C, D, P 为 $hash$ 参数, 可自选.

9.21.14 竞赛图 Landau's Theorem

n 个点竞赛图点按出度按升序排序, 前 i 个点的出度之和不小于 $\frac{i(i-1)}{2}$, 度数总和等于 $\frac{n(n-1)}{2}$. 否则可以用优先队列构造出方案.

9.21.15 Ramsey Theorem $R(3,3)=6, R(4,4)=18$

6 个人中存在 3 人相互认识或者相互不认识.

9.21.16 树的计数 Prufer序列

树和其prufer编码一一对应, 一颗 n 个点的树, 其prufer编码长度为 $n-2$, 且度数为 d_i 的点在prufer 编码中出现 d_i-1 次.

由树得到序列: 总共需要 $n-2$ 步, 第 i 步在当前的树中寻找具有最小标号的叶子节点, 将与其相连的点的标号设为Prufer序列的第 i 个元素 p_i , 并将此叶子节点从树中删除, 直到最后得到一个长度为 $n-2$ 的Prufer 序列和一个只有两个节点的树.

由序列得到树: 先将所有点的度赋初值为 1, 然后加上它的编号在Prufer序列中出现的次数, 得到每个点的度; 执行 $n-2$ 步, 第 i 步选取具有最小标号的度为 1 的点 u 与 $v = p_i$ 相连, 得到树中的一条边, 并将 u 和 v 的度减一. 最后再把剩下的两个度为 1 的点连边, 加入到树中.

相关结论: n 个点完全图, 每个点度数依次为 $d_1, d_2, \dots, d_n$, 这样生成树的棵数为: $\frac{(n-2)!}{(d_1-1)!(d_2-1)!\dots(d_n-1)!}$.

左边有 n_1 个点, 右边有 n_2 个点的完全二分图的生成树棵数为 $n_1^{n_2-1} \times n_2^{n_1-1}$.

m 个连通块, 每个连通块有 c_i 个点, 把他们全部连通的生成树方案数: $(\sum c_i)^{m-2} \prod c_i$

9.21.17 有根树计数 1,1,2,4,9,20,48,115,286,719,1842,4766

无标号 $a_{n+1} = 1/n \sum_{k=1}^n (\sum_{d|k} d \cdot a(d)) \cdot a(n-k+1)$

9.21.18 无根树计数

n 是奇数时, 有 $a_n - \sum_{i=1}^{n/2} a_i a_{n-i}$ 种不同的无根树.

n 时偶数时, 有 $a_n - \sum_{i=1}^{n/2} a_i a_{n-i} + \frac{1}{2} a_{n/2} (a_{n/2} + 1)$ 种不同的无根树.

9.21.19 生成树计数 Kirchhoff's Matrix-Tree Theorem

Kirchhoff Matrix $T = \text{Deg} - A$, Deg 是度数对角阵, A 是邻接矩阵. 无向图度数矩阵是每个点度数; 有向图度数矩阵是每个点入度.

邻接矩阵 $A[u][v]$ 表示 $u \rightarrow v$ 边个数, 重边按照边数计算, 自环不计入度数.

无向图生成树计数: $c = |K|$ 的任意 1 个 $n-1$ 阶主子式

有向图外向树计数: $c = |$ 去掉根所在的那阶得到的主子式 $|$

9.21.20 有向图欧拉回路计数 BEST Theorem

$$\text{ec}(G) = t_w(G) \prod_{v \in V} (\deg(v) - 1)!$$

其中 $\deg$ 为入度 (欧拉图中等于出度), $t_w(G)$ 为以 w 为根的外向树的个数. 相关计算参考生成树计数.

欧拉连通图中任意两点外向树个数相同: $t_v(G) = t_w(G)$.

9.21.21 Tutte Matrix

Tutte matrix A of a graph $G = (V, E)$:

$$A_{ij} = \begin{cases} x_{ij} & \text{if } (i, j) \in E \text{ and } i < j \\ -x_{ij} & \text{if } (i, j) \in E \text{ and } i > j \\ 0 & \text{otherwise} \end{cases}$$

where x_{ij} are indeterminates. The determinant of this skew-symmetric matrix is then a polynomial (in the variables $x_{ij}, i < j$): this coincides with the square of the pfaffian of the matrix A and is non-zero (as a polynomial) if and only if a perfect matching exists.

9.21.22 Edmonds Matrix

Edmonds matrix A of a balanced $(|U| = |V|)$ bipartite graph $G = (U, V, E)$:

$$A_{ij} = \begin{cases} x_{ij} & (u_i, v_j) \in E \\ 0 & (u_i, v_j) \notin E \end{cases}$$

where the x_{ij} are indeterminates. G 有完美匹配当且仅当关于 x_{ij} 的多项式 $\det(A_{ij})$ 不恒为 0. 完美匹配的个数等于多项式中单项式的个数.

9.21.23 有向图无环定向, 色多项式

图的色多项式 $P_G(q)$ 对图 G 的 q -染色计数。

Triangle K_3 : $x(x-1)(x-2)$

Complete graph K_n : $x(x-1)(x-2)\cdots(x-(n-1))$

Tree with n vertices: $x(x-1)^{n-1}$

Cycle C_n : $(x-1)^n + (-1)^n(x-1)$

acyclic orientations of an n -vertex graph G is $(-1)^n P_G(-1)$.

9.21.24 拟阵交问题

拟阵定义: S, T 是独立集, 则 S 子集是, 若 $|S| > |T|$, 则 S 能扩充 T . 最大带权拟阵交问题: 全集 U 中每个元素都有权 w_i . 设同一个全集 U 上有两个满足拟阵性质的集族 $\mathcal{F}_1, \mathcal{F}_2$. 对于 $k = 1..|U|$, 分别求出一个集合 S , 满足 $S \in \mathcal{F}_1 \cap \mathcal{F}_2$ 且 $|S|$ 恰好为 k 的前提下, S 中元素权值和最小. 设集合大小为 k 时已经求出了答案 S . 现在希望求出集合大小为 $k+1$ 的答案. U 中所有元素分为两个集合: 当前答案集合 S , 和剩余集合 $T = U \setminus S$. 考虑 T 中的某个元素 x_i . 记 $A = \{x_i | S \cup \{x_i\} \in \mathcal{F}_1\}$, $B = \{x_i | S \cup \{x_i\} \in \mathcal{F}_2\}$. 如果 T 中某个元素 $x_i \notin A$, 说明 x_i 加进 S 中形成了某个“环”, 从而不满足 $\mathcal{F}_1$ 的限制. 考虑这个“环”上每个元素 y_j , 满足 $S \setminus \{y_j\} \cup \{x_i\} \in \mathcal{F}_1$, 将 y_j 向每个 x_i 连边. 如果 T 中某个元素 $x_i \notin B$, 同理找出 S 中每一个元素 y_j 使得 $S \setminus \{y_j\} \cup \{x_i\} \in \mathcal{F}_2$, 将 x_i 向 y_j 连边. 现在求出从 A 到 B 的多源多汇最短路, 权值在点上, 若点属于 T 则权值为正, 否则属于 S , 权值为负. 最短路上每个 T 中的点放进 S , S 中的点放进 T , 则完成了一次增广. 由于每次增广路的起点和终点都在 T 中, 所以每次增广都会使得 $|S|$ 增加1.

最大拟阵交问题可以去掉权值直接求增广路.

9.21.25 双极定向

```
1 //无向图每条边定向后成为DAG, 使得s可达所有点可达t
2 //topo为定向后DAG的拓扑序, u->v: 拓扑序中u在v的前面.
3 int n, dfn[N], low[N], stp, p[N], ord[N], topo[N];
4 bool fail = 0, sign[N]; vector<int> e[N];
5 void dfs(int x, int fa, int s, int t){
6     dfn[x] = low[x] = ++stp;
7     ord[stp] = x, p[x] = fa;
8     if (x == s) dfs(t, x, s, t);
9     for (int y : e[x]){
10         if (x == s && y == t) continue;
11         if (!dfn[y]){
12             if (x == s) fail = true;
13             dfs(y, x, s, t);
14             low[x] = min(low[x], low[y]);
15         } else if (dfn[y] < dfn[x] && y != fa)
16             low[x] = min(low[x], dfn[y]);
17     }
18     bool bipolar_orientation(int s, int t){
19         e[s].push_back(t), e[t].push_back(s);
20         stp = fail = 0, dfs(s, s, s, t);
21         for (int i = 1; i <= n; i++){
22             if (i != s && i != t && (!dfn[i] || low[i] >= dfn[i]))
23                 fail = true;
24             if (fail) return false;
25             sign[s] = 0; //memset sign[] is not necessary
26             int pre[n+5], suf[n+5]; // list
27             suf[0] = s; pre[s] = 0, suf[s] = t;
28             pre[t] = s, suf[t] = n+1; pre[n+1] = t;
29             for (int i = 3; i <= n; i++){
30                 int v = ord[i];
31                 if (!sign[ord[low[v]])] { // insert before p[v]
32                     int P = pre[p[v]];
33                     pre[v] = P, suf[v] = p[v];
34                     suf[P] = pre[p[v]] = v;
35                 } else { // insert after p[v]
36                     int S = suf[p[v]];
37                     pre[v] = p[v], suf[v] = S;
38                     suf[p[v]] = pre[S] = v;
39                 }
40                 sign[p[v]] = !sign[ord[low[v]]];
41             }
42             for (int x = s, cnt = 0; x != n+1; x = suf[x])
43                 topo[++cnt] = x;
44             return true;
45         }
46     }
47 }
```

9.21.26 图中的环

没有奇环的图是二分图, 没有偶环的图是仙人掌. 判定没有奇环仅用深度奇偶性判即可; 判定没有偶环的图需要记录覆盖次数判定是否存在奇环有交.

10. 离线算法 (如 CDQ、莫队)

10.1 莫队二次离线

```
1 for(int i=1;i<=n;i++){sum[i] = query(arr[i]) + sum[i-1];
2     add(i);}
3 sort(Q+1, Q+1+m, cmp);
4 int l = 1, r = 0;
5 for(int i=1;i<=m;i++){
6     int ql = Q[i].l, qr = Q[i].r;
7     if(l > ql) v[r].push_back({ql, l-1, i}), Q[i].ans -=
8         sum[l-1] - sum[ql-1], l = ql;
```

```
7 | if(r < qr) v[l-1].push_back({r+1, qr, -i}), Q[i].ans +=
8     sum[qr] - sum[r], r = qr;
9 | if(l < ql) v[r].push_back({l, ql-1, -i}), Q[i].ans +=
10     sum[ql-1] - sum[l-1], l = ql;
11 | if(r > qr) v[l-1].push_back({qr+1, r, i}), Q[i].ans -=
12     sum[r] - sum[qr], r = qr;
13 }
14 // 清空贡献
15 for(int i=1;i<=n;i++){
16     add(i);
17     for(auto& [l, r, id] : v[i]){
18         ll tmp = 0;
19         for(int j=l;j<=r;j++){
20             tmp += query(arr[j]);
21             // 注意如果j <= i且会产生重复贡献的时候要扣去
22         }
23         if(id > 0) Q[id].ans += tmp;
24         else Q[-id].ans -= tmp;
25     }
26 }
27 for(int i=1;i<=m;i++) Q[i].ans += Q[i-1].ans;
28 for(int i=1;i<=m;i++) ans[Q[i].id] = Q[i].ans;
```

11. 随机化与博弈论

11.1 Python Hint

```
1 from itertools import combinations, combinations_with_replacement,
2     permutations, product
3
4 def RandomAndList():
5     import random
6     random.normalvariate(0.5, 0.1)
7     l = [str(i) for i in range(9)]
8     sorted(l, min(1), max(1), len(1))
9     random.shuffle(l)
10    l.sort(key=lambda x: ord(x)^1, reverse=True)
11    from functools import cmp_to_key
12    l.sort(key=cmp_to_key(lambda x, y: (ord(y)^1)-(ord(x)^1)))
13    for i in product('ABCD', repeat=2):
14        pass # AA AB AC AD BA BB BC BD CA CB CC CD DA DB DC DD
15    for i in permutations('ABCD', 2):
16        pass # AB AC AD BA BC BD CA CB CD DA DB DC
17    for i in combinations('ABCD', 2):
18        pass # AB AC AD BC BD CD
19    for i in combinations_with_replacement('ABCD', 2):
20        pass # AA AB AC AD BB BC BD CC CD DD
21 def FractionOperation():
22     from fractions import Fraction
23     a = Fraction(0.233).limit_denominator(1000)
24     a.numerator, a.denominator, str(a)
25 def DecimalOperation():
26     import decimal
27     from decimal import Decimal, getcontext, FloatOperation
28     getcontext().prec = 100
29     getcontext().rounding = getattr(decimal, 'ROUND_HALF_EVEN')
30     # default; other: FLOOR, CEILING, DOWN, ...
31     getcontext().traps[FloatOperation] = True
32     Decimal((0, (1, 4, 1, 4), -3)) # 1.414
33     a = Decimal(1<<31) / Decimal(100000)
34     print(f"{a:.9f}") # 21474.83648
35     print(a.sqrt(), a.ln(), a.log10(), a.exp(), a ** 2)
36 def Complex():
37     a = 1-2j
38     print(a.real, a.imag, a.conjugate())
39 def FastIO():
40     import sys, atexit, io
41     _INPUT_LINES = sys.stdin.read().splitlines()
42     input = iter(_INPUT_LINES).__next__
43     _OUTPUT_BUFFER = io.StringIO()
44     sys.stdout = _OUTPUT_BUFFER
45     @atexit.register
46     def write():
47         sys.__stdout__.write(_OUTPUT_BUFFER.getvalue())
```

11.2 Hacks: 指令集, 读入优化, Bitset, builtin, 随机种子

```
1 //fast = O3 + ffast-math + fallow-store-data-races
2 #pragma GCC optimize("Ofast")
3 2: 加include后面| 位运算 |浮点| SIMD | 优化寄存器|
4 #pragma GCC target("abm,bmi,bmi2,fma,sse2,avx2,tune=native")
5 const int SZ = 1 << 16; int getc() {
6     static char buf[SZ], *ptr = buf, *top = buf;
7     if (ptr == top) {
8         ptr = buf, top = buf + fread(buf, 1, SZ, stdin);
9         if (top == buf) return -1;
10    }
11    return *ptr++;
12 }
13 idx=b._Find_first();idx!=b.size();idx=b._Find_next(idx);
```

```

12 struct HashFunc{size_t operator()(const KEY &key)const{}};
13 __builtin_uaddll_overflow(a, b, &c) // binary big int
14 void GopersHack(int k, int n) {
15     | if (k <= 0 || k > n || n >= 31) return;
16     | for (int s = (1 << k) - 1, c, r; s < (1 << n);
17     |   c = s & -s, r = s + c, s = (((r ^ s) >> 2) / c) | r); }
18 chrono::steady_clock::now().time_since_epoch().count();

```

11.3 试机赛与纪律文件

- 入场检查— 检查身份证件：胸牌。
- 检查：手机、智能手表、金属（钥匙）等等。
- 完整测试键盘、鼠标、显示器和座椅。热身赛发现联系工作人员。
- 环境检查— 测试所有可能的提交方式，找到所有题面、排行榜位置。
- 测试 -fsanitize address, undefined, define _GLIBCXX_DEBUG
- 测试 time 命令是否能显示内存占用。/usr/bin/time -v ./a.out
- OJ检查— 测试编译器版本。C++20 cin >> (s + 1); C++17 auto [x, y]: a; C++14 [](auto x, auto y); C++11 auto; bits/stdc++.h; pb_ds。

```

#include <ext/rope>
using namespace __gnu_cxx;
rope <int> R; R.insert(y, x); R[x]; R.erase(x, 1);
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace __gnu_pbds;
tree <int, null_type, less<int>, rb_tree_tag,
tree_order_statistics_node_update> s;
s.insert(1); s.find_by_order(0); s.order_of_key(5);

```

- __int128, __float128, long double。
- pragma target CE; 代码长度限制 (256K), 尝试 NO-OUTPUT, OLE, MLE。
- for i in 1..11; do pc2submit a.cpp -y; done 有没有提交限制。默认一分钟提交 10 次; 优先队列, 上一发没有测完, 下一发优先级 -1。
- clock() 是否能够正常工作; 测试本地性能与提交性能。

```

const int N = 1 << 20;
for (int T = 4; T; T--) {
    int a[N], b[N], c[N];
    for (int i = 0; i < N; i++) a[i] = i, b[i] = N - i;
    ntt_init(N); ntt(a, N, 1); ntt(b, N, 1);
    for (int i = 0; i < N; i++) c[i] = a[i] * b[i];
    ntt(c, N, -1); assert(c[0] == 0xad223d); }
// 机房: 190ms; (1s T = 21)
// QOJ: 340+-40ms; (1s T = 11.9)
// CF: 1050+-100ms; (1s T = 3.9)

```

```

import sys
sys.set_int_max_str_digits(0) # >= Python 3.11
T = 3
for _ in range(T): # 本地 assert failed 检查下数字抄错过
    assert str(3 ** 100000 * 10 ** 100000)[:9] == "133497141"
# 机房 Pypy: 130ms; (1s T = 39)
# QOJ: 201ms; (1s T = 16)
# CF Python3: 1100ms; (1s T = 2.9) Pypy3: 310ms; (1s T = 12)

```

- 提交前检查— 保存, 编译, 测过样例, 测过 n=0/1;
- 数组大小: N/M; 时间空间限制; 输入实数? MOD 多少? 开 LL
- 调用 init, 清空时数组大小 (0, 1, n 还是 m, n+1)
- 输出取模取到正值, 输出格式
- 多组没读入完就 break
- 关流同步, 注意不要关流同步混用
- 加上需要怀疑的 assert
- 开 Ofast, 位运算开 target("abm")

- 提交后检查— 变量复用后忘记改回去
- sum 多组输入: 应只用了输入内容 (memset TL, 错下标 WA), 用前清空还是用后
- int/LL 溢出, INF/-1 大小, 浮点数 eps 和 = 0, __builtin_popcountll
- 类似 pair <LL, int> x = pair <int, int>(); 不会报警告
- 离散化与二分: lower_bound upper_bound +-1, begin(), end()
- 自定义排序: 排序方向, 比较函数为小于, 考虑坏点: 如叉积 (0, 0)
- 样例是否为对称/回文的? 考虑构造不对称的情况, 正序逆序
- 复制过的代码对应位置正确修改
- 检查 shadow 变量重名
- 分类讨论: if 嵌套, break 位置
- 图: 边标号初始是否为 1, 单双向边, 反向边边权
- 几何: 共线, sqrt(-0.0), nan / inf, 重点, 除零 det/dot, 旋转方向
- 浮点数: eps 量纲: 应当至少取到 (最短线段 × 精度要求)。
- 二分, 三分: 次数。

n	$\log_{10} n$	$n!$	$C(n, n/2)$	$\text{LCM}(1 \dots n)$	P_n
2	0.30102999	2	2	2	2
3	0.47712125	6	3	6	3
4	0.60205999	24	6	12	5
5	0.69897000	120	10	60	7
6	0.77815125	720	20	60	11
7	0.84509804	5040	35	420	15
8	0.90308998	40320	70	840	22
9	0.95424251	362880	126	2520	30
10	1	3628800	252	2520	42
11	1.04139269	39916800	462	27720	56
12	1.07918125	479001600	924	27720	77
15	1.17609126	1.31e12	6435	360360	176
20	1.30103000	2.43e18	184756	232792560	627
25	1.39794001	1.55e25	5200300	26771144400	1958
30	1.47712125	2.65e32	155117520	1.444e14	5604
P_n	37338 ₄₀	204226 ₅₀	966467 ₆₀	190569292 ₁₀₀	1e9 ₁₁₄

$n \leq$	10	100	1e3	1e4	1e5	1e6
$\max \omega(n)$	2	3	4	5	6	7
$\max d(n)$	4	12	32	64	128	240
$\arg \max d$	6	60	840	7560	83160	720720
$\pi(n)$	4	25	168	1229	9592	78498
$n \leq$	1e7	1e8	1e9	1e10	1e11	1e12
$\max \omega(n)$	8	8	9	10	10	11
$\max d(n)$	448	768	1344	2304	4032	6720
$\arg \max d$	8648640	73513440	$\times 10$	$\times 9.5$	$\times 14$	...
$\pi(n)$	664579	5761455	5.08e7	4.55e8	4.12e9	3.7e10
$n \leq$	1e13	1e14	1e15	1e16	1e17	1e18
$\max \omega(n)$	12	12	13	13	14	15
$\max d(n)$	10752	17280	26880	41472	64512	103680
$\arg \max d$	1e18: 897612484786617600 = $2^8 3^4 5^2 7^2 11 \dots 37$					
$\pi(n)$	Prime number theorem: $\pi(x) \sim x/\log(x)$					

```

1 export CXXFLAGS='-g -Wall -Wextra -Wconversion -Wshadow
   ↪ -std=gnu++20'
2 ulimit -s 1048576

```